

INTEGRATION BASED ON STANDARD RESULTS

1. Which one of the following is TRUE :

निम्न में से कौनसा एक सत्य है :

(A) $x \cdot \int \frac{dx}{x} = x \ln |x| + C$

(B) $x \cdot \int \frac{dx}{x} = x \ln |x| + Cx$

(C) $\frac{1}{\cos x} \cdot \int \cos x \, dx = \tan x + C$

(D) $\frac{1}{\cos x} \cdot \int \cos x \, dx = x + C$

Ans. B

Sol. $x \cdot \int \frac{dx}{x} = x (\ln |x| + C) = x \ln |x| + Cx$

$$\frac{1}{\cos x} \int \cos x \, dx = \frac{1}{\cos x} (\sin x + C)$$

$$= \tan x + \frac{C}{\cos x}$$

2. If $\int \frac{\sin x}{\sin(x-\alpha)} dx = Ax + B \log \sin(x-\alpha) + C$, then value of (A,B) is :

यदि $\int \frac{\sin x}{\sin(x-\alpha)} dx = Ax + B \log \sin(x-\alpha) + C$, तो (A,B) बराबर है :

(A) $(\sin \alpha, \cos \alpha)$ (B) $(\cos \alpha, \sin \alpha)$ (C) $(-\sin \alpha, \cos \alpha)$ (D) $(-\cos \alpha, \sin \alpha)$

Ans. B

Sol. $\int \frac{\sin x}{\sin(x-\alpha)} dx = \int \frac{\sin(x-\alpha+\alpha)}{\sin(x-\alpha)} dx$

$$= \int \frac{\sin(x-\alpha)\cos\alpha + \cos(x-\alpha)\sin\alpha}{\sin(x-\alpha)} dx$$

$$= \cos\alpha \int dx + \sin\alpha \int \frac{\cos(x-\alpha)}{\sin(x-\alpha)} dx$$

$$= \cos\alpha x + \sin\alpha \ln |\sin(x-\alpha)| + C$$

$$\Rightarrow A = \cos \alpha, B = \sin \alpha$$

$$(A, B) = (\cos \alpha, \sin \alpha)$$

3. $\int \frac{dx}{\cos x + \sqrt{3} \sin x} =$

(A) $\frac{1}{2} \log \tan \left(\frac{x}{2} + \frac{\pi}{12} \right) + C$

(B) $\frac{1}{2} \log \tan \left(\frac{x}{2} - \frac{\pi}{12} \right) + C$

(C) $\log \tan \left(\frac{x}{2} + \frac{\pi}{12} \right) + C$

(D) $\log \tan \left(\frac{x}{2} - \frac{\pi}{12} \right) + C$

Ans. A

Sol.
$$\int \frac{dx}{\cos x + \sqrt{3} \sin x} = \frac{1}{2} \int \frac{dx}{\frac{1}{2} \cos x + \frac{\sqrt{3}}{2} \sin x}$$

$$= \frac{1}{2} \int \sec \left(x - \frac{\pi}{3} \right) dx = \frac{1}{2} \ln \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} - \frac{\pi}{6} \right) \right| + C$$

$$= \frac{1}{2} \ln \left| \tan \left(\frac{\pi}{12} + \frac{x}{2} \right) \right| + C \quad \left\{ \int \sec \theta d\theta = \ln \left| \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) \right| + C \right\}$$

INTEGRATION BY SUBSTITUTION

4. If $\int \frac{(2x+3)dx}{x(x+1)(x+2)(x+3)+1} = C - \frac{1}{f(x)}$ where $f(x)$ is of the form of $ax^2 + bx + c$ then $(a+b+c)$ equals :

यदि $\int \frac{(2x+3)dx}{x(x+1)(x+2)(x+3)+1} = C - \frac{1}{f(x)}$ जहाँ $f(x)$, $ax^2 + bx + c$ रूप का है तब $(a+b+c)$ बराबर :

(A) 4 (B) 5 (C) 6 (D) none

Ans. B

Sol.
$$\int \frac{2x+3}{(x^2+3x)(x^2+3x+2)+1} dx$$

put $x^2 + 3x = t \Rightarrow (2x+3) dx = dt$

$$\int \frac{dt}{t(t+2)+1}; \quad \int \frac{dt}{(t+1)^2} = C - \frac{1}{t+1} = C - \frac{1}{x^2+3x+1}$$

$\Rightarrow a=1, b=3, c=1 \Rightarrow a+b+c=5$

5. Consider $f(x) = \frac{x^2}{1+x^3}$; $g(t) = \int f(t) dt$. If $g(1) = 0$ then $g(x)$ equals :

माना $f(x) = \frac{x^2}{1+x^3}$; $g(t) = \int f(t) dt$. If $g(1) = 0$ तब $g(x)$ बराबर :

(A) $\frac{1}{3} \ln(1+x^3)$ (B) $\frac{1}{3} \ln\left(\frac{1+x^3}{2}\right)$ (C) $\frac{1}{2} \ln\left(\frac{1+x^3}{3}\right)$ (D) $\frac{1}{3} \ln\left(\frac{1+x^3}{3}\right)$

Ans. B

Sol. $g(t) = \int f(t) dt = \int \frac{t^2}{1+t^3} dt$

let $1+t^3 = v$

$\Rightarrow 3t^2 dt = dv$

$= \int \frac{dv}{3v} = \frac{1}{3} \ln|v| + C$

$\Rightarrow g(t) = \frac{1}{3} \ln|1+t^3| + C$

Given $g(1) = 0$

$\Rightarrow \frac{1}{3} \ln 2 + C = 0 \Rightarrow C = -\frac{1}{3} \ln 2$

$\Rightarrow g(t) = \frac{1}{3} \ln \left| \frac{t^3+1}{2} \right|$

$\Rightarrow g(x) = \frac{1}{3} \ln \left| \frac{1+x^3}{2} \right|$

INTEGRATION BY PARTS

6. $\int \frac{\cot^{-1}(e^x)}{e^x} dx =$

(A) $\frac{1}{2} \ln(e^{2x} + 1) - \frac{\cot^{-1}(e^x)}{e^x} + x + c$

(B) $\frac{1}{2} \ln(e^{2x} + 1) + \frac{\cot^{-1}(e^x)}{e^x} + x + c$

$$(C) \frac{1}{2} \ln(e^{2x} + 1) - \frac{\cot^{-1}(e^x)}{e^x} - x + c$$

$$(D) \frac{1}{2} \ln(e^{2x} + 1) + \frac{\cot^{-1}(e^x)}{e^x} - x + c$$

Ans. C

Sol.
$$\int \frac{\cot^{-1}(e^x)}{e^x} dx = \int \underbrace{\cot^{-1}(e^x)}_I \underbrace{e^{-x}}_{II} dx$$

$$= \cot^{-1}(e^x) \cdot (-1)e^{-x} + \int \frac{-1}{1+e^{2x}} e^{-x} \cdot e^x dx$$

$$= -\cot^{-1}(e^x) e^x - \int \frac{dx}{1+(e^x)^2}$$

$$= \frac{-\cot^{-1}(e^x)}{e^x} - \frac{1}{2} \int \frac{2dx}{(1+(e^x)^2)}$$

$$= \frac{-\cot^{-1}(e^x)}{e^x} - \frac{1}{2} \int \frac{2e^{-2x}}{e^{-2x} + 1} dx$$

$$= \frac{-\cot^{-1}(e^x)}{e^x} + \frac{1}{2} \ln(e^{-2x} + 1) + C$$

$$= \frac{-\cot^{-1}(e^x)}{e^x} + \frac{1}{2} \ln(1 + e^{2x}) - x + C$$

7.
$$\int x \cdot \frac{\ln(x + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx =$$

$$(A) \sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) - x + c$$

$$(B) \frac{x}{2} \cdot \ln^2(x + \sqrt{1+x^2}) - \frac{x}{\sqrt{1+x^2}} + c$$

$$(C) \frac{x}{2} \cdot \ln^2(x + \sqrt{1+x^2}) + \frac{x}{\sqrt{1+x^2}} + c$$

$$(D) \sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) + x + c$$

Ans. A

Sol. $\int \underbrace{\ln(x + \sqrt{1+x^2})}_I \cdot \underbrace{\frac{x}{\sqrt{1+x^2}}}_{II} dx$

$$= \ln(x + \sqrt{1+x^2}) \sqrt{1+x^2} - \int \sqrt{x^2+1} \cdot \frac{1}{\sqrt{x^2+1}} dx$$

$$= \ln(x + \sqrt{1+x^2}) \sqrt{1+x^2} - x + C$$

8. $\int \frac{(\log x)}{x^2} dx =$

(A) $\frac{1}{2}(\log x + 1) + c$ (B) $-\frac{1}{x}(\log x + 1) + c$ (C) $\frac{1}{x}(\log x - 1) + c$ (D) $\log(x + 1) + c$

Ans. B

Sol. $\int \frac{\log x}{x^2} dx$

let $\ln x = t \Rightarrow x = e^t$

$$\frac{1}{x} dx = dt$$

$$= \int t e^{-t} dt = -te^{-t} - e^{-t} + C$$

$$= -e^{-t}(t + 1) + C$$

$$= -\frac{1}{x}(1 + \ln x) + C$$

9. If $\int f(x) dx = \Psi(x)$, then $\int x^5 f(x^3) dx$ is equal to :

[JEE(Main)2013]

यदि $\int f(x) dx = \Psi(x)$, तो $\int x^5 f(x^3) dx$ बराबर है :

[JEE(Main)2013]

(A) $\frac{1}{3} \left[x^3 \Psi(x^3) - \int x^2 \Psi(x^3) dx \right] + C$

(B) $\frac{1}{3} x^3 \Psi(x^3) - 3 \int x^3 \Psi(x^3) dx + C$

(C) $\frac{1}{3} x^3 \Psi(x^3) - \int x^2 \Psi(x^3) dx + C$

(D) $\frac{1}{3} \left[x^3 \Psi(x^3) - \int x^3 \Psi(x^3) dx \right] + C$

Ans. C

Sol. put $x^3 = t \Rightarrow 3x^2 dx = dt$

$$\begin{aligned}
 I &= \frac{1}{3} \int f(t) t dt \\
 &= \frac{1}{3} \left[t \int f(t) dt - \int (1 \cdot \int f(t) dt) dt \right] \\
 &= \frac{1}{3} x^3 \Psi(x^3) - \int \Psi(x^3) x^2 dx
 \end{aligned}$$

TWO CLASSIC INTEGRALS

10. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} (x + \sqrt{x}) dx =$
- (A) $2e^{\sqrt{x}} [x - \sqrt{x} + 1] + C$ (B) $e^{\sqrt{x}} [x - 2\sqrt{x} + 1]$
- (C) $e^{\sqrt{x}} (x + \sqrt{x}) + C$ (D) $e^{\sqrt{x}} (x + \sqrt{x} + 1) + C$

Ans. A

Sol. $I = \frac{e^{\sqrt{x}}}{\sqrt{x}} (x + \sqrt{x}) dx$

let $\sqrt{x} = t$

$\Rightarrow \frac{1}{2\sqrt{x}} dx = dt$

$I = 2 \int \underbrace{e^t}_{\text{II}} \underbrace{(t^2 + t)}_{\text{I}} dt$

$= 2 \left[e^t (t^2 + t) - \int (2t + 1) e^t dt \right]$

$= 2e^t (t^2 + t) - 2(2t + 1)e^t + 2 \int 2e^t dt$

$= 2 \left[e^t (t^2 + t) - (2t + 1)e^t + 2e^t \right] + C$

$= 2 \left[e^{\sqrt{x}} (x + \sqrt{x} - 2\sqrt{x} - 1 + 2) \right] + C$

$= 2e^{\sqrt{x}} (x + 1 - \sqrt{x}) + C$

11. $\int \left\{ \frac{(\log x - 1)}{1 + (\log x)^2} \right\}^2 dx =$

(A) $\frac{\log x}{(\log x)^2 + 1} + C$

(B) $\frac{x}{x^2 + 1} + C$

(C) $\frac{xe^x}{1 + x^2} + C$

(D) $\frac{x}{(\log x)^2 + 1} + C$

Ans. D

Sol. $\int \left\{ \frac{\ell nx - 1}{1 + (\ell nx)^2} \right\}^2 dx$

let $\ell nx = t \Rightarrow \frac{1}{x} dx = dt$

$= \int \left[\frac{(t-1)}{1+t^2} \right]^2 \cdot e^t dt$

$= \int e^t \left[\frac{t^2 - 2t + 1}{(1+t^2)^2} \right] dt$

$= \int e^t \left[\underbrace{\frac{1}{1+t^2}}_{f(t)} - \underbrace{\frac{2t}{(1+t^2)^2}}_{f'(t)} \right] dt$

$= e^t \cdot \left(\frac{1}{1+t^2} \right) + C = \frac{x}{1 + (\ell nx)^2} + C$

12. $\int \left(\int e^x \left(\ell nx + \frac{2}{x} - \frac{1}{x^2} \right) dx \right) dx =$

(A) $e^x \ell nx + C_1 x + C_2$

(B) $e^x \ell nx + \frac{1}{x} + C_1 x + C_2$

(C) $\frac{\ell nx}{x} + C_1 x + C_2$

(D) None of these

Ans. A

Sol.
$$\int \left[\int e^x \left(\ln x + \frac{2}{x} - \frac{1}{x^2} \right) dx \right] dx$$

$$\Rightarrow \int \left(\int \left(e^x \left(\ln x + \frac{1}{x} \right) + e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) \right) dx \right) dx$$

$$\Rightarrow \int \left(e^x \ln x + \frac{e^x}{x} + C_1 \right) dx$$

$$= e^x \ln x + C_1 x + C_2$$

13. The integral $\int \left(1 + x - \frac{1}{x} \right) e^{x+\frac{1}{x}} dx$ is equal to : [JEE(Main)2014]

समाकलन $\int \left(1 + x - \frac{1}{x} \right) e^{x+\frac{1}{x}} dx$ बराबर है : [JEE(Main)2014]

(A) $(x+1)e^{x+\frac{1}{x}} + c$ (B) $-xe^{x+\frac{1}{x}} + c$ (C) $(x-1)e^{x+\frac{1}{x}} + c$ (D) $xe^{x+\frac{1}{x}} + c$

Ans. D

Sol.
$$\int \left(1 + x - \frac{1}{x} \right) e^{x+\frac{1}{x}} dx$$

$$= \int e^{x+\frac{1}{x}} \left(1 + x \left(1 - \frac{1}{x^2} \right) \right) dx$$

$$= \int \underbrace{1 \cdot e^{x+\frac{1}{x}}}_I dx + \int x e^{x+\frac{1}{x}} \left(1 - \frac{1}{x^2} \right) dx$$

$$= x e^{x+\frac{1}{x}} - \int x e^{x+\frac{1}{x}} \left(1 - \frac{1}{x^2} \right) dx + \int x e^{x+\frac{1}{x}} \left(1 - \frac{1}{x^2} \right) dx$$

$$= x e^{x+\frac{1}{x}} + C$$

INTEGRATION OF TRIGONOMETRIC FUNCTION

14. $\int \frac{\sin 2x + \sin 4x - \sin 6x}{1 + \cos 2x + \cos 4x + \cos 6x} dx$ is equal to : (where C is indefinite integration constant)

$\int \frac{\sin 2x + \sin 4x - \sin 6x}{1 + \cos 2x + \cos 4x + \cos 6x} dx$ बराबर होगा : (जहाँ C समाकलन अचर है)

(A) $\frac{1}{3} \ln |\sec 3x| + \frac{1}{2} \ln |\sec 2x| + \ln |\sec x| + C$

(B) $\frac{1}{3} \ln |\sec 3x| - \frac{1}{2} \ln |\sec 2x| - \ln |\sec x| + C$

(C) $C - \frac{1}{3} \ln |\sec 3x| + \frac{1}{2} \ln |\sec 2x| + \ln |\sec x|$

(D) $C - \frac{1}{3} \ln |\sec 3x| - \frac{1}{2} \ln |\sec 2x| - \ln |\sec x|$

Ans. B

Sol. $I = \int \frac{\sin 2x + \sin 4x - \sin 6x}{1 + \cos 2x + \cos 4x + \cos 6x} dx = \int \frac{2 \sin 3x \cos x - 2 \sin 3x \cos 3x}{2 \cos^2 x + 2 \cos x \cos 5x} dx$
 $= \int \frac{\sin 3x (\cos x - \sin 3x)}{\cos x (\cos x + \cos 5x)} dx = \int \frac{2 \sin 3x \sin 2x \sin x}{2 \cos x \cos 2x \cos 3x} dx = \int \tan x \tan 2x \tan 3x dx$

$\therefore 3x = 2x + x \Rightarrow \tan 3x = \frac{\tan 2x + \tan x}{1 - \tan 2x \tan x} \Rightarrow \tan 3x - \tan 3x \tan 2x \tan x = \tan 2x + \tan x$

$\Rightarrow \tan 3x \tan 2x \tan x = \tan 3x - \tan 2x - \tan x$

$\therefore I = \int (\tan 3x - \tan 2x - \tan x) dx = \frac{1}{3} \ln |\sec 3x| - \frac{1}{2} \ln |\sec 2x| - \ln |\sec x| + C$

15. $\int \sec^2 \theta (\sec \theta + \tan \theta)^2 d\theta :$

(A) $\frac{(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$

(B) $\frac{(\sec \theta + \tan \theta)}{3} [2 + 4 \tan \theta (\sec \theta + \tan \theta)] + C$

(C) $\frac{(\sec \theta + \tan \theta)}{3} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$

(D) $\frac{3(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$

Ans. C

Sol. $\sec \theta + \tan \theta = y$ (1)
 $\sec \theta (\tan \theta + \sec \theta) d\theta = dy$

now, $\sec \theta - \tan \theta = \frac{1}{y}$ (2)

from (1) and (2)

$$2 \sec \theta = y + \frac{1}{y}$$

$$\therefore I = \frac{1}{2} \int y \left(y + \frac{1}{y} \right) dy = \frac{1}{2} \left[\frac{y^3}{3} + y \right] + C$$

$$\frac{1}{2} \left[\frac{(\sec \theta + \tan \theta)^3}{3} + (\sec \theta + \tan \theta) \right] = \frac{(\sec \theta + \tan \theta)}{6} [(\sec \theta + \tan \theta)^2 + 3]$$

16. $I = \int \frac{(\sin 2x)^{1/3} d(\tan^{1/3} x)}{\sin^{2/3} x + \cos^{2/3} x} =$

(A) $\frac{1}{2^{2/3}} \ln(1 + \tan^{1/3} x) + C$

(B) $\ln(1 + \tan^{2/3} x) + C$

(C) $2^{1/3} \ln(1 + \tan^{2/3} x) + C$

(D) $\frac{1}{2^{2/3}} \ln(1 + \tan^{2/3} x) + C$

Ans. D

Sol. $\int \frac{(\sin 2x)^{\frac{1}{3}}}{\sin^{\frac{2}{3}} x + \cos^{\frac{2}{3}} x} d\left(\tan^{\frac{1}{3}} x\right)$

$$= 2^{\frac{1}{3}} \int \frac{\sin^{\frac{1}{3}} x \cos^{\frac{1}{3}} x}{\left(\tan^{\frac{2}{3}} x + 1\right) \cos^{\frac{2}{3}} x} d\left(\tan^{\frac{1}{3}} x\right)$$

$$= \frac{2^{\frac{1}{3}}}{2} \int \frac{2 \tan^{\frac{1}{3}} x}{\left(1 + \left(\tan^{\frac{1}{3}} x\right)^2\right)} d\left(\tan^{\frac{1}{3}} x\right)$$

$$= \frac{1}{2^{\frac{2}{3}}} \ln\left(1 + \tan^{\frac{2}{3}} x\right) + C$$

17. If the integral $\int \frac{5 \tan x}{\tan x - 2} dx = x + a \ln |\sin x - 2 \cos x| + k$ then a is equal to : [AIEEE - 2012]

यदि समाकल $\int \frac{5 \tan x}{\tan x - 2} dx = x + a \ln |\sin x - 2 \cos x| + k$ है, तो a बराबर है : [AIEEE - 2012]

- (A) 2 (B) -1 (C) -2 (D) 1

Ans. A

Sol. $\int \frac{5 \tan x}{\tan x - 2} dx = \int \frac{5 \sin x}{\sin x - 2 \cos x} dx$

$$5 \sin x = A(\sin x - 2 \cos x) + B \frac{d}{dx} (\sin x - 2 \cos x)$$

$$= A(\sin x - 2 \cos x) + B (\cos x + 2 \sin x)$$

$$\left. \begin{array}{l} A + 2B = 5 \\ -2A + B = 0 \end{array} \right\} A = 1, B = 2$$

Now $\int \frac{5 \sin x}{\sin x - 2 \cos x} dx$

$$= \int \left(\frac{(\sin x - 2 \cos x)}{\sin x - 2 \cos x} + \frac{2(\cos x + 2 \sin x)}{\sin x - 2 \cos x} \right) dx$$

$$= x + 2 \log |\sin x - 2 \cos x| + K$$

$$a = 2$$

KUTUR PUTUR

18. $\int \frac{1-x^7}{x(1+x^7)} dx =$

(A) $\ln x + \frac{2}{7} \ln (1+x^7) + c$

(B) $\ln x - \frac{2}{7} \ln (1-x^7) + c$

(C) $\ln x - \frac{2}{7} \ln (1+x^7) + c$

(D) $\ln x + \frac{2}{7} \ln (1-x^7) + c$

Ans. C

Sol. $I = \int \frac{dx}{x} - 2 \int \frac{x^7}{x(1+x^7)} dx$

$$= \ln x - \frac{2}{7} \ln (1+x^7) + C$$

19. Primitive of $\frac{3x^4 - 1}{(x^4 + x + 1)^2}$ w.r.t. x is :

$\frac{3x^4 - 1}{(x^4 + x + 1)^2}$ का x के सापेक्ष समाकलन है :

(A) $\frac{x}{x^4 + x + 1} + c$ (B) $-\frac{x}{x^4 + x + 1} + c$ (C) $\frac{x+1}{x^4 + x + 1} + c$ (D) $-\frac{x+1}{x^4 + x + 1} + c$

Ans. B

Sol. $\int \frac{(3x^4 - 1)}{x^2(x^3 + 1 + x^{-1})^2} dx = \int \frac{(3x^2 - x^{-2})}{(x^3 + 1 + x^{-1})^2} dx = -\frac{x}{x^4 + x + 1} + c$

20. The evaluation of $\int \frac{px^{p+2q-1} - qx^{q-1}}{x^{2p+2q} + 2x^{p+q} + 1} dx$ is :

$\int \frac{px^{p+2q-1} - qx^{q-1}}{x^{2p+2q} + 2x^{p+q} + 1} dx$ बराबर :

(A) $-\frac{x^p}{x^{p+q} + 1} + C$ (B) $\frac{x^q}{x^{p+q} + 1} + C$ (C) $-\frac{x^q}{x^{p+q} + 1} + C$ (D) $\frac{x^p}{x^{p+q} + 1} + C$

Ans. C

Sol. $\int \frac{px^{p+2q-1} - qx^{q-1}}{(x^{p+q} + 1)^2} dx = \int \frac{px^{p+2q-1} - qx^{q-1}}{x^{2q}(x^p + x^{-q})^2} dx = \int \frac{px^{p-1} - qx^{-q-1}}{(x^p + x^{-q})^2} dx$

taking x^q as x^{2q} common from Denominator and take it in N^r

21. $\int \frac{dx}{\sqrt[3]{x^{5/2} (x+1)^{7/2}}}$

(A) $-\left(\frac{x+1}{x}\right)^{1/6} + C$ (B) $6\left(\frac{x+1}{x}\right)^{-1/6} + C$ (C) $\left(\frac{x}{x+1}\right)^{5/6} + C$ (D) $-\left(\frac{x}{x+1}\right)^{5/6} + C$

Ans. B

Sol. $\int \frac{dx}{\left[x^6 \left(1 + \frac{1}{x}\right)^{7/2}\right]^{1/3}} = \int \frac{dx}{x^2 \left(1 + \frac{1}{x}\right)^{7/6}}$

now put $1 + \frac{1}{x} = t^6$

$$= 6 \left(\frac{x+1}{x} \right)^{-1/6} + C$$

22. The integral $\int \frac{2x^{12} + 5x^9}{(x^5 + x^3 + 1)^3} dx$ is equal to : (where c is an arbitrary constant) [JEE(Main)2016]

समाकल $\int \frac{2x^{12} + 5x^9}{(x^5 + x^3 + 1)^3} dx$ बराबर है : (जहाँ c एक स्वेच्छ अचर है) [JEE(Main)2016]

(A) $\frac{x^5}{2(x^5 + x^3 + 1)^2} + C$ (B) $\frac{-x^{10}}{2(x^5 + x^3 + 1)^2} + C$ (C) $\frac{-x^5}{(x^5 + x^3 + 1)^2} + C$ (D) $\frac{x^{10}}{2(x^5 + x^3 + 1)^2} + C$

Ans. D

Sol. $\int \frac{(2x^{12} + 5x^9) dx}{x^{15} (1 + x^{-2} + x^{-5})^3}$

$$1 + x^{-2} + x^{-5} = t$$

$$I = - \int \frac{dt}{t^3}$$

$$= \frac{1}{2t^2} + C$$

$$= \frac{x^{10}}{2(x^5 + x^3 + 1)^2} + C$$

23. The integral $\int \frac{\sin^2 x \cos^2 x}{(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x)^2} dx$ is equal to : (Where C is a constant of integration) [JEE(Main)2018]

समाकलन $\int \frac{\sin^2 x \cos^2 x}{(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x)^2} dx$ बराबर है : (जहाँ c एक स्वेच्छ अचर है)

[JEE(Main)2018]

(A) $\frac{1}{1 + \cot^3 x} + C$ (B) $\frac{-1}{1 + \cot^3 x} + C$ (C) $\frac{1}{3(1 + \tan^3 x)} + C$ (D) $\frac{-1}{3(1 + \tan^3 x)} + C$

Ans. D

Sol. Divide numerator & denominator by $\cos^{10}x$

$$\int \frac{\tan^2 x \sec^6 x \, dx}{(\tan^5 x + \tan^2 x + \tan^3 x + 1)^2}$$

$$\int \frac{\tan^2 x \sec^6 x \, dx}{(1 + \tan^3 x)^2 (1 + \tan^2 x)^2}$$

$$\int \frac{\tan^2 x \sec^6 x \, dx}{(1 + \tan^3 x)^2}$$

Put $1 + \tan^3 x = t$

$$I = \frac{1}{3} \int \frac{dt}{t^2}$$

$$= -\frac{1}{3t} + C$$

$$= -\frac{1}{3(1 + \tan^3 x)} + C$$

MISCELLANEOUS

24. Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int x^{n-2} g(x) \, dx$ equals :

माना $f(x) = \frac{x}{(1+x^n)^{1/n}}$ तो $n \geq 2$ के लिये $g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$. तो $\int x^{n-2} g(x) \, dx$ का मान होगा :

(A) $\frac{1}{n(n-1)}(1+nx^n)^{1-\frac{1}{n}} + K$

(B) $\frac{1}{(n-1)}(1+nx^n)^{1-\frac{1}{n}} + K$

(C) $\frac{1}{n(n+1)}(1+nx^n)^{1+\frac{1}{n}} + K$

(D) $\frac{1}{(n+1)}(1+nx^n)^{1+\frac{1}{n}} + K$

Ans. A

Sol. Here $f(x) = \frac{f(x)}{[1+f(x)^n]^{1/n}} = \frac{x}{(1+2x^n)^{1/n}}$

and $f(x) = \frac{x}{(1+3x^n)^{1/n}}$

$\therefore g(x) = (f \circ f \circ \dots \circ f)(x) = \frac{x}{(1+nx^n)^{1/n}}$

$$\text{Let } I = \int x^{n-2} g(x) dx = \int \frac{x^{n-1}}{(1+nx^n)^{1/n}} dx$$

$$= \frac{1}{n^2} \int \frac{n^2 x^{n-1}}{(1+nx^n)^{1/n}} dx = \frac{1}{n^2} \int \frac{\frac{d}{dx}(1+nx^n)}{(1+nx^n)^{1/n}} dx$$

$$= \frac{1}{n(n-1)} (1+nx^n)^{1-\frac{1}{n}} + K$$

25. $\int \frac{x^2(1-\ln x)}{\ln^4 x - x^4} dx$ equals :

$$\int \frac{x^2(1-\ln x)}{\ln^4 x - x^4} dx \text{ बराबर :}$$

(A) $\frac{1}{2} \ln\left(\frac{x}{\ln x}\right) - \frac{1}{4} \ln(\ln^2 x - x^2) + C$

(B) $\frac{1}{4} \ln\left(\frac{\ln x - x}{\ln x + x}\right) - \frac{1}{2} \tan^{-1}\left(\frac{\ln x}{x}\right) + C$

(C) $\frac{1}{4} \ln\left(\frac{\ln x + x}{\ln x - x}\right) + \frac{1}{2} \tan^{-1}\left(\frac{\ln x}{x}\right) + C$

(D) $\frac{1}{4} \left(\ln\left(\frac{\ln x - x}{\ln x + x}\right) + \tan^{-1}\left(\frac{\ln x}{x}\right) \right) + C$

Ans. B

Sol. $I = \int \frac{x^2(1-\ln x)}{x^4 \left(\left(\frac{\ln x}{x} \right)^4 - 1 \right)} dx = \int \frac{1-\ln x}{x^2 \left(\left(\frac{\ln x}{x} \right)^4 - 1 \right)} dx$

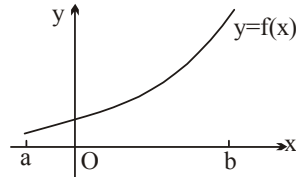
put $\frac{\ln x}{x} = t \Rightarrow \frac{1-\ln x}{x^2} = dt$

$$I = \int \frac{dt}{(t^4 - 1)} = \int \frac{dt}{(t^2 + 1)(t^2 - 1)} = \frac{1}{2} \int \frac{(t^2 + 1) - (t^2 - 1)}{(t^2 + 1)(t^2 - 1)} dt$$

$$I = \frac{1}{2} \left(\int \frac{dt}{t^2 - 1} - \int \frac{dt}{t^2 + 1} \right) = \frac{1}{2} \left(\frac{1}{2} \ln \frac{t-1}{t+1} - \tan^{-1} t \right)$$

$$= \frac{1}{4} \ln\left(\frac{\ln x - x}{\ln x + x}\right) - \frac{1}{2} \tan^{-1}\left(\frac{\ln x}{x}\right) + C$$

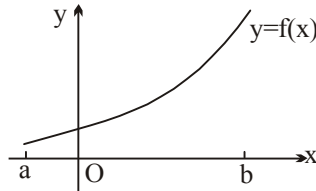
26. The graph of the function $y = f(x)$, where $f: [a, b] \rightarrow \mathbb{R}$ and a is a negative constant and b a positive constant, is shown. The function $F: [a, b] \rightarrow \mathbb{R}$ is defined by $F(t) = \int_0^t f(x) dx$. Then $F(t) > 0$ for :



- (A) $t \in [a, b] \setminus \{0\}$ (B) $t \in [a, 0)$ only
(C) $t \in [0, b]$ only (D) $t \in (0, b]$ only

वक्र $y = f(x)$, जहाँ $f: [a, b] \rightarrow \mathbb{R}$ एवं a एक ऋणात्मक नियतांक है तथा b एक धनात्मक नियतांक है। फलन $F: [a, b] \rightarrow \mathbb{R}$

परिभाषित है $F(t) = \int_0^t f(x) dx$ हो तो $F(t) > 0$ के लिये :



- (A) $t \in [a, b] \setminus \{0\}$ (B) केवल $t \in [a, 0)$
(C) केवल $t \in [0, b]$ (D) केवल $t \in (0, b]$

Ans. D

Sol. for negative values of t

- (A) will not hold good for $(0, a)$ and will be -ve
(B) will not be correct as $F(t) < 0$.
(C) at $t = 0$, $F(t) = 0 \Rightarrow$ (D) is correct

27. If $\frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} = 0$, where C_0, C_1, C_2 are all real, the equation $C_2x^2 + C_1x + C_0 = 0$ has :

- (A) atleast one root in $(0, 1)$
(B) one root in $(1, 2)$ and other in $(3, 4)$
(C) one root in $(-1, 1)$ & the other in $(-5, -2)$
(D) both roots imaginary

यदि $\frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} = 0$, जहाँ C_0, C_1, C_2 वास्तविक संख्याएँ हो, तो समीकरण $C_2x^2 + C_1x + C_0 = 0$ का मान :

- (A) कम से कम एक मूल $(0, 1)$ में होगा
(B) एक मूल $(1, 2)$ में तथा दूसरा $(3, 4)$ में होगा

- (C) एक मूल $(-1, 1)$ में तथा दूसरा $(-5, -2)$ में होगा
 (D) दोनों मूल काल्पनिक हैं

Ans. A

Sol. $\frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} = 0; C_1, C_2, C_3 \in \mathbb{R}$

$$C_2x^2 + C_1x + C_0 = 0$$

$$\Rightarrow \int C_2x^2 dx + \int C_1x dx + \int C_0 dx = C$$

$$\Rightarrow \frac{C_2x^3}{3} + \frac{C_1x^2}{2} + C_0x = 0$$

$$\text{let } f(x) = C_0x + \frac{C_1x^2}{2} + \frac{C_2x^3}{3}$$

$$\Rightarrow f(0) = 0$$

$$f(1) = 0$$

& $f(x)$ is continuous in $[0, 1]$

& diff. in $(0, 1)$

$\Rightarrow$ By Rolle's Theorem $\exists$ at least one real in $(0, 1)$ such that $f'(c) = 0$

$$\Rightarrow C_2x^2 + C_1x + C_0 = 0 \text{ have at least one real root in } (0, 1).$$

28. Let $I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx$, $J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$. Then, for an arbitrary constant C, the value of $J - I$ equals:

[JEE-2008]

माना $I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx$, $J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$ तो C अचर है तो $J - I$ का मान होगा :

[JEE-2008]

(A) $\frac{1}{2} \ln \left(\frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^{2x} + 1} \right) + C$

(B) $\frac{1}{2} \ln \left(\frac{e^{2x} + e^x + 1}{e^{2x} - e^x + 1} \right) + C$

(C) $\frac{1}{2} \ln \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + C$

(D) $\frac{1}{2} \ln \left(\frac{e^{4x} + e^{2x} + 1}{e^{4x} - e^{2x} + 1} \right) + C$

Ans. C

Sol. $J - I = \int \left(\frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} - \frac{e^x}{e^{4x} + e^{2x} + 1} \right) dx$

$$= \int \frac{e^{3x} - e^x}{e^{4x} + e^{2x} + 1} dx = \int \frac{e^x (e^{2x} - 1)}{e^{4x} + e^{2x} + 1} dx$$

Let $e^x = t$ $e^x dx = dt$

$$= \int \frac{t^2 - 1}{t^4 + t^2 + 1} dt = \int \frac{1 - 1/t^2}{(t + 1/t)^2 - 1} dt$$

$$= \frac{1}{2} \ln \left| \frac{t + \frac{1}{t} - 1}{t + \frac{1}{t} + 1} \right| + C = \frac{1}{2} \ln \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + C$$

DEFINITE INTEGRATION WITH THE HELP OF INDEFINITE INTEGRATION

29. If $g(x) = \int_0^x \cos^4 t \, dt$, then $g(x + \pi)$ equals :

यदि $g(x) = \int_0^x \cos^4 t \, dt$, तो $g(x + \pi)$ का मान है :

- (A) $g(x) + g(\pi)$ (B) $g(x) - g(\pi)$ (C) $g(x) g(\pi)$ (D) $[g(x)/g(\pi)]$

Ans. A

Sol. $g(x + \pi) = \int_0^{x+\pi} \cos^4 t \, dt = \int_0^x \cos^4 t \, dt + \int_x^{x+\pi} \cos^4 t \, dt = g(x) + \int_0^\pi \cos^4 t \, dt = g(x) + g(\pi)$ Ans.

30. If $g(x) = \int_1^x e^{t^2} \, dt$ then the value of $\int_3^{x^3} e^{t^2} \, dt$ equals :

यदि $g(x) = \int_1^x e^{t^2} \, dt$ तो $\int_3^{x^3} e^{t^2} \, dt$ का मान है :

- (A) $g(x^3) - g(3)$ (B) $g(x^3) + g(3)$ (C) $g(x^3) - 3$ (D) $g(x^3) - 3g(x)$

Ans. A

Sol. $\int_3^{x^3} e^{t^2} \, dt = \int_1^1 e^{t^2} \, dt + \int_1^{x^3} e^{t^2} \, dt = \int_1^{x^3} e^{t^2} \, dt - \int_1^3 e^{t^2} \, dt = g(x^3) - g(3)$

31. Given $\int_0^{\pi/2} \frac{dx}{1 + \sin x + \cos x} = \ln 2$, then the value of the def. integral. $\int_0^{\pi/2} \frac{\sin x}{1 + \sin x + \cos x} dx$ is equal to :

दिया गया है $\int_0^{\pi/2} \frac{dx}{1 + \sin x + \cos x} = \ln 2$, तो $\int_0^{\pi/2} \frac{\sin x}{1 + \sin x + \cos x} dx$ का मान है :

- (A) $\frac{1}{2} \ln 2$ (B) $\frac{\pi}{2} - \ln 2$ (C) $\frac{\pi}{4} - \frac{1}{2} \ln 2$ (D) $\frac{\pi}{2} + \ln 2$

Ans. C

Sol. Let $I = \int_0^{\pi/2} \frac{\sin x \, dx}{1 + \sin x + \cos x}$

$$I = \int_0^{\pi/2} \frac{\cos x}{1 + \sin x + \cos x} \Rightarrow 2I = \int_0^{\pi/2} \frac{\sin x + \cos x + 1 - 1}{\sin x + \cos x + 1} dx$$

$$\Rightarrow 2I = \frac{\pi}{2} - \ln 2 \Rightarrow I = \frac{\pi}{4} - \frac{1}{2} \ln 2$$

32. Let $\frac{d}{dx} F(x) = \left(\frac{e^{\sin x}}{x} \right)$, $x > 0$. If $\int_1^4 \frac{3}{x} e^{\sin x^3} dx = F(k) - F(1)$, then one of the possible values of k, is :

माना कि $\frac{d}{dx} F(x) = \left(\frac{e^{\sin x}}{x} \right)$, $x > 0$ है यदि $\int_1^4 \frac{3}{x} e^{\sin x^3} dx = F(k) - F(1)$, तो k का एक संभव मान होगा :

- (A) 64 (B) 15 (C) 16 (D) 63

Ans. A

Sol. $\frac{d}{dx} F(x) = \frac{e^{\sin x}}{x} \quad x > 0$

$$\Rightarrow \int_1^4 \frac{3}{x} e^{\sin x^3} dx$$

$$\text{let } x^3 = t$$

$$3x^2 dx = dt$$

$$= \int_1^{64} \frac{1}{t} e^{\sin t} dt = [F(t)]_1^{64} = F(64) - F(1)$$

$$= \int_1^{64} \frac{1}{t} e^{\sin t} dt = [F(t)]_1^{64} = F(64) - F(1)$$

33. Let $F(x) = f(x) + f\left(\frac{1}{x}\right)$, where $f(x) = \int_1^x \frac{\log t}{1+t} dt$. Then $F(e)$ equals :

माना $F(x) = f(x) + f\left(\frac{1}{x}\right)$, जहाँ $f(x) = \int_1^x \frac{\log t}{1+t} dt$, तब $F(e)$ बराबर है :

- (A) $\frac{1}{2}$ (B) 0 (C) 1 (D) 2

Ans. A

Sol. $F(x) = f(x) + f(1/x)$ put $x = e$

$$F(e) = \int_1^e \frac{\log t}{1+t} dt + \int_1^{1/e} \frac{\log t}{1+t} dt$$

$$\text{let } t = \frac{1}{z} \Rightarrow \frac{dt}{dz} = \left(\frac{-1}{z^2}\right)$$

$$= \int_1^e \frac{\ln t}{(1+t)} dt + \int_1^e \frac{\ln 1/z}{(1+1/z)} \left(\frac{-1}{z^2}\right) dz$$

$$\text{by property } \int_a^b f(x) dx = \int_a^b f(t) dt$$

$$\int_1^e \frac{\ln t}{(1+t)} dt + \int_1^e \frac{\ln t}{t(1+t)} dt = \int_1^e \frac{\ln t}{t} dt = \frac{1}{2}$$

34. The value of $\int_0^1 \left(\prod_{r=1}^n (x+r) \right) \left(\sum_{k=1}^n \frac{1}{x+k} \right) dx$ equals :

$$\int_0^1 \left(\prod_{r=1}^n (x+r) \right) \left(\sum_{k=1}^n \frac{1}{x+k} \right) dx \text{ का मान होगा :}$$

- (A) n (B) n! (C) (n+1)! (D) n · n!

Ans. D

Sol. The given integrand is perfect differential coeff. of $\prod_{r=1}^n (x+r)$

$$\Rightarrow I = \left[\prod_{r=1}^n (x+r) \right]_0^1 = (n+1)! - n! = n \cdot n!$$

Alternatively: $(x+1)(x+2)(x+3).....(x+n) = t$

so that when $x = 1$, $t = (n + 1)!$

if $x = 0$ then $t = n!$

$$[\ln(x + 1) + \ln(x + 2) + \dots + \ln(x + n)] = \ln t$$

$$\therefore \left(\frac{1}{(x+1)} + \frac{1}{(x+2)} + \dots + \frac{1}{(x+n)} \right) dx = \frac{1}{t} dt \Rightarrow I = \int_{n!}^{(n+1)!} \frac{1}{t} dt$$

$$\text{Hence } I = \int_{\ln n!}^{(n+1)!} e^t dt = e^t \Big|_{\ln n!}^{\ln(n+1)!} = e^{\ln(n+1)!} - e^{\ln n!} = (n+1)! - n! = n \cdot n! \text{ Ans.}$$

35. The value(s) of $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$ is(are) : [JEE-2010]

$\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$ का मान होगा : [JEE-2010]

(A) $\frac{22}{7} - \pi$ (B) $\frac{2}{105}$ (C) 0 (D) $\frac{71}{15} - \frac{3\pi}{2}$

Ans. A

Sol.
$$I = \int_0^1 \frac{x^4(1-2x+x^2)^2}{1+x^2} dx$$

$$I = \int_0^1 \frac{x^4 \left\{ (1+x^2)^2 - 4x(1+x^2) + 4x^2 \right\}}{1+x^2} dx$$

$$= \int_0^1 (1+x^2)x^4 dx - \int_0^1 4x^5 dx + 4 \int_0^1 \frac{(x^6+1)-1}{1+x^2} dx$$

$$= \frac{1}{5} + \frac{1}{7} - 4 \cdot \frac{1}{6} + 4 \int_0^1 \frac{(x^2+1)^3 - 3x^2(1+x^2)}{1+x^2} dx - 4 \int_0^1 \frac{dx}{1+x^2}$$

$$= \frac{12}{35} - \frac{2}{3} + 4 \int_0^1 (x^4 + 2x^2 + 1) dx - 12 \int_0^1 x^2 dx - \pi$$

$$= \frac{12}{35} - \frac{2}{3} + 4 \left(\frac{1}{5} + \frac{2}{3} + 1 \right) - 4 - \pi$$

$$= \frac{12}{35} - \frac{2}{3} + \frac{52}{15} - \pi = \frac{22}{7} - \pi$$

PROPERTY FOR DISCONTINUOUS NON UNIFORM FUNCTION

36. $\int_{-1/2}^{1/2} \left[\left(\frac{x+1}{x-1} \right)^2 + \left(\frac{x-1}{x+1} \right)^2 - 2 \right]^{1/2} dx$ has the value equal to :

$\int_{-1/2}^{1/2} \left[\left(\frac{x+1}{x-1} \right)^2 + \left(\frac{x-1}{x+1} \right)^2 - 2 \right]^{1/2} dx$ का मान होगा :

- (A) 4 (B) $4 \ln \frac{3}{4}$ (C) $3 \ln \frac{3}{4}$ (D) $4 \ln \frac{4}{3}$

Ans. D

Sol. $\int_{-1/2}^{1/2} \left| \frac{x+1}{x-1} - \frac{x-1}{x+1} \right| dx$

$$= \int_{-\frac{1}{2}}^{\frac{1}{2}} \left| \frac{(x+1) - (x-1)^2}{x^2-1} \right| dx$$

$$= \int_{-\frac{1}{2}}^{\frac{1}{2}} \left| \frac{4x}{x^2-1} \right| dx$$

$$= - \int_{-\frac{1}{2}}^0 \frac{4x}{x^2-1} + \int_0^{\frac{1}{2}} \frac{4x}{x^2-1} dx$$

$$= 4 \ln \frac{4}{3}$$

37. Let $[.]$ denote the greatest integer function then the value of $\int_0^{1.5} x[x^2] dx$ is : [AIEEE-2011]

माना $[.]$ एक महत्तम पूर्णांक फलन दर्शाता है, तो $\int_0^{1.5} x[x^2] dx$ का मान है : [AIEEE-2011]

- (A) $\frac{5}{4}$ (B) 0 (C) $\frac{3}{2}$ (D) $\frac{3}{4}$

Ans. D

Sol. $\int_0^{1.5} x[x^2] dx$

$$\int_0^1 0 dx + \int_1^{\sqrt{2}} x dx + \int_{\sqrt{2}}^{1.5} 2x dx$$

$$\left[\frac{x^2}{2} \right]_1^{\sqrt{2}} + [x^2]_{\sqrt{2}}^{1.5}$$

$$\left(\frac{2}{2} - \frac{1}{2} \right) + (2.25 - 2)$$

$$\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

38. The value of $\int_1^a [x] f'(x) dx$, $a > 1$, where $[x]$ denotes the greatest integer not exceeding x is :

$\int_1^a [x] f'(x) dx$, $a > 1$, जहाँ $[x]$ महत्तम पूर्णांक संख्या है जो x से बड़ी नहीं है, का मान है :

(A) $[a] f(a) - \{f(1) + f(2) + \dots + f([a])\}$

(B) $[a] f([a]) - \{f(1) + f(2) + \dots + f(a)\}$

(C) $a f([a]) - \{f(1) + f(2) + \dots + f(a)\}$

(D) $a f(a) - \{f(1) + f(2) + \dots + f([a])\}$

Ans. A

Sol. $\int_1^2 1 \cdot f'(x) dx + \int_2^3 2 \cdot f'(x) dx + \dots + \int_{[a]}^a [a] f'(x) dx$

$$= [f(2) - f(1)] + 2[f(3) - f(2)] + \dots + [a] [f(a) - f([a])]$$

$$= [a] f(a) - \{f(1) + f(2) + \dots + f([a])\}$$

39. The integral $\int_0^\pi \sqrt{1 + 4 \sin^2 \frac{x}{2} - 4 \sin \frac{x}{2}} dx$ equals :

[JEE(Main)2014]

समाकलन $\int_0^\pi \sqrt{1 + 4 \sin^2 \frac{x}{2} - 4 \sin \frac{x}{2}} dx$ बराबर है :

[JEE(Main)2014]

(A) $4\sqrt{3} - 4$

(B) $4\sqrt{3} - 4 - \frac{\pi}{3}$

(C) $\pi - 4$

(D) $\frac{2\pi}{3} - 4 - 4\sqrt{3}$

Ans. B

Sol. $I = \int_0^\pi \sqrt{1 + 4 \sin^2 \frac{x}{2} - 4 \sin \frac{x}{2}} dx$

$$\Rightarrow I = \int_0^\pi \sqrt{\left(2 \sin \frac{x}{2} - 1\right)^2} dx$$

$$\Rightarrow I = \int_0^\pi \left| 2 \sin \frac{x}{2} - 1 \right| dx$$

$$\begin{aligned}
\Rightarrow I &= \int_0^{\pi/3} \left(1 - 2 \sin \frac{x}{2}\right) dx + \int_{\pi/3}^{\pi} \left(2 \sin \frac{x}{2} - 1\right) dx \\
&= x + 4 \cos \frac{x}{2} \Big|_0^{\pi} + \left[- \left(4 \cos \frac{x}{2} + x \right) \right]_{\pi/3}^{\pi} \\
&= \left[\left(\frac{\pi}{3} + 4 \frac{\sqrt{3}}{2} \right) - (0 + 4) \right] - \left[(0 + \pi) - \left(4 \frac{\sqrt{3}}{2} + \frac{\pi}{3} \right) \right] \\
&= 4\sqrt{3} - 4 - \frac{\pi}{3}
\end{aligned}$$

PROPERTY FOR ODD AND EVEN FUNCTIONS

40. The integral $\int_{-\frac{1}{2}}^{\frac{1}{2}} \left([x] + \ln \left(\frac{1+x}{1-x} \right) \right) dx$ equals, where $[]$ denotes greatest integer function :

समाकलन $\int_{-\frac{1}{2}}^{\frac{1}{2}} \left([x] + \ln \left(\frac{1+x}{1-x} \right) \right) dx$ का मान होगा, जहाँ $[]$ महत्तम पूर्णांक फलन है।

(A) $-\frac{1}{2}$

(B) 0

(C) 1

(D) $2 \ln \left(\frac{1}{2} \right)$

Ans. A

Sol. $\int_{-\frac{1}{2}}^{\frac{1}{2}} \left([x] + \ln \left(\frac{1+x}{1-x} \right) \right) dx$

$$= \int_{-\frac{1}{2}}^{\frac{1}{2}} [x] dx + \underbrace{\int_{-\frac{1}{2}}^{\frac{1}{2}} \ln \left(\frac{1+x}{1-x} \right) dx}_{\text{Odd Function}}$$

$$= \int_{-\frac{1}{2}}^0 [x] dx + \int_0^{\frac{1}{2}} [x] dx + 0$$

$$= -[x]_{-\frac{1}{2}}^0 + 0 = -\frac{1}{2}$$

41. $\int_{-1}^1 \frac{x^3 + |x| + 1}{x^2 + 2|x| + 1} dx = a \ln 2 + b$ then :

$\int_{-1}^1 \frac{x^3 + |x| + 1}{x^2 + 2|x| + 1} dx = a \ln 2 + b$ तो :

- (A) $a = 2 ; b = 1$ (B) $a = 2 ; b = 0$ (C) $a = 3 ; b = -2$ (D) $a = 4 ; b = -1$

Ans. B

Sol. $I = \int_{-1}^1 \frac{x^3}{x^2 + 2|x| + 1} dx + \int_{-1}^1 \frac{|x| + 1}{(|x| + 1)^2} dx = 2 \int_0^1 \frac{dx}{1 + x} = 2 \ln 2$

42. The value of $\int_{-2}^3 |1 - x^2| dx$ is :

$\int_{-2}^3 |1 - x^2| dx$ का मान है :

- (A) $28/3$ (B) $14/3$ (C) $7/3$ (D) $1/3$

Ans. A

Sol. $\int_{-2}^3 |1 - x^2| dx$ $|1 - x^2| = \begin{cases} x^2 - 1 & x < -1 \\ 1 - x^2 & -1 \leq x \leq 1 \\ x^2 - 1 & x > 1 \end{cases}$

$= \int_{-2}^{-1} (x^2 - 1) dx + \int_{-1}^1 (1 - x^2) dx + \int_1^3 (x^2 - 1) dx$

$= \left[\frac{x^3}{3} - x \right]_{-2}^{-1} + \left[x - \frac{x^3}{3} \right]_{-1}^1 + \left[\frac{x^3}{3} - x \right]_1^3$

$= -\frac{1}{3} + 3 + \frac{8}{3} - 2 + \frac{2}{3} + \frac{2}{3} + 9 - 3 - \frac{1}{3} + 1$

$= \frac{10}{3} + 6 = \frac{28}{3}$

43. $\int_{-3\pi/2}^{-\pi/2} [(x + \pi)^3 + \cos^2(x + 3\pi)] dx$ is equal to :

$$\int_{-3\pi/2}^{-\pi/2} [(x + \pi)^3 + \cos^2(x + 3\pi)] dx \text{ बराबर है :}$$

- (A) $(\pi^4/32) + (\pi/2)$ (B) $\pi/2$ (C) $(\pi/4) - 1$ (D) $\pi^4/32$

Ans. B

Sol.
$$\int_{-\frac{3\pi}{2}}^{-\frac{\pi}{2}} \left((x + \pi)^3 + \cos^2(x + 3\pi) \right) dx$$

$$= \int_{-\frac{3\pi}{2}}^{-\frac{\pi}{2}} (x + \pi)^3 dx + \int_{-\frac{3\pi}{2}}^{-\frac{\pi}{2}} \cos^2 x dx$$

$$= \left[\frac{(x + \pi)^4}{4} \right]_{-\frac{3\pi}{2}}^{-\frac{\pi}{2}} + \frac{1}{2} \left[x + \frac{\sin 2x}{2} \right]_{-\frac{3\pi}{2}}^{-\frac{\pi}{2}}$$

$$= \frac{1}{4} \left[\left(\frac{\pi}{2} \right)^4 - \left(\frac{\pi}{2} \right)^4 \right] + \frac{1}{2} \left[\pi + \frac{1}{2}(0 - 0) \right] = \frac{\pi}{2}$$

KING & QUEEN PROPERTY

44. If $I = \int_0^{\pi/2} \ln(\sin x) dx$ then $\int_{-\pi/4}^{\pi/4} \ln(\sin x + \cos x) dx =$

यदि $I = \int_0^{\pi/2} \ln(\sin x) dx$, तो $\int_{-\pi/4}^{\pi/4} \ln(\sin x + \cos x) dx =$

- (A) $\frac{I}{2}$ (B) $\frac{I}{4}$ (C) $\frac{I}{\sqrt{2}}$ (D) I

Ans. A

Sol.
$$I_1 = \int_{-\pi/4}^{\pi/4} \ln(\sin x + \cos x) dx = \int_{-\pi/4}^{\pi/4} \ln(\cos x - \sin x) dx \text{ (using king)}$$

$$\Rightarrow 2 I_1 = \int_{-\pi/4}^{\pi/4} \ln \cos 2x dx = 2 \int_0^{\pi/4} \ln(\cos 2x) dx = \int_0^{\pi/2} \ln(\cos t) dt \text{ where } 2x = t$$

$$\int_0^{\pi/2} \ln(\sin t) dt = I \Rightarrow I_1 = I/2$$

45. The value of the definite integral $\int_0^{\pi/2} \sin x \sin 2x \sin 3x \, dx$ is equal to :

निश्चित समाकलन $\int_0^{\pi/2} \sin x \sin 2x \sin 3x \, dx$ का मान है :

- (A) $\frac{1}{3}$ (B) $-\frac{2}{3}$ (C) $-\frac{1}{3}$ (D) $\frac{1}{6}$

Ans. D

Sol. $I = \int_0^{\pi/2} \sin x \sin 2x \sin 3x \, dx$

Apply king property

$$I = - \int_0^{\pi/2} \cos x \sin 2x \cos 3x \, dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \sin 2x \cos 4x \, dx$$

$$I = \frac{1}{4} \int_0^{\pi/2} (\sin 6x - \sin 2x) \, dx$$

$$I = \frac{1}{6}$$

46. The value of $\int_{\pi}^{2\pi} [2 \cos x] \, dx$ where $[\cdot]$ represents the greatest integer function is :

- (A) $-\frac{5\pi}{6}$ (B) $-\frac{\pi}{2}$ (C) $-\pi$ (D) none

$\int_{\pi}^{2\pi} [2 \cos x] \, dx$ का मान होगा जहाँ $[\cdot]$ महत्तम पूर्णांक फलन है :

- (A) $-\frac{5\pi}{6}$ (B) $-\frac{\pi}{2}$ (C) $-\pi$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\int_{\pi}^{2\pi} [2 \cos x] \, dx$

If $x \in (\pi, 2\pi)$ then $\cos x \in [-1, 1]$

$$\Rightarrow 2\cos x \in [-2, 2].$$

$$\begin{aligned} &= \int_{\pi}^{\pi+\frac{\pi}{3}} -2dx + \int_{\pi+\frac{\pi}{3}}^{\frac{3\pi}{2}} -1dx + \int_{\frac{3\pi}{2}}^{\frac{5\pi}{3}} 0dx + \int_{\frac{5\pi}{3}}^{2\pi} 1dx \\ &= -\frac{2\pi}{3} - \frac{\pi}{6} + \frac{\pi}{3} \\ &= -\frac{2\pi}{3} + \frac{\pi}{6} = \frac{-4\pi + \pi}{6} = -\frac{\pi}{2} \end{aligned}$$

47. If $f(a+b-x) = f(x)$, then $\int_a^b x f(x) dx$ is equal to :

यदि $f(a+b-x) = f(x)$, तो $\int_a^b x f(x) dx$ बराबर है :

(A) $\frac{a+b}{2} \int_a^b f(a+b-x) dx$

(B) $\frac{a+b}{2} \int_a^b f(b-x) dx$

(C) $\frac{a+b}{2} \int_a^b f(x) dx$

(D) $\frac{b-a}{2} \int_a^b f(x) dx$

Ans. C

Sol. $\int_a^b x f(x) dx = I$

Apply King Property

$$I = \int_a^b (a+b-x) f(x) dx$$

$$\Rightarrow 2I = (a+b) \int_a^b f(x) dx$$

$$\Rightarrow I = \frac{a+b}{2} \int_a^b f(x) dx = \frac{a+b}{2} \int_a^b f(a+b-x) dx$$

48. If $f(y) = e^y$, $g(y) = y$; $y > 0$ and $F(t) = \int_0^t f(t-y) g(y) dy$, then :

यदि $f(y) = e^y$, $g(y) = y$; $y > 0$ एवं $F(t) = \int_0^t f(t-y) g(y) dy$, तो :

$$(A) F(t) = t e^{-t}$$

$$(B) F(t) = 1 - e^{-1} (1+t)$$

$$(C) F(t) = e^t - (1+t)$$

$$(D) F(t) = t e^t$$

Ans. C

So. $f(y) = e^y, g(y) = y; y > 0$

$$\text{and } F(t) = \int_0^t f(t-y)g(y)dy$$

$$= \int_0^t e^{t-y} y dy = e^t \int_0^t e^{-y} y dy = e^t [-ye^{-y} - e^{-y}]_0^t$$

$$= -e^t [te^{-t} + e^{-t} - 0 - 1] = e^t - (1+t)$$

49. If $f(x) = \frac{e^x}{1+e^x}, I_1 = \int_{f(-a)}^{f(a)} xg\{x(1-x)\} dx$ and $I_2 = \int_{f(-a)}^{f(a)} g\{x(1-x)\} dx$, then the value of $\frac{I_2}{I_1}$ is :

यदि $f(x) = \frac{e^x}{1+e^x}, I_1 = \int_{f(-a)}^{f(a)} xg\{x(1-x)\} dx$ एवं $I_2 = \int_{f(-a)}^{f(a)} g\{x(1-x)\} dx$, तो $\frac{I_2}{I_1}$ का मान है :

$$(A) 2$$

$$(B) -3$$

$$(C) -1$$

$$(D) 1$$

Ans. A

Sol. $f(x) = \frac{e^x}{1+e^x} \quad I_1 = \int_{f(-a)}^{f(a)} xg[x(1-x)]dx$

$$I_2 = \int_{f(-a)}^{f(a)} g[x(1-x)]dx$$

$$f(a) = \frac{e^a}{1+e^a}, f(-a) = \frac{e^{-a}}{1+e^{-a}}$$

$$\therefore f(a) + f(-a) = 1$$

$$2I_1 = \int_{f(-a)}^{f(a)} xg\{x(1-x)\}dx + \int_{f(-a)}^{f(a)} \{f(a) + f(-a) - x\}g(1-x)(x)dx$$

$$2I_1 = \int_{f(-a)}^{f(a)} g\{x(1-x)\}dx = I_2$$

$$\therefore f(a) + f(-a) = 1$$

$$2I_1 = I_2$$

$$\frac{I_2}{I_1} = 2$$

50. The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x} dx$, $a > 0$, is :

$\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x} dx$, $a > 0$, का मान है :

- (A) $a\pi$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{a}$ (D) 2π

Ans. B

Sol. Putting $-x$ for x

$$I = \int_{\pi}^{-\pi} \frac{\cos^2 x}{1+a^{-x}} (-dx) = \int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^{-x}} dx$$

$$I + I = \int_{-\pi}^{\pi} \cos^2 x \left(\frac{1}{1+a^x} + \frac{1}{1+a^{-x}} \right) dx$$

$$= \int_{-\pi}^{\pi} \cos^2 x dx \Rightarrow 2I = 2 \int_0^{\pi} \cos^2 x dx$$

$$= \int_0^{\pi} (1 + \cos 2x) dx$$

$$2I = \left[x + \frac{\sin 2x}{2} \right]_0^{\pi}$$

$$2I = \pi \Rightarrow I = \frac{\pi}{2}$$

51. The value of the integral, $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x} + \sqrt{x}} dx$ is :

समाकल $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x} + \sqrt{x}} dx$ का मान है :

- (A) $\frac{3}{2}$ (B) 2 (C) 1 (D) $\frac{1}{2}$

Ans. A

Sol. $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x} + \sqrt{x}} = I$

Apply King Property

$$I = \int_3^6 \frac{\sqrt{9-x}}{\sqrt{x} + \sqrt{9-x}} dx$$

$$\Rightarrow 2I = \int_3^6 dx \Rightarrow I = \frac{3}{2}$$

52. The value of $I = \int_0^{\pi/2} \frac{(\sin x + \cos x)^2}{\sqrt{1 + \sin 2x}} dx$ is :

$$I = \int_0^{\pi/2} \frac{(\sin x + \cos x)^2}{\sqrt{1 + \sin 2x}} dx \text{ का मान है :}$$

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $I = \int_0^{\frac{\pi}{2}} \frac{(\sin x + \cos x)^2}{\sqrt{1 + \sin 2x}} dx$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{(\sin x + \cos x)^2}{|\sin x + \cos x|} dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} (\sin x + \cos x) dx$$

$$\Rightarrow I = [-\cos x + \sin x]_0^{\frac{\pi}{2}}$$

$$\Rightarrow I = 2$$

53. If $\int_0^{\pi} x f(\sin x) dx = A \int_0^{\pi/2} f(\sin x) dx$, then A is :

$$\text{यदि } \int_0^{\pi} x f(\sin x) dx = A \int_0^{\pi/2} f(\sin x) dx, \text{ तो } A \text{ का मान है :}$$

(A) 0 (B) π (C) $\pi/4$ (D) 2π

Ans. B

Sol. $\int_0^{\pi} x f(\sin x) dx = I \text{ (let)}$

Apply King Property

$$I = \int_0^{\pi} (\pi - x) f(\sin(\pi - x)) dx$$

$$= \int_0^{\pi} (\pi - x) f(\sin x) dx$$

$$\Rightarrow 2I = \pi \int_0^{\pi} f(\sin x) dx$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$$

Apply queen property

$$\Rightarrow I = \frac{\pi}{2} \times 2 \int_0^{\frac{\pi}{2}} f(\sin x) dx$$

$$\Rightarrow A = \pi$$

54. $\int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x} dx =$

(A) π^2

(B) $\pi^2/4$

(C) $\pi/8$

(D) $\pi^2/8$

Ans. A

Sol. $I = \int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x} dx$

$$= \int_{-\pi}^{\pi} \underbrace{\frac{2x}{1+\cos^2 x}}_{\text{Odd Function}} dx + \int_{-\pi}^{\pi} \frac{2x \sin x}{1+\cos^2 x} dx$$

$$= \int_{-\pi}^{\pi} \frac{2x \sin x}{1+\cos^2 x} dx$$

Apply King Property

$$I = \int_{-\pi}^{\pi} \frac{2(\pi - x)}{1+\cos^2 x} \sin x dx$$

$$\Rightarrow 2I = 2\pi \int_{-\pi}^{\pi} \frac{\sin x}{1+\cos^2 x} dx$$

Apply Queen Property

$$\Rightarrow 2I = 4\pi \int_{-\pi}^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

$$= -4\pi \left[\tan^{-1}(\cos x) \right]_0^{\pi} = -4\pi \left[-\frac{\pi}{2} \right] = 2\pi^2$$

$$\Rightarrow I = \pi^2$$

55. The integral $\int_2^4 \frac{\log x^2}{\log x^2 + \log(36 - 12x + x^2)} dx$ is equal to : [JEE(Main)2015]

समाकल $\int_2^4 \frac{\log x^2}{\log x^2 + \log(36 - 12x + x^2)} dx$ बराबर है : [JEE(Main)2015]

(A) 2 (B) 4 (C) 1 (D) 6

Ans. C

Sol. $I = \int_2^4 \frac{\ln(x^2)}{\ln(x^2) + \ln(6-x)^2} dx$

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

$$I = \int_2^4 \frac{\ln(6-x)^2}{\ln(6-x)^2 + \ln(x)^2} dx$$

$$2I = \int_2^4 dx \Rightarrow 2I = 2 \Rightarrow I = 1$$

56. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1 + 2^x} dx$ is : [JEE(Main)2018]

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1 + 2^x} dx$ का मान है : [JEE(Main)2018]

(A) 4π (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{2}$

Ans. B

Sol. $I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1 + 2^x} dx$

Use king & Add

$$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \, dx$$

$$= 2 \int_0^{\frac{\pi}{2}} \sin^2 x \, dx$$

$$I = \frac{\pi}{4}$$

57. The value of $\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\ln 6 - x^2)} dx$ is : [JEE-2011]

$\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\ln 6 - x^2)} dx$ का मान होगा : [JEE-2011]

(A) $\frac{1}{4} \ln \frac{3}{2}$ (B) $\frac{1}{2} \ln \frac{3}{2}$ (C) $\ln \frac{3}{2}$ (D) $\frac{1}{6} \ln \frac{3}{2}$

Ans. A

Sol. $I = \int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\ln 6 - x^2)} dx$; put $x^2 = t$

$$\Rightarrow 2x dx = dt$$

$$\Rightarrow I = \frac{1}{2} \int_{\ln 2}^{\ln 3} \frac{\sin t}{\sin t + \sin(\ln 6 - t)} dt \quad \dots\dots\dots(i)$$

$$\Rightarrow I = \frac{1}{2} \int_{\ln 2}^{\ln 3} \frac{\sin(\ln 6 - t)}{\sin(\ln 6 - t) + \sin t} dt \quad \dots\dots\dots(ii)$$

Adding equation (i) & (ii)

$$\Rightarrow 2I = \frac{1}{2} \int_{\ln 2}^{\ln 3} dt \quad \Rightarrow I = \frac{1}{4} \ln \left(\frac{3}{2} \right)$$

PROPERTY FOR PERIODIC FUCTIONS

58. Let $T > 0$ be a fixed real number. Suppose f is a continuous function such that for all $x \in \mathbb{R}$

$f(x+T) = f(x)$. If $I = \int_0^T f(x) \, dx$ then the value of $\int_3^{3+3T} f(2x) \, dx$ is :

माना $T > 0$ एक निश्चित वास्तविक संख्या है तथा x के प्रत्येक मान के लिये f एक सतत फलन है तथा

$f(x+T)=f(x)$. यदि $I = \int_0^T f(x) dx$ तो $\int_3^{3+3T} f(2x) dx$ का मान होगा :

- (A) $\frac{3}{2}I$ (B) $2I$ (C) $3I$ (D) $6I$

Ans. C

Sol. $I = \int_0^T f(x) dx \quad \left\{ \because \int_a^{a+nT} f(x) dx = n \int_0^T f(x) dx; n \in \mathbb{Z}, a \in \mathbb{R} \right\}$

$$\begin{aligned} \Rightarrow \int_3^{3+3T} f(2x) dx &= \int_0^{3T} f(2x) dx = \frac{1}{2} \int_0^{6T} f(t) dt \\ &= \frac{6}{2} \int_0^T f(t) dt \quad \left\{ \because f(x+t) = f(x); \forall x \in \mathbb{R} \right\} \\ &= 3I \end{aligned}$$

59. The value of the definite integral $\int_{19}^{37} (\{x\}^2 + 3(\sin 2\pi x)) dx$ where $\{x\}$ denotes the fractional part function :

$\int_{19}^{37} (\{x\}^2 + 3(\sin 2\pi x)) dx$ का मान होगा जहाँ $\{x\}$ भिन्नात्मक फलन है :

- (A) 0 (B) 6 (C) 9 (D) 18

Ans. B

Sol. put $x = 19 + y \Rightarrow dx = dy,$

$$I = \int_0^{18} (\{19+y\}^2 + \sin 2\pi y) dy = \int_0^{18} (\{y\}^2 + \underbrace{\sin 2\pi y}_{\text{zero}}) dy = 18 \int_0^1 y^2 dy = 6$$

60. $\int_{\pi}^{10\pi} |\sin x| dx =$

- (A) 9 (B) 10 (C) 18 (D) 20

Ans. C

Sol. $\int_{\pi}^{10\pi} |\sin x| dx \quad \left[\because \int_{aT}^{bT} f(x) dx = (b-a) \int_0^T f(x) dx \right]$

$$\begin{aligned} &= 9 \int_0^{\pi} |\sin x| dx = -9 [\cos x]_0^{\pi} \\ &= 18 \end{aligned}$$

DIFFERENTIATION OF INTEGRATION BY LEIBNITZ

61. Let $f(x) = \int_1^x \sqrt{2-t^2} dt$. Then the real roots of the equation $x^2 - f'(x) = 0$ are :

- (A) ± 1 (B) $\pm \frac{1}{\sqrt{2}}$ (C) $\pm \frac{1}{2}$ (D) 0 and 1

यदि $f(x) = \int_1^x \sqrt{2-t^2} dt$ तो समीकरण $x^2 - f'(x) = 0$ के वास्तविक मूल हैं :

- (A) ± 1 (B) $\pm \frac{1}{\sqrt{2}}$ (C) $\pm \frac{1}{2}$ (D) 0 तथा 1

Ans. A

Sol. $f(x) = \int_1^x \sqrt{2-t^2} dt$

Apply Leibnitz rule

$$f'(x) = \sqrt{2-x^2} \cdot 1 - 0 = \sqrt{2-x^2}$$

$$\Rightarrow f'(x) = x^2$$

$$\Rightarrow \sqrt{2-x^2} = x^2$$

$$\text{let } x^2 = v$$

$$\Rightarrow \sqrt{2-v} = v$$

$$\Rightarrow v^2 + v - 2 = 0$$

$$\Rightarrow (v-1)(v+2) = 0$$

$$\Rightarrow v = 1 \text{ or } v = -2$$

$$\Rightarrow x^2 = 1 \quad \text{or } x^2 = -2 \text{ (not possible)}$$

$$\Rightarrow x = \pm 1$$

62. Variable x and y are related by equation $x = \int_0^y \frac{dt}{\sqrt{1+t^2}}$. The value of $\frac{d^2y}{dx^2}$ is equal to :

चर x तथा y सम्बन्ध $x = \int_0^y \frac{dt}{\sqrt{1+t^2}}$ से जुड़ा है, तो $\frac{d^2y}{dx^2}$ का मान है :

- (A) $\frac{y}{\sqrt{1+y^2}}$ (B) y (C) $\frac{2y}{\sqrt{1+y^2}}$ (D) $4y$

Ans. B

Sol. $\frac{dx}{dy} = \frac{1}{\sqrt{y^2+1}}$

$$\frac{dy}{dx} = \sqrt{y^2+1}; \quad \frac{d^2y}{dx^2} = \frac{y}{\sqrt{y^2+1}} \sqrt{y^2+1} = y$$

63. Let $f(x) = \int_{-1}^x e^{t^2} dt$ and $h(x) = f(1+g(x))$, where $g(x)$ is defined for all x , $g'(x)$ exists for all x , and $g(x) < 0$ for $x > 0$. If $h'(1) = e$ and $g'(1) = 1$, then the possible values which $g(1)$ can take :

माना $f(x) = \int_{-1}^x e^{t^2} dt$ तथा $h(x) = f(1+g(x))$, जहाँ $g(x)$, x के सभी मानों के लिए परिभाषित तथा $g'(x)$, x के सभी मानों के लिए अस्तित्व में है तथा $g(x) < 0$, $x > 0$ के लिए। यदि $h'(1) = e$ तथा $g'(1) = 1$ हो, तो $g(1)$ के सम्भव मान होंगे :

- (A) 0 (B) -1 (C) -2 (D) -4

Ans. C

Sol. Given $f(x) = \int_{-1}^x e^{t^2} dt$; $h(x) = f(1+g(x))$; $g(x) < 0$ for $x > 0$

$$\text{now } h(x) = \int_{-1}^{1+g(x)} e^{t^2} dt = f(1+g(x)) \quad (\text{given})$$

differentiating $h'(x) = e^{(1+g(x))^2} \cdot g'(x)$

$$h'(1) = e \quad (\text{given})$$

$$e^{(1+g(1))^2} \cdot g'(1) = e$$

$$\therefore (1+g(1))^2 = 1$$

$$1+g(1) = \pm 1 \Rightarrow g(1) = 0 \text{ (not possible) or } g(1) = -2 \Rightarrow (C)$$

64. Let $f: [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 0$, then $\lim_{n \rightarrow \infty} n^2 \int_0^{\frac{1}{n}} f(t) dt$ equals :

माना $f: [0, 1] \rightarrow \mathbb{R}$ एक अवकलनीय फलन है जहाँ $f(0) = 0$, तो $\lim_{n \rightarrow \infty} n^2 \int_0^{\frac{1}{n}} f(t) dt$ का मान है :

- (A) $\frac{1}{2} f'(0)$ (B) $f'(0)$ (C) $2 f'(0)$ (D) 0

Ans. A

Sol. Let $I = \lim_{n \rightarrow \infty} n^2 \int_0^{\frac{1}{n}} f(t) dt = \lim_{n \rightarrow \infty} \frac{\int_0^{\frac{1}{n}} f(t) dt}{\frac{1}{n^2}}$ $\left(\frac{0}{0} \text{ form, as } n \rightarrow \infty \right)$

Using L'Hospital's rule and Using Leibnitz rule, we get

$$I = \lim_{n \rightarrow \infty} \frac{n^3 f\left(\frac{1}{n}\right) \left(-\frac{1}{n^2}\right)}{-2} = \lim_{n \rightarrow \infty} \frac{n}{2} f\left(\frac{1}{n}\right)$$

Put $n = \frac{1}{h}$, hence

$$I = \frac{1}{2} \lim_{h \rightarrow 0} \left(\frac{f(0+h) - f(0)}{h} \right) = \frac{1}{2} f'(0)$$

65. If a, b and c are real numbers then the value of $\lim_{t \rightarrow 0} \ln \left(\frac{1}{t} \int_0^t (1 + a \sin bx)^{c/x} dx \right)$ equals :

यदि a, b तथा c वास्तविक संख्या है, तो $\lim_{t \rightarrow 0} \ln \left(\frac{1}{t} \int_0^t (1 + a \sin bx)^{c/x} dx \right)$ का मान होगा :

- (A) abc (B) $\frac{ab}{c}$ (C) $\frac{bc}{a}$ (D) $\frac{ca}{b}$

Ans. A

Sol. $I = \ln \lim_{t \rightarrow 0} \frac{\int_0^t (1 + a \sin bx)^{c/x} dx}{t} = \ln \lim_{t \rightarrow 0} (1 + a \sin bt)^{c/t}$ (using L'Hospital's rule)

$$= \ln e^{\lim_{t \rightarrow 0} \frac{c}{t} (a \sin bt)} = \lim_{t \rightarrow 0} \frac{abc \sin bt}{bt} = abc$$

66. $\lim_{x \rightarrow 0} \frac{\int_0^x \sin t^2 dt}{x(1 - \cos x)}$ equals :

$$\lim_{x \rightarrow 0} \frac{\int_0^x \sin t^2 dt}{x(1 - \cos x)} \text{ का मान होगा :}$$

(A) $\frac{1}{3}$

(B) 2

(C) $\frac{1}{2}$

(D) $\frac{2}{3}$

Ans. D

Sol. $\lim_{x \rightarrow 0} \frac{\int_0^x \sin t^2 dt}{x^3 \frac{(1 - \cos x)}{x^2}} \quad \left(\text{using } \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \right)$

$$= 2 \lim_{x \rightarrow 0} \frac{\int_0^x \sin t^2 dt}{x^3} \quad (\text{Using L'Hospital Rule})$$

$$= 2 \lim_{x \rightarrow 0} \frac{\sin x^2}{3x^2} = \frac{2}{3}$$

67. The value of $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sec^2 t dt}{x \sin x}$ is :

$$\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sec^2 t dt}{x \sin x} \text{ का मान है :}$$

(A) 0

(B) 3

(C) 2

(D) 1

Ans. D

Sol. $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sec^2 t dt}{x \sin x} \left(\frac{0}{0} \text{ form} \right) = \lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sec^2 t dt}{x^2} \left(\frac{0}{0} \text{ form} \right)$

Apply L' Hospital Rule & Leibnitz Rule

$$= \lim_{x \rightarrow 0} \frac{\sec^2(x^2) \cdot 2x}{2x} = 1$$

68. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function having $f(2) = 6$, $f'(2) = \left(\frac{1}{48}\right)$. Then $\lim_{x \rightarrow 2} \int_6^{f(x)} \frac{4t^3}{x-2} dt$ equals :

माना $f: \mathbb{R} \rightarrow \mathbb{R}$ एक अवकलनीय फलन है, जिसके लिए $f(2) = 6$, $f'(2) = \left(\frac{1}{48}\right)$ है, तो $\lim_{x \rightarrow 2} \int_6^{f(x)} \frac{4t^3}{x-2} dt$ बराबर है :

(A) 24

(B) 36

(C) 12

(D) 18

Ans. D

Sol. $f : \mathbb{R} \xrightarrow{\text{diff.}} \mathbb{R}$

$$f(2) = 6, f'(2) = \frac{1}{48}$$

$$\lim_{x \rightarrow 2} \int_6^{f(x)} \frac{4t^3}{x-2} dt = \lim_{x \rightarrow 2} \frac{\int_6^{f(x)} 4t^3 dt}{x-2} \left(\frac{0}{0} \text{ form} \right)$$

Apply L'Hospital & Leibnitz Rule

$$= \lim_{x \rightarrow 2} \frac{4(f(x))^3 \cdot f'(x)}{1}$$

$$= 4(f(2))^3 \cdot f'(2)$$

$$= 4 \cdot 216 \cdot \frac{1}{48} = 18$$

$$69. \quad \lim_{x \rightarrow \frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(t) dt}{x^2 - \frac{\pi^2}{16}} =$$

$$(A) \frac{8}{\pi} f(2)$$

$$(B) \frac{2}{\pi} f(2)$$

$$(C) \frac{2}{\pi} f\left(\frac{1}{2}\right)$$

$$(D) 4f(2)$$

Ans. A

$$\text{Sol.} \quad \lim_{x \rightarrow \frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(t) dt}{x^2 - \frac{\pi^2}{16}} \left(\frac{0}{0} \text{ form} \right)$$

Apply Leibnitz & L'Hospital Rule

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{f(\sec^2 x) \cdot 2 \sec^2 x \tan x}{2x} = \frac{f(2) \cdot 4}{\frac{\pi}{2}} = \frac{8}{\pi} f(2)$$

$$70. \quad \text{The value of } \lim_{x \rightarrow 0} \frac{1}{x^3} \int_0^x \frac{t \ln(1+t)}{t^4 + 4} dt \text{ is :}$$

[JEE-2010]

$\lim_{x \rightarrow 0} \frac{1}{x^3} \int_0^x \frac{t \ln(1+t)}{t^4 + 4} dt$ का मान होगा :

[JEE-2010]

(A) 0

(B) $\frac{1}{12}$

(C) $\frac{1}{24}$

(D) $\frac{1}{64}$

Ans. B

Sol. Applying L-Hospital rule,

$$\lim_{x \rightarrow 0} \frac{\int_0^x \frac{t \ln(1+t)}{t^4 + 4} dt}{x^3} = \lim_{x \rightarrow 0} \frac{\frac{x \ln(1+x)}{x^4 + 4}}{3x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\ln(1+x)}{3x(x^3 + 4)} = \frac{1}{12}$$

DETERMINATION OF FUNCTION

71. If $f'(x) = f(x) + \int_0^1 f(x) dx$ and given $f(0) = 1$, then $\int f(x) dx$ is equal to :

यदि $f'(x) = f(x) + \int_0^1 f(x) dx$ तथा दिया गया है कि $f(0) = 1$ है, तो $\int f(x) dx$ बराबर होगा :

(A) $\frac{2}{3-e} e^x + \left(\frac{3-e}{1-e} \right) x + C$

(B) $\frac{2}{3-e} e^x + \left(\frac{1-e}{3-e} \right) x + C$

(C) $\frac{3}{2-e} e^x + \left(\frac{1+e}{3+e} \right) x + C$

(D) $\frac{2}{2-e} e^x + \left(\frac{1-e}{3+e} \right) x + C$

Ans. B

Sol. $f'(x) = f(x) + \int_0^1 f(x) dx$; $f(0) = 1$

$$\Rightarrow f''(x) = f'(x)$$

$$\Rightarrow \ln |f'(x)| = x + \ln C$$

$$\Rightarrow f'(x) = C \cdot e^x$$

$$\Rightarrow f(x) = C e^x + d$$

$$\text{Given } f(0) = 1$$

$$\Rightarrow c + d = 1$$

$$\Rightarrow f(x) = c(e^x - 1) + 1$$

$$\therefore f'(x) = f(x) + \int_0^1 f(x) dx$$

$$\Rightarrow ce^x = c(e^x - 1) + 1 + 1[ce^x - cx + x]_0^1$$

$$\Rightarrow C - 1 = [ce - c + 1 - c]$$

$$\Rightarrow C - 1 = ce - 2c + 1$$

$$\Rightarrow 3c = ce + 2$$

$$\Rightarrow c = \frac{2}{3-e}$$

$$\therefore f(x) = \frac{2}{3-e}(e^x - 1) + 1$$

$$\Rightarrow \int f(x) dx = \frac{2}{3-e}e^x + x\left(\frac{1-e}{3-e}\right) + \lambda; \lambda \in \mathbb{R}$$

72. The set of values of 'a' which satisfy the equation $\int_0^2 (t - \log_2 a) dt = \log_2 \left(\frac{4}{a^2}\right)$ is :

'a' के मानों का समुच्चय जो समीकरण $\int_0^2 (t - \log_2 a) dt = \log_2 \left(\frac{4}{a^2}\right)$ को सन्तुष्ट करता है :

- (A) $a \in \mathbb{R}$ (B) $a \in \mathbb{R}^+$ (C) $a < 2$ (D) $a > 2$

Ans. B

Sol. $\frac{t^2}{2} - \log_2 a \cdot t \Big|_0^2 = 2 - \log_2(a^2)$ (note $a > 0$)

$$(2 - 2 \log_2 a) = 2 - 2 \log_2 a \quad (\text{an identity})$$

$$2 \log_2 a = 2 \log_2 a \Rightarrow a \in \mathbb{R}^+$$

73. If $\int_{\sin x}^1 t^2(f(t)) dt = (1 - \sin x)$, then $f\left(\frac{1}{\sqrt{3}}\right)$ is :

यदि $\int_{\sin x}^1 t^2(f(t)) dt = (1 - \sin x)$, तो $f\left(\frac{1}{\sqrt{3}}\right)$ होगा :

- (A) $1/3$ (B) $1/\sqrt{3}$ (C) 3 (D) $\sqrt{3}$

Ans. C

Sol. $\int_{\sin x}^1 t^2 f(t) dt = 1 - \sin x$

diff. w.r.t. x

$$\Rightarrow -\sin^2 x \cdot f(\sin x) \cos x = -\cos x$$

$$\text{If } \sin x = \frac{1}{\sqrt{3}} \Rightarrow \cos x = \pm \sqrt{\frac{2}{3}}$$

$$\Rightarrow -\frac{1}{3} f\left(\frac{1}{\sqrt{3}}\right) = -1$$

$$\Rightarrow f\left(\frac{1}{\sqrt{3}}\right) = 3$$

74. Let $p(x)$ be a function defined on R such that $p'(x) = p'(1-x)$, for all $x \in [0, 1]$, $p(0) = 1$ and $p(1) = 41$. Then $\int_0^1 p(x) dx$ equals : [AIEEE-2010]

माना $p(x)$, R पर परिभाषित एक ऐसा फलन है कि $p'(x) = p'(1-x)$, प्रत्येक $x \in [0, 1]$, $p(0) = 1$ तथा $p(1) = 41$ है। तो $\int_0^1 p(x) dx$: [AIEEE-2010]

- (A) $\sqrt{41}$ (B) 21 (C) 41 (D) 42

Ans. B

Sol. $\int P'(x) dx = \int P'(1-x) dx$

$$P(x) = -P(1-x) + C$$

$$C = P(x) + P(1-x)$$

Put $x = 1 \therefore P(1) + P(0) = 42$ (i)

Let $I = \int_0^1 P(x) dx$ (ii)

$I = \int_0^1 P(1-x) dx$ (iii)

(ii) + (iii)

$$2I = \int_0^1 P(x) + P(1-x) dx$$

$$= 42 \int_0^1 dx = 42$$

$\therefore I = 21$

75. Let f be a non-negative function defined on the interval $[0, 1]$. If $\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt$, $0 \leq x \leq 1$, and $f(0) = 0$, then : [JEE-2009]

- (A) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$ (B) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$
 (C) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$ (D) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

यदि f अन्तर्गण्य फलन है जो कि $[0, 1]$ में परिभाषित है यदि $\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt$, $0 \leq x \leq 1$, तथा $f(0) = 0$, तो :

[JEE-2009]

- (A) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ तथा $f\left(\frac{1}{3}\right) > \frac{1}{3}$ (B) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ तथा $f\left(\frac{1}{3}\right) > \frac{1}{3}$
 (C) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ तथा $f\left(\frac{1}{3}\right) < \frac{1}{3}$ (D) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ तथा $f\left(\frac{1}{3}\right) < \frac{1}{3}$

Ans. C

Sol. $\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt$, $0 \leq x \leq 1$

differentiating both the sides & squaring

$$\Rightarrow 1 - (f'(x))^2 = f^2(x) \Rightarrow \frac{f'(x)}{\sqrt{1 - f^2(x)}} = 1$$

$$\Rightarrow \sin^{-1} f(x) = x + c$$

$$f(0) = 0$$

$$\Rightarrow f(x) = \sin x \Rightarrow \therefore \sin x \leq x \text{ for } x \in [0, 1]$$

$$\Rightarrow f\left(\frac{1}{2}\right) < \frac{1}{2} \text{ and } f\left(\frac{1}{3}\right) < \frac{1}{3}$$

76. Let f be a real-valued function defined on the interval $(-1, 1)$ such that $e^{-x} f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt$, for all $x \in (-1, 1)$, and let f^{-1} be the inverse function of f . Then $(f^{-1})'(2)$ is equal to : [JEE-2010]

यदि अन्तराल $(-1, 1)$ में f एक वास्तविक फलन है जहाँ $e^{-x} f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt$, सभी $x \in (-1, 1)$, एवं f^{-1} फलन f का प्रतिलोम है तो $(f^{-1})'(2)$ का मान होगा : [JEE-2010]

- (A) 1 (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{e}$

Ans. B

Sol. $e^{-x}f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt$

$$e^{-x}f'(x) - e^{-x}f(x) = \sqrt{x^4 + 1}$$

$$\Rightarrow f'(x) - f(x) = e^x \sqrt{x^4 + 1}$$

$$\Rightarrow \frac{dy}{dx} = y + e^x \sqrt{x^4 + 1} \quad (\text{say}) \quad \dots\dots(i)$$

considering $y = f(x)$. so that $x = f'(y)$

$$f^{-1}'(2) = \left(\frac{dx}{dy} \right)_{y=2} \quad \dots\dots(ii)$$

$$\begin{aligned} \text{for } x = 0 & \Rightarrow f(x) = 2 \text{ i.e. } y = 2 \\ & \Rightarrow f^{-1}(2) = 0 \end{aligned}$$

$$\frac{dy}{dx} = 2 + 1\sqrt{1} = 3$$

$$\text{from (2), } f^{-1}'(2) = \frac{1}{3}$$

DEFINITE INTEGRAL AS A LIMIT OF A SUM

77. $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \dots\dots + \frac{1}{\sqrt{n}\sqrt{4n}}$ is equal to :

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \dots\dots + \frac{1}{\sqrt{n}\sqrt{4n}} \text{ का मान है :}$$

(A) 2 (B) 4 (C) $2(\sqrt{2} - 1)$ (D) $2\sqrt{2} - 1$

Ans. A

Sol. $T_n = \frac{1}{\sqrt{n}\sqrt{n+r}}$ where r varies from 1 to $3n$

$$\Rightarrow \frac{1}{n} < \frac{r}{n} < 3 \quad \Rightarrow 0 < x < 3 \text{ as } n \rightarrow \infty$$

$$\lim_{n \rightarrow \infty} \sum_{r=1}^{3n} \frac{1}{n\sqrt{1 + \frac{r}{n}}} = \int_0^3 \frac{dx}{\sqrt{1+x}} = \int_0^3 (1+x)^{-1/2} dx = 2\sqrt{1+x} \Big|_0^3 = 2[2 - 1] = 2$$

78. $f(x) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 + k^2 x^2}$, $x > 0$ is equal to :

$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 + k^2 x^2}$, $x > 0$ का मान है :

(A) $x \tan^{-1}(x)$ (B) $\tan^{-1}(x)$ (C) $\frac{\tan^{-1}(x)}{x}$ (D) $\frac{\tan^{-1}(x)}{x^2}$

Ans. C

Sol. $T_n = \frac{n}{n^2 + k^2 x^2} = \frac{1}{n(1 + (k/n)^2 x^2)}$;

$$S = \frac{1}{n} \sum_{k=1}^n \frac{1}{1 + \underbrace{(k/n)^2}_{t} x^2} = \int_0^1 \frac{dt}{1 + t^2 x^2} = \frac{1}{x^2} \int_0^1 \frac{dt}{t^2 + (1/x^2)} = \frac{1}{x} \tan^{-1} tx \Big|_0^1 = \frac{\tan^{-1} x}{x}$$

79. $\lim_{n \rightarrow \infty} \frac{1^p + 2^p + 3^p + \dots + n^p}{n^{p+1}} =$

(A) 1 (B) $\frac{1}{p+1}$ (C) $\frac{1}{p+2}$ (D) p^2

Ans. B

Sol. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r^p}{n^{p+1}} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{r}{n}\right)^p \cdot \frac{1}{n}$

$$= \int_0^1 x^p \cdot dx = \left[\frac{x^{p+1}}{p+1} \right]_0^1 = \frac{1}{p+1}$$

80. $\lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sec^2 \frac{1}{n^2} + \frac{2}{n^2} \sec^2 \frac{4}{n^2} + \dots + \frac{1}{n} \sec^2 1 \right]$ equals :

$\lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sec^2 \frac{1}{n^2} + \frac{2}{n^2} \sec^2 \frac{4}{n^2} + \dots + \frac{1}{n} \sec^2 1 \right]$ का मान है :

(A) $\frac{1}{2} \sec 1$ (B) $\frac{1}{2} \operatorname{cosec} 1$ (C) $\tan 1$ (D) $\frac{1}{2} \tan 1$

Ans. D

Sol. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r}{n^2} \sec^2 \frac{r^2}{n^2}$

$$= \int_0^1 x \sec^2 x^2 dx$$

$$\text{Put } x^2 = t ; 2x dx = dt ; x dx = \frac{dt}{2}$$

$$= \frac{1}{2} \int_0^1 \sec^2 t dt$$

$$= \frac{1}{2} (\tan t)_0^1 = \frac{1}{2} \tan 1$$

81. $\lim_{n \rightarrow \infty} \left(\frac{(n+1)(n+2) \dots 3n}{n^{2n}} \right)^{1/n}$ is equal to : [JEE(Main)2016]

$\lim_{n \rightarrow \infty} \left(\frac{(n+1)(n+2) \dots 3n}{n^{2n}} \right)^{1/n}$ बराबर है : [JEE(Main)2016]

(A) $\frac{9}{e^2}$ (B) $3 \log 3 - 2$ (C) $\frac{18}{e^4}$ (D) $\frac{27}{e^2}$

Ans. D

Sol. $L = \prod_{r=1}^{2n} \left(\frac{n+r}{n} \right)^{\frac{1}{n}}$

$$\ln L = \frac{1}{n} \sum_{r=1}^{2n} \ln \left(1 + \frac{r}{n} \right)$$

$$= \int_0^2 \ln(1+x) dx$$

$$\ln L = 3 \ln 3 - 2$$

$$L = \frac{27}{e^2}$$

DEFINITE INTEGRATION BY ESTIMATION

82. If $I_1 = \int_0^1 2^{x^2} dx$, $I_2 = \int_0^1 2^{x^3} dx$, $I_3 = \int_1^2 2^{x^2} dx$ and $I_4 = \int_1^2 2^{x^3} dx$ then :

यदि $I_1 = \int_0^1 2^{x^2} dx$, $I_2 = \int_0^1 2^{x^3} dx$, $I_3 = \int_1^2 2^{x^2} dx$ तथा $I_4 = \int_1^2 2^{x^3} dx$ तो :

(A) $I_2 > I_1$ (B) $I_1 > I_2$ (C) $I_3 = I_4$ (D) $I_3 > I_4$

Ans. B

Sol. for $0 < x < 1$, $x^2 > x^3$ and
for $1 < x < 2$, $x^3 > x^2$
 $\therefore$ for, $0 < x < 1$, $2^{x^2} > 2^{x^3}$ and
for $1 < x < 2$, $2^{x^2} < 2^{x^3}$

$$\therefore \int_0^1 2^{x^2} dx > \int_0^1 2^{x^3} dx \text{ and } \int_1^2 2^{x^2} dx < \int_1^2 2^{x^3} dx$$

$$\therefore I_1 > I_2 \text{ and } I_3 < I_4$$

83. Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}} dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}} dx$. Then which one of the following is true? [AIEEE-2008]

$$(A) 1 > \frac{2}{3} \text{ and } J > 2 \quad (B) I < \frac{2}{3} \text{ and } J < 2 \quad (C) I < \frac{2}{3} \text{ and } J > 2 \quad (D) I > \frac{2}{3} \text{ and } J < 2$$

माना $I = \int_0^1 \frac{\sin x}{\sqrt{x}} dx$ तथा $J = \int_0^1 \frac{\cos x}{\sqrt{x}} dx$ तब निम्न में से कौनसा कथन सत्य है? [AIEEE-2008]

$$(A) 1 > \frac{2}{3} \text{ तथा } J > 2 \quad (B) I < \frac{2}{3} \text{ तथा } J < 2 \quad (C) I < \frac{2}{3} \text{ तथा } J > 2 \quad (D) I > \frac{2}{3} \text{ तथा } J < 2$$

Ans. B

Sol. Now

$$\sin x < x \Rightarrow \frac{\sin x}{\sqrt{x}} < \sqrt{x}$$

$$\int_0^1 \frac{\sin x}{\sqrt{x}} dx < \int_0^1 \sqrt{x} dx$$

$$I < \left[\frac{2}{3} x^{3/2} \right]_0^1$$

$$I < \frac{2}{3}$$

$$\therefore \cos x < 1$$

$$\frac{\cos x}{\sqrt{x}} < \frac{1}{\sqrt{x}} \Rightarrow \int_0^1 \frac{\cos x}{\sqrt{x}} dx < \int_0^1 \frac{1}{\sqrt{x}} dx < \left[2\sqrt{x} \right]_0^1 < 2$$

$$J < 2$$

INTEGRATION BY REDUCTION

84. If $I_n = \int (\sin x)^n dx$ $n \in \mathbb{N}$, then $5I_4 - 6I_6$ is equal to :

यदि $I_n = \int (\sin x)^n dx$ $n \in \mathbb{N}$, तब $5I_4 - 6I_6$ बराबर :

(A) $\sin x \cdot (\cos x)^5 + C$

(B) $\sin 2x \cdot \cos 2x + C$

(C) $\frac{\sin 2x}{8} [\cos^2 2x + 1 - 2 \cos 2x] + C$

(D) $\frac{\sin 2x}{8} [\cos^2 2x + 1 + 2 \cos 2x] + C$

Ans. C

Sol. reduction formula of $(\sin x)^n$

85. Let $I_n = \int \tan^n x dx$, ($n > 1$). If $I_4 + I_6 = a \tan^5 x + bx^5 + C$, where C is a constant of integration, then the ordered pair (a, b) is equal to : **[JEE(Main)-2017]**

माना $I_n = \int \tan^n x dx$, ($n > 1$) है। यदि $I_4 + I_6 = a \tan^5 x + bx^5 + C$ है, जहाँ C एक समाकलन अचर है, तो क्रमित युग्म (a, b) बराबर है : **[JEE(Main)-2017]**

(A) $\left(\frac{1}{5}, -1\right)$

(B) $\left(-\frac{1}{5}, 0\right)$

(C) $\left(-\frac{1}{5}, 1\right)$

(D) $\left(\frac{1}{5}, 0\right)$

Ans. D

Sol. $I_4 + I_6 = \int \tan^4(x) (1 + \tan^2 x) dx$

$$= \frac{(\tan x)^5}{5} + C$$

$$a = \frac{1}{5}, b = 0$$

JEE EXERCISES

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS**INTEGRATION BASED ON STANDARD RESULTS**

1. $\int \frac{\sin 2x - \sin 2k}{\sin x - \sin k + \cos x - \cos k} dx :$

Ans. $(\sin x - \cos x) + (\sin k + \cos k)x + C$

Sol.
$$\begin{aligned} & \int \frac{\sin 2x - \sin 2k}{\sin x - \sin k + \cos x - \cos k} dx \\ &= \int \frac{(\sin x + \cos x)^2 - (\sin k + \cos k)^2}{\sin x - \sin k + \cos x - \cos k} dx \\ &= \int (\sin x + \cos x + \sin k + \cos k) dx \\ &= (\sin x - \cos x) + x(\sin k + \cos k) + C \end{aligned}$$

2. $\int \frac{x^2 + 3}{x^6(x^2 + 1)} dx :$

Ans. $C - \frac{2}{x} + \frac{2}{3} \frac{1}{x^3} - \frac{3}{5} \frac{1}{x^5} - 2\tan^{-1}x$

Sol.
$$\begin{aligned} \int \frac{x^2 + 3}{x^6(x^2 + 1)} dx &= \int \frac{(3x^2 + 3 - 2x^2)}{x^6(x^2 + 1)} dx \\ &= 3 \int \frac{dx}{x^6} - 2 \int \frac{dx}{x^4(x^2 + 1)} \\ &= -\frac{3}{5} \frac{1}{x^5} - \int \frac{(2 + 2x^2 - 2x^2)}{x^4(x^2 + 1)} dx \\ &= -\frac{3}{5x^5} - \int \frac{2dx}{x^4} + 2 \int \frac{dx}{x^2(x^2 + 1)} \end{aligned}$$

$$= -\frac{3}{5x^5} + \frac{2}{3x^3} + 2 \int \left(\frac{1}{x^2} - \frac{1}{x^2+1} \right) dx$$

$$= -\frac{3}{5x^5} + \frac{2}{3x^3} - \frac{2}{x} - 2 \tan^{-1} x + C$$

3. $\int \sin x \cos x \cos 2x \cos 4x \, dx :$

Ans. $-\frac{1}{64} \cos 8x + C$

Sol. $\int \sin x \cos x \cos 2x \cos 4x \, dx$

$$= \frac{1}{8} \int \sin 8x \, dx$$

$$= -\frac{1}{64} \cos 8x + C$$

INTEGRATION BY SUBSTITUTION

4. $\int \frac{\ln \left(\ln \left(\frac{1+x}{1-x} \right) \right)}{1-x^2} dx :$

Ans. $\frac{1}{2} \left[\ln \left(\frac{1+x}{1-x} \right) \cdot \ln \left(\ln \frac{1+x}{1-x} \right) - \ln \left(\frac{1+x}{1-x} \right) \right] + C$

Sol. $\int \frac{\ln \left(\ln \left(\frac{1+x}{1-x} \right) \right)}{1-x^2} dx$

$$\text{let } \ln \left(\frac{1+x}{1-x} \right) = t$$

$$\Rightarrow \frac{1-x}{1+x} \times \frac{(1-x) + (1+x)}{(1-x)^2} dx = dt$$

$$\Rightarrow \frac{2}{1-x^2} dx = dt$$

$$= \frac{1}{2} \int \ln(t) dt = \frac{1}{2} [t(\ln t) - t] + C$$

$$= \frac{1}{2} \left[\ln \left(\frac{1+x}{1-x} \right) \cdot \ln \left(\ln \left(\frac{1+x}{1-x} \right) \right) - \ln \left(\frac{1+x}{1-x} \right) \right] + C$$

5. $\int \left[\left(\frac{x}{e} \right)^x + \left(\frac{e}{x} \right)^x \right] \ell_{nx} dx :$

Ans. $\left(\frac{x}{e} \right)^x - \left(\frac{e}{x} \right)^x + C$

Sol. $I = \int \left[\left(\frac{x}{e} \right)^x + \left(\frac{e}{x} \right)^x \right] \ell_{nx} dx = \int \left(1 + \frac{1}{\left(\frac{x}{e} \right)^{2x}} \right) \ell_{nx} \left(\frac{x}{e} \right)^x dx$

$$\left\{ \text{Put } \left(\frac{x}{e} \right)^x = t \Rightarrow \ell_{nx} \left(\frac{x}{e} \right)^x dx = dt \right\}$$

$$I = \int \left(1 + \frac{1}{t^2} \right) dt = t - \frac{1}{t} + C = \left(\frac{x}{e} \right)^x - \left(\frac{e}{x} \right)^x + C$$

6. $\int \frac{\log_6 \sqrt[6]{(\sin x) 6^{\cos x}} \cos x}{\sin x} dx :$

Ans. $\frac{1}{6} \left(\frac{\ln^2(\sin x)}{\ln 36} + \ln \left(\tan \frac{x}{2} \right) + \cos x \right) + C$

Sol. $\int \frac{\log_6 \left((\sin x) 6^{\cos x} \right)^{\frac{1}{6}}}{\sin x} \cdot \cos x dx$

$$= \frac{1}{6} \int \frac{(\log_6 \sin x + \cos x) \cos x}{\sin x} dx$$

let $\sin x = t \Rightarrow \cos x dx = dt$

$$= \frac{1}{6} \int \left(\frac{\log_6 t}{t} + \frac{\sqrt{1-t^2}}{t} \right) dt$$

$$\begin{aligned}
&= \frac{1}{6} \int \left[\frac{\ln t}{\ln 6t} + \frac{\sqrt{1-t^2}}{t} \right] dt \\
&= \frac{1}{6 \ln 6} \int \ln t \cdot \frac{1}{t} dt + \frac{1}{6} \int \frac{\sqrt{1-t^2}}{t} dt \\
&= \frac{1}{6 \ln 6} \frac{[\ln t]^2}{2} + \frac{1}{6} \int \frac{\cos^2 x}{\sin x} dx \\
&= \frac{1}{6 \ln 36} (\ln \sin x)^2 + \frac{1}{6} \left[\int (\operatorname{cosec} x - \sin x) dx \right] \\
&= \frac{1}{6 \ln 36} (\ln \sin x)^2 + \frac{1}{6} \left[\ln \left| \tan \frac{x}{2} \right| + \cos x \right] + C
\end{aligned}$$

7. $\int \frac{x^2 + x}{(e^x + x + 1)^2} dx :$

Ans. $C - \ln(1 + (x+1)e^{-x}) - \frac{1}{1 + (x+1)e^{-x}}$

Sol. $\int \frac{x^2 + x}{(e^x + x + 1)^2} dx = \int \frac{e^{-2x} x (x+1)}{((x+1)e^{-x} + 1)^2} dx$

Let $(x+1)e^{-x} + 1 = t$

$(e^{-x}[1] - e^{-x}(x+1))dx = dt$

$= \int \frac{(1-t)}{t^2} dt = -\frac{1}{t} - \ln|t| + C$

$= \frac{-1}{(x+1)e^{-x} + 1} - \ln|(x+1)e^{-x} + 1| + C$

8. Let $\int \frac{f'(x)g(x) - g'(x)f(x)}{(f(x) + g(x))\sqrt{f(x)g(x) - g^2(x)}} dx = \sqrt{m} \tan^{-1} \left(\sqrt{\frac{f(x) - g(x)}{ng(x)}} \right) + C$, where $m, n \in \mathbb{N}$ and 'C' is constant of integration ($g(x) > 0$). Find the value of $(m^2 + n^2)$.

माना $\int \frac{f'(x)g(x) - g'(x)f(x)}{(f(x) + g(x))\sqrt{f(x)g(x) - g^2(x)}} dx = \sqrt{m} \tan^{-1} \left(\sqrt{\frac{f(x) - g(x)}{ng(x)}} \right) + C$, जहाँ $m, n \in \mathbb{N}$ तथा 'C', समाकलन

का अचर है ($g(x) > 0$) तो $(m^2 + n^2)$ का मान बताओं।

Ans. 8

Sol.
$$\int \frac{f'(x)g(x) - g'(x)f(x)}{(f(x) + g(x))\sqrt{f(x)g(x) - g^2(x)}} dx$$

$$= \int \frac{1}{\left(\frac{f(x)}{g(x)} + 1\right)\sqrt{\frac{f(x)}{g(x)} - 1}} \cdot \frac{f'(x)g(x) - g'(x)f(x)}{g'(x)} dx$$

Let $\frac{f(x)}{g(x)} - 1 = t^2$

$$\Rightarrow \left(\frac{f'(x)g(x) - g'(x)f(x)}{g'(x)} \right) dx = 2t dt$$

$$= \int \frac{2t dt}{t(t^2 + 2)} = 2 \int \frac{dt}{t^2 + 2}$$

$$= \sqrt{2} \left(\tan^{-1} \left(\frac{\sqrt{\frac{f(x)}{g(x)} - 1}}{\sqrt{2}} \right) \right) + C$$

$$\Rightarrow m = 2, n = 2$$

$$\Rightarrow m^2 + n^2 = 4 + 4 = 8$$

9. $\int \frac{\sin^p x}{\cos^{p+2} x} dx :$

Ans. $\frac{\tan^{p+1} x}{p+1} + c$

Sol. $\int \frac{\sin^p x}{\cos^{p+2} x} dx = \int \tan^p x \sec^2 x dx$

Let $\tan x = t$

$$\sec^2 x \, dx = dt$$

$$= \int t^p dt = \frac{1}{p+1} \tan^{p+1} x + C$$

10. $\int x^x \ln(ex) dx :$

Ans. $x^x + C$

Sol.
$$= \int x^x (\ln(ex)) dx = \int x^x (1 + \ln x) dx$$
$$= x^x + C$$

INTEGRATION BY TRIGONOMETRIC SUBSTITUTION

11. $\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx :$

Ans. $(a+x) \arctan \sqrt{\frac{x}{a}} - \sqrt{ax} + C$

Sol.
$$\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx = \int \sin^{-1} \left(\frac{\sqrt{x}}{\sqrt{a+x}} \right) dx$$

Let $x = a \tan^2 \theta$

$$dx = 2a \tan \theta \sec^2 \theta \, d\theta$$

$$= \int \sin^{-1} \left(\frac{\sqrt{a + \tan^2 \theta}}{\sqrt{a} \sec \theta} \right) \cdot 2a \tan \theta \sec^2 \theta \, d\theta$$

$$= 2a \int \underbrace{\theta}_{\text{I}} \underbrace{(\tan \theta \cdot \sec^2 \theta)}_{\text{II}} d\theta$$

$$= 2a\theta \frac{\tan^2 \theta}{2} - a \int \tan^2 \theta \, d\theta$$

$$= a\theta \tan^2 \theta - a \tan \theta + a\theta + C$$

$$= (x+a) \tan^{-1} \sqrt{\frac{x}{a}} - \sqrt{ax} + C$$

12. $\int \sin^{-1}\left(\frac{2x+2}{\sqrt{4x^2+8x+13}}\right) dx :$

Ans. $(x+1) \tan^{-1} \frac{2(x+1)}{3} - \frac{3}{4} \ln(4x^2+8x+13) + C$

Sol. $\int \sin^{-1}\left(\frac{2x+2}{\sqrt{4x^2+8x+13}}\right) dx$

$$= \int \sin^{-1}\left(\frac{2x+2}{\sqrt{(2x+2)^2+9}}\right) dx$$

Let $(2x+2) = 3 \tan \theta \Rightarrow 2dx = 3 \sec^2 \theta d\theta$

$$= \int \sin^{-1}\left(\frac{3 \tan \theta}{3 \sec \theta}\right) \cdot \frac{3}{2} \cdot \sec^2 \theta d\theta$$

$$= \frac{3}{2} \int \underbrace{\theta}_{\text{I}} \cdot \underbrace{\sec^2 \theta d\theta}_{\text{II}}$$

$$= \frac{3}{2} \left[\theta \tan \theta - \int \tan \theta d\theta \right]$$

$$= \frac{3}{2} \frac{(2x+2)}{3} \tan^{-1}\left(\frac{2x+2}{3}\right) + \frac{3}{2} \ln \cos \theta + C$$

$$= (x+1) \tan^{-1}\left(\frac{2x+2}{3}\right) - \frac{3}{2} \ln(\sqrt{1+\tan^2 \theta}) + C$$

$$= (x+1) \tan^{-1}\left(\frac{2x+2}{3}\right) - \frac{3}{4} \ln\left(1 + \frac{(2x+2)^2}{9}\right) + C$$

$$= (x+1) \tan^{-1}\left(\frac{2x+2}{3}\right) - \frac{3}{4} \ln(4x^2+8x+13) + C$$

INTEGRATION BY PARTS

13. $\int \frac{x \ln x}{(x^2 - 1)^{3/2}} dx$

Ans. $\arcsin x - \frac{\ln x}{\sqrt{x^2 - 1}} + C$

Sol. $\int \frac{x \ln x}{(x^2 - 1)^{3/2}} dx = \int \frac{x}{(x^2 - 1)^{3/2}} \cdot \ln x dx$

Let $x = \sec \theta$

$dx = \sec \theta \tan \theta d\theta$

$$= \int \frac{\sec \theta \ln(\sec \theta) \cdot \sec \theta \tan \theta d\theta}{\tan^3 \theta}$$

$$= \int \frac{\sec^2 \theta \ln(\sec \theta)}{\tan^2 \theta} d\theta$$

$$= \int \underbrace{\frac{\sec^2 \theta}{\tan^2 \theta}}_{II} \cdot \underbrace{\ln(\sec \theta)}_I d\theta$$

$$= \ln(\sec \theta) \cdot \frac{-1}{\tan \theta} + \int \frac{d\theta}{\sec \theta \tan \theta} \times \sec \theta \tan \theta$$

$$= -\frac{\ln(\sec \theta)}{\tan \theta} + \int d\theta$$

$$= -\frac{\ln(x)}{\sqrt{x^2 - 1}} + \sec^{-1} x + C$$

14. $\int x^3 e^{x^2} dx$

Ans. $\frac{1}{2}(x^2 - 1)e^{x^2} + c$

Sol. $\int x^3 e^{x^2} dx$

Let $x^2 = t \Rightarrow 2x dx = dt$

$$= \frac{1}{2} \int \underbrace{t}_{\text{I}} \underbrace{e^t}_{\text{II}} dt = \frac{1}{2} [te^t - e^t] + C$$

$$= \frac{1}{2} [x^2 - 1] e^{x^2} + C$$

INTEGRATION OF RATIONAL FUNCTIONS

15. $\int \frac{x^5 + 3x^4 - x^3 + 8x^2 - x + 8}{x^2 + 1} dx$

Ans. $\frac{x^4}{4} + x^3 - x^2 + 5x + \frac{1}{2} \ln(x^2 + 1) + 3 \tan^{-1}x + C$

Sol. $\int \frac{x^5 + 3x^4 - x^3 + 8x^2 - x + 8}{x^2 + 1} dx$

$$= \int \frac{(x^3 + 3x^2 - 2x + 5)(x^2 + 1) + (x + 3)dx}{(x^2 + 1)}$$

$$= \int (x^3 + 3x^2 - 2x + 5) dx + \int \frac{(x + 3)}{x^2 + 1} dx$$

$$= \frac{x^4}{4} + x^3 - x^2 + 5x + \frac{1}{2} \ln(x^2 + 1) + 3 \tan^{-1}x + C$$

16. $\int \frac{(\sqrt{x} + 1)dx}{\sqrt{x}(\sqrt[3]{x} + 1)}$

Ans. $6 \left[\frac{t^4}{4} - \frac{t^2}{2} + t + \frac{1}{2} \ln(1 + t^2) - \tan^{-1}t \right] + C$ where $t = x^{1/6}$

Sol. $\int \frac{(\sqrt{x} + 1)}{\sqrt{x}(\sqrt[3]{x} + 1)} dx = \int \frac{(\sqrt{x} + 1)}{\sqrt{x}(\sqrt[3]{x} + 1)} dx$

Let $x^{1/6} = t$

$$\Rightarrow \frac{1}{6} x^{-5/6} dx = dt$$

$$= \int \frac{(t^3 + 1)}{t^3(t^2 + 1)} \cdot \frac{6t^5 dt}{5}$$

$$\begin{aligned}
&= 6 \int \frac{t^2(t^3+1)}{t^2+1} dt = 6 \int \frac{(t^5+t^2)}{t^2+1} dt \\
&= 6 \int \frac{(t^2+1+t^5-1)}{t^2+1} dt \\
&= 6 \left[t + \int \frac{t^5-1}{t^2+1} dt \right] \\
&= 6 \left[t - \tan^{-1} t + \int \frac{t^4 \cdot t}{\underbrace{t^2+1}_{\text{let } t^2+1=\theta}} dt \right] \\
&= 6 \left[t - \tan^{-1} t + \frac{1}{2} \int \frac{(\theta-1)^2}{\theta} d\theta \right] \\
&= 6 \left[t - \tan^{-1} t + \frac{\theta^2}{4} - \theta + \frac{1}{2} \ln |\theta| \right] + C \\
&= 6 \left[t - \tan^{-1} t + \frac{(t^2+1)^2}{4} - t^2 - 1 + \frac{1}{2} \ln(1+t^2) \right] + C \\
&= 6 \left[\frac{t^4}{4} - \frac{t^2}{2} + t + \frac{1}{2} \ln(1+t^2) - \tan^{-1} t \right] + C; t = x^{1/6}
\end{aligned}$$

17. $\int \sqrt{x^2 - 8x + 7} dx$

Ans. $-\frac{9}{2} \log [x-4 + \sqrt{x^2 - 8x + 7}] + \frac{1}{2}(x-4) \sqrt{x^2 - 8x + 7} + c$

Sol. $\int \sqrt{x^2 - 8x + 7} dx$

$$= \int \sqrt{(x-4)^2 - 9} dx$$

$$= \frac{x-4}{2} \sqrt{x^2 - 8x + 7} - \frac{9}{2} \log |x-4 + \sqrt{x^2 - 8x + 7}| + C$$

TWO CLASSIC INTEGRALS

18. $\int \frac{e^x(2-x^2)}{(1-x)\sqrt{1-x^2}} dx$

Ans. $e^x \sqrt{\frac{1+x}{1-x}} + c$

Sol. $\int \frac{e^x(2-x^2)}{(1-x)\sqrt{1-x^2}} dx$

$$= \int e^x \left(\frac{1+(1+x)(1-x)}{(1-x)\sqrt{1-x^2}} \right) dx$$

$$= \int e^x \left(\underbrace{\frac{\sqrt{1+x}}{\sqrt{1-x}}}_{f(x)} + \underbrace{\frac{1}{(1-x)\sqrt{1-x^2}}}_{f'(x)} \right) dx$$

$$= e^x \sqrt{\frac{1+x}{1-x}} + C$$

19. $\int \frac{e^{\cos x}(x \sin^3 x + \cos x)}{\sin^2 x} dx$

Ans. $C - e^{\cos x}(x + \operatorname{cosec} x)$

Sol. $\int e^{\cos x} \left(\frac{x \sin^3 x + \cos x}{\sin^2 x} \right) dx$

$$= \int e^{\cos x} (x \sin x + \cot x \cdot \operatorname{cosec} x) dx$$

$$= \int x e^{\cos x} \sin x dx + \int e^{\cos x} \operatorname{cosec} x \cot x dx$$

$$= x \int e^{\cos x} \sin x dx - \int \left(\int e^{\cos x} \sin x dx \right) dx + e^{\cos x} \int \operatorname{cosec} x \cot x dx$$

$$+ \int e^{\cos x} \sin x \left[\int \operatorname{cosec} x \cot x dx \right] dx + C$$

$$= -x e^{\cos x} + \int e^{\cos x} dx - e^{\cos x} \operatorname{cosec} x + \int e^{\cos x} dx + C$$

$$= -(x + \operatorname{cosec} x)e^{\cos x} + C$$

INTEGRATION OF TRIGONOMETRIC FUNCTION

20. (a) $\int \frac{dx}{\cot \frac{x}{2} \cdot \cot \frac{x}{3} \cdot \cot \frac{x}{6}};$ (b) $\int \frac{\tan(\ln x) \tan\left(\ln \frac{x}{2}\right) \tan(\ln 2)}{x} dx$

Ans. (a) $2\ln\left(\sec \frac{x}{2}\right) - 3\ln\left(\sec \frac{x}{3}\right) - 6\ln\left(\sec \frac{x}{6}\right) + C;$ (b) $\ln\left(\frac{\sec(\ln x)}{\sec(\ln(x/2))x^{\tan(\ln 2)}}\right) + C$

Sol. (a) $\int \frac{dx}{\cot\left(\frac{x}{2}\right) \cot\left(\frac{x}{3}\right) \cot\left(\frac{x}{6}\right)}$

$$= \int \tan \frac{x}{2} \cdot \tan \frac{x}{3} \cdot \tan \frac{x}{6} dx$$

$$\because \frac{x}{2} - \frac{x}{3} = \frac{x}{6}$$

$$\Rightarrow \tan\left(\frac{x}{2} - \frac{x}{3} - \frac{x}{6}\right) = 0$$

$$\Rightarrow \tan \frac{x}{2} + \tan\left(\frac{-x}{3}\right) + \tan\left(\frac{-x}{6}\right) = \tan \frac{x}{2} \cdot \tan\left(\frac{-x}{3}\right) \cdot \tan\left(\frac{-x}{6}\right)$$

$$= \int \left(\tan \frac{x}{2} - \tan \frac{x}{3} - \tan \frac{x}{6} \right) dx$$

$$= 2\ln \left| \sec \frac{x}{2} \right| - 3\ln \left| \sec \frac{x}{3} \right| - 6\ln \left| \sec \frac{x}{6} \right| + C$$

(b) $\int \frac{\tan(\ln x) \tan\left(\ln \frac{x}{2}\right) \tan(\ln 2)}{x} dx$

Let $\ln x = t$

$$= \int \tan t \cdot \tan(t - \ln 2) \cdot \tan(\ln 2) dx$$

$$= \int (\tan t - \tan(\ln 2) - \tan(t - \ln 2)) dt$$

$$= \ln |\sec t| - t \tan (\ln 2) - \ln |\sec t - \ln 2| + C$$

$$= \ln \left| \frac{\sec(\ln x)}{\sec\left(\ln\left(\frac{x}{2}\right)\right) \cdot x^{\tan(\ln 2)}} \right| + C$$

$$21. \int \sqrt{\frac{\operatorname{cosec} x - \cot x}{\operatorname{cosec} x + \cot x}} \cdot \frac{\sec x}{\sqrt{1 + 2 \sec x}} dx$$

$$\text{Ans. } \sin^{-1} \left(\frac{1}{2} \sec^2 \frac{x}{2} \right) + C$$

$$\begin{aligned} \text{Sol. } & \int \sqrt{\frac{\operatorname{cosec} x - \cot x}{\operatorname{cosec} x + \cot x}} \cdot \frac{\sec x}{\sqrt{1 + 2 \sec x}} dx \\ &= \int \frac{1}{\operatorname{cosec} x + \cot x} \cdot \frac{1}{\sqrt{\cos^2 x + 2 \cos x + 1 - 1}} dx \\ &= \int \frac{\sin x}{1 + \cos x} \cdot \frac{1}{\sqrt{\cos^2 x + 2 \cos x + 1 - 1}} dx \end{aligned}$$

$$\text{Let } 1 + \cos x = t$$

$$- \sin x dx = dt$$

$$= - \int \frac{dt}{t \sqrt{t^2 - 1}}$$

$$= + \operatorname{cosec}^{-1}(t) + C$$

$$= \sin^{-1} \left(\frac{1}{2 \cos^2 \frac{x}{2}} \right) + C$$

$$= \sin^{-1} \left(\frac{1}{2} \sec^2 \frac{x}{2} \right) + C$$

$$22. \int \frac{\tan 2\theta}{\sqrt{\cos^6 \theta + \sin^6 \theta}} d\theta$$

Ans. $\ln\left(\frac{1+\sqrt{1+3\cos^2 2\theta}}{\cos 2\theta}\right) + C$

Sol.
$$\begin{aligned} & \int \frac{\tan 2\theta}{\sqrt{\cos^6 \theta + \sin^6 \theta}} d\theta \\ &= \int \frac{\sin 2\theta}{\cos 2\theta \sqrt{1-3\sin^2 \theta \cos^2 \theta}} d\theta \\ &= \int \frac{\sin 2\theta}{\cos 2\theta \sqrt{\sin^4 \theta + \cos^4 \theta - \sin^2 \theta \cos^2 \theta}} d\theta \\ &= \int \frac{\sin 2\theta}{\cos 2\theta} \cdot \frac{1}{\sqrt{(\cos^2 \theta - \sin^2 \theta)^2 + \sin^2 \theta \cos^2 \theta}} d\theta \\ &= \int \frac{2\sin 2\theta}{\cos 2\theta} \cdot \frac{1}{\sqrt{\cos^2 2\theta + \frac{1}{4}\sin^2 2\theta}} d\theta \\ &= \int \frac{2\sin 2\theta}{\cos 2\theta} \cdot \frac{1}{\sqrt{3\cos^2 2\theta + 1}} d\theta \end{aligned}$$

Let $\cos 2\theta = t$

$\Rightarrow -2\sin 2\theta d\theta = dt$

$= -\int \frac{dt}{t\sqrt{3t^2+1}} = -\int \frac{t dt}{t^2\sqrt{3t^2+1}}$

Let $3t^2 + 1 = \theta^2$

$\Rightarrow 6t dt = 2\theta d\theta$

$= -\int \frac{2\theta}{6\left(\frac{\theta^2-1}{3}\right) \cdot \theta} d\theta = -\int \frac{d\theta}{\theta^2-1}$

$= \frac{-1}{2} \ln \left| \frac{\theta-1}{\theta+1} \right| + C$

$= \frac{-1}{2} \ln \left| \frac{\theta+1}{\theta-1} \right| + C$

$$= \frac{1}{2} \ln \left| \frac{1 + \sqrt{1 + 3 \cos^2 2\theta}}{\sqrt{1 + 3 \cos^2 2\theta} - 1} \right| + C$$

$$= \ln \left| \frac{1 + \sqrt{1 + 3 \cos^2 2\theta}}{\sqrt{3} \cos 2\theta} \right| + C$$

$$= \ln \left| \frac{1 + \sqrt{1 + 3 \cos^2 2\theta}}{\cos 2\theta} \right| + C$$

23. $\int \frac{a^2 \sin^2 x + b^2 \cos^2 x}{a^4 \sin^2 x + b^4 \cos^2 x} dx$

Ans. $\frac{1}{a^2 + b^2} \left(x + \tan^{-1} \left(\frac{a^2 \tan x}{b^2} \right) \right) + C$

Sol. $\int \frac{a^2 \sin^2 x + b^2 \cos^2 x}{a^4 \sin^2 x + b^4 \cos^2 x} dx$

$$= \int \frac{(a^2 + \tan^2 x + b^2)}{(a^4 \tan^2 x + b^4)} dx$$

$$= \frac{1}{a^2 + b^2} = \int \frac{(a^4 + \tan^2 x + b^4 + b^2 a^2 + a^2 \tan^2 x)}{a^4 \tan^2 x + b^4} dx$$

$$= \frac{1}{a^2 + b^2} \left[\int dx + \int \frac{a^2 b^2 + a^2 \tan^2 x}{a^4 \tan^2 x + b^4} dx \right]$$

$$= \frac{1}{a^2 + b^2} \left[x + \int \frac{\frac{a^2}{b^2} + \frac{a^2}{b^2} \tan^2 x}{\left(\frac{a^2}{b^2} \tan x \right)^2 + 1} dx \right]$$

$$= \frac{1}{a^2 + b^2} \left[x + \tan^{-1} \left(\frac{a^2 \tan x}{b^2} \right) \right] + C$$

24. $\int \frac{\cos^2 x}{1 + \tan x} dx$

Ans. $\frac{1}{4} \ln(\cos x + \sin x) + \frac{x}{2} + \frac{1}{8} (\sin 2x + \cos 2x) + C$

Sol. $\int \frac{\cos^2 x}{1 + \tan x} dx = \int \frac{\sec^2 x dx}{(1 + \tan^2 x)^2 (1 + \tan x)}$

Let $\tan x = t$

$\sec^2 x dx = dt$

$= \int \frac{dt}{(1+t)(1+t^2)^2}$

$\therefore \frac{1}{(1+t)(1+t^2)^2} = \frac{A}{1+t} + \frac{B}{(1+t^2)} + \frac{Ct+D}{(t^2+1)^2}$

$\Rightarrow A = \frac{1}{4}, A+B=0$

$\Rightarrow A = \frac{1}{4}, B = -\frac{1}{4}, E = \frac{1}{4}, C = -\frac{1}{2}, D = \frac{1}{2}$

$\Rightarrow \int \frac{dt}{(1+t)(1+t^2)^2} = \frac{1}{4} \int \left(\frac{1}{1+t} \right) + \frac{1-t}{4(1+t^2)} + \frac{1-t}{2(t^2+1)^2} dt$

$= \frac{1}{4} \int \frac{dt}{1+t} + \frac{1}{4} \int \frac{dt}{1+t^2} - \frac{1}{4} \int \frac{t}{1+t^2} dt$

$+ \frac{1}{4} \int \frac{(1-t)}{(t^2+1)^2} dt$

$= \frac{-1}{8} \log(\sec^2 x) + \frac{1}{4} \ln |1 + \tan x| + \frac{1}{4} x$

$+ \frac{1}{2} I_1 \quad ; \quad I_1 = \int \frac{1-t}{(t^2+1)^2} dt$

Now, $I_1 = \int \frac{(1-t)}{(1+t^2)^2} dt$

Let $\tan \theta = t$

$$\sec^2 \theta d\theta = dt$$

$$= \int \frac{1 - \tan \theta}{\sec^4 \theta} \cdot \sec^2 \theta d\theta$$

$$= \int \cos^2 \theta (1 - \tan \theta) d\theta$$

$$= \int (1 - \sin^2 \theta + \sin^2 \theta + \tan \theta - \tan \theta) d\theta$$

$$= \frac{\theta}{2} + \frac{1}{4} \sin 2\theta + \frac{\cos^2 \theta}{2}$$

$$\Rightarrow \int \frac{\cos^2 x}{1 + \tan x} dx$$

$$= \frac{x}{2} + \frac{1}{8} (\sin 2x + \cos 2x)$$

$$+ \frac{1}{4} \log (\sin x + \cos x) + C$$

25. $\int (\sin x)^{-11/3} (\cos x)^{-1/3} dx$

Ans. $-\frac{3(1+4\tan^2 x)}{8(\tan x)^{8/3}} + C$

Sol. $\int (\sin x)^{-11/3} (\cos x)^{-1/3} dx$

$$= \int \frac{dx}{(\sin^{11} x \cos x)^{1/3}} = \int \frac{1}{(\sin^{12} x \cot x)^{1/3}} dx$$

Let $\cot x = t$

$$\operatorname{cosec}^2 x dx = -dt$$

$$= -\int \left(\frac{1+t^2}{t^{1/3}} \right) dt = -\int t^{-1/3} dt - \int t^{5/3} dt$$

$$= -\left[\frac{t^{2/3}}{2/3} + \frac{t^{8/3}}{8/3} \right] + C$$

$$= \frac{-3}{8} \left[4 \cos^{2/3} x + \cot^{8/3} x \right] + C$$

26. $\int \frac{dx}{\sec x + \cos x} dx$

Ans. $\frac{1}{2} \left[\sin x - \cos x - \frac{1}{\sqrt{2}} \ln \tan \left(\frac{x}{2} + \frac{\pi}{8} \right) \right] + C$

Sol. $\int \frac{dx}{\sec x + \cos x} = \frac{1}{2} \int \frac{\sin 2x}{\sin x + \cos x} dx$

$$= \frac{1}{2} \int \left(\frac{(\sin x + \cos x)^2 - 1}{\sin x \cos x} \right) dx$$

$$\frac{1}{2} \left[\int (\sin x + \cos x) dx - \frac{1}{\sqrt{2}} \int \frac{dx}{\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x} \right]$$

$$= \frac{1}{2} \left[\sin x - \cos x - \frac{1}{\sqrt{2}} \int \frac{dx}{\sin \left(x + \frac{\pi}{4} \right)} \right]$$

$$= \frac{1}{2} \left[\sin x - \cos x - \frac{1}{\sqrt{2}} \ln \left| \tan \left(\frac{x}{2} + \frac{\pi}{8} \right) \right| \right] + C$$

27. $\int \frac{dx}{\sin x + \sec x}$

Ans. $\frac{1}{2\sqrt{3}} \ln \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - \sin x + \cos x} + \arctan (\sin x + \cos x) + C$

Sol. $\int \frac{dx}{\sin x + \sec x} = \int \frac{\cos x dx}{\sin x \cos x + 1}$

$$= \int \frac{2 \cos x dx}{\sin 2x + 2} = \int \frac{(\cos x + \sin x) + (\cos x - \sin x)}{\sin 2x + 1 + 1} dx$$

$$\begin{aligned}
&= \int \frac{(\cos x + \sin x) + (\cos x - \sin x)}{(\sin x + \cos x)^2 + 1} dx \\
&= \int \frac{\cos x - \sin x}{1 + (\sin x + \cos x)^2} dx + \int \frac{(\cos x + \sin x)}{1 + (\cos x + \sin x)^2} dx \\
&= \tan^{-1}(\sin x + \cos x) + \int \frac{(\cos x + \sin x)}{3 - (\sin x - \cos x)^2} dx \\
&= \tan^{-1}(\sin x + \cos x) + \frac{1}{2\sqrt{3}} \ln \left| \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - \sin x + \cos x} \right| + C
\end{aligned}$$

28. $\int \frac{\cos x - \sin x}{7 - 9\sin 2x} dx$

Ans. $\frac{1}{24} \ln \frac{(4 + 3\sin x + 3\cos x)}{(4 - 3\sin x - 3\cos x)} + C$

Sol. $\int \frac{\cos x - \sin x}{7 - 9\sin 2x} dx$

$$\begin{aligned}
&= \int \frac{\cos x - \sin x}{16 - 9(\sin x + \cos x)^2} dx \\
&= \frac{1}{24} \ln \left| \frac{4 + 3\sin x + 3\cos x}{4 - 3\sin x - 3\cos x} \right| + C
\end{aligned}$$

29. $\int \frac{\sqrt{\cot x} - \sqrt{\tan x}}{1 + 3\sin 2x} dx$

Ans. $\tan^{-1} \left(\frac{\sqrt{2\sin 2x}}{\sin x + \cos x} \right) + C$

Sol. $\int \frac{\sqrt{\cot x} - \sqrt{\tan x}}{1 + 3\sin 2x} dx$

$$= \int \frac{\cos x - \sin x}{\sqrt{\sin x \cos x} (3(\sin x + \cos x)^2 - 2)} dx$$

$$= \sqrt{2} \int \frac{\cos x - \sin x}{\sqrt{\sin 2x} (3(\sin x + \cos x)^2 - 2)} dx$$

$$\text{Let } \sin x + \cos x = t \Rightarrow 1 + \sin 2x = t^2$$

$$\Rightarrow (\cos x - \sin x) dx = dt$$

$$= \sqrt{2} \int \frac{dt}{\sqrt{t^2 - 1} (3t^2 - 2)}$$

$$\text{Let } t = \frac{1}{\theta}$$

$$\Rightarrow dt = \frac{-1}{\theta^2} d\theta$$

$$= \sqrt{2} \int \frac{\frac{-1}{\theta^2} d\theta}{\frac{\sqrt{1 - \theta^2}}{\theta} \cdot \left(\frac{3 - 2\theta^2}{\theta^2} \right)} = \sqrt{2} \int \frac{-\theta}{\sqrt{1 - \theta^2} (3 - 2\theta^2)} d\theta$$

$$\text{Let } \sqrt{1 - \theta^2} = \alpha$$

$$\Rightarrow \frac{\theta d\theta}{\sqrt{1 - \theta^2}} = d\alpha$$

$$= \sqrt{2} \int \frac{d\alpha}{1 + 2\alpha^2} = \frac{\sqrt{2}}{\sqrt{2}} \tan^{-1}(\sqrt{2}\alpha) + C$$

$$= \tan^{-1}(\sqrt{2} \cdot \sqrt{1 - \theta^2}) + C$$

$$= \tan^{-1}\left(\frac{\sqrt{2}\sqrt{t^2 - 1}}{t}\right) + C$$

$$= \tan^{-1}\left(\frac{\sqrt{2}\sin 2x}{\sin x + \cos x}\right) + C$$

INTEGRATION BY PARTIAL FRACTIONS

30. $\int \frac{3x^2+1}{(x^2-1)^3} dx$

Ans. $C - \frac{x}{(x^2-1)^2}$

Sol.
$$\begin{aligned}\int \frac{3x^2+1}{(x^2-1)^3} dx &= \int \frac{3x^2+1}{(x-1)^3(x+1)^3} dx \\&= \int -\frac{1}{2} \left[\frac{1}{(x+1)^3} - \frac{1}{(x-1)^3} \right] dx \\&= \frac{1}{2} \left[\frac{(x-1)^{-2}}{-2} + \frac{(x+1)^{-2}}{-2} \right] = C - \frac{x}{(x^2-1)^2}\end{aligned}$$

31. Let $f(x)$ is a quadratic function such that $f(0) = 1$ and $\int \frac{f(x)dx}{x^2(x+1)^3}$ is a rational function, find the value of $f'(0)$.

माना $f(x)$ एक द्विघातीय फलन है जहाँ $f(0) = 1$ तथा $\int \frac{f(x)dx}{x^2(x+1)^3}$ एक परिमेय फलन है तो $f'(0)$ का मान बताइये।

Ans. 3

Sol.
$$\begin{aligned}\int \frac{f(x)dx}{x^2(x+1)^3} &= \int \left(\frac{1}{x^2} - \frac{x}{(x+1)^3} \right) dx \\ \int \frac{f(x)dx}{x^2(x+1)^3} &= \int \frac{3x^2+3x+1}{x^2(x+1)^3} dx \\ \Rightarrow f(x) &= 3x^2+3x+1 \\ f'(0) &= 3\end{aligned}$$

32. $\int \frac{4x^5-7x^4+8x^3-2x^2+4x-7}{x^2(x^2+1)^2} dx$

Ans. $4 \ln x + \frac{7}{x} + 6 \tan^{-1}(x) + \frac{6x}{1+x^2} + C$

Sol. $I = \int \frac{4x^5 - 7x^4 + 8x^3 - 2x^2 + 4x - 7}{x^2(x^2 + 1)^2} dx$

We can split into partial fraction as $\frac{A}{x^2} + \frac{B}{x} + \frac{Cx + D}{x^2 + 1} + \frac{Ex + F}{(x^2 + 1)^2}$ _____(1)

Multiplying both sides by x^2 and putting $x = 0$ we get $A = -7$

Similaly multiplying by $(x^2 + 1)$ and substituting $x = i$ we get $Ei + F = 12$

$\therefore E = 0, F = 12$

to solve it substituting different and convenient values of x .

put $x = 1$ we get $2B + C + D = 8$ _____(2)

$x = -1$ we get $-2B - C + D = -8$ _____(3)

$x = 2$ we get $5B + 4C = 20$ _____(4)

From (2), (3) and (4) we get $D = 0, C = 0$ and $B = 4$

$$I = \int \left(\frac{-7}{x^2} + \frac{4}{x} + \frac{12}{(x^2 + 1)^2} \right) dx$$

$$I = \frac{7}{x} + 4 \log|x| + \int \frac{12}{(x^2 + 1)^2} dx$$

Put $x = \tan \theta$

$$\frac{7}{x} + 4 \log|x| + 12 \int \cos^2 \theta d\theta$$

$$I = 4 \ln x + \frac{7}{x} + 6 \tan^{-1}(x) + \frac{6x}{1 + x^2} + C$$

KUTUR PUTUR

33. $\int \frac{(ax^2 - b) dx}{x \sqrt{c^2 x^2 - (ax^2 + b)^2}}$

Ans. $\sin^{-1} \left(\frac{ax^2 + b}{cx} \right) + k$

Sol.
$$\int \frac{(ax^2 - b)}{cx^2 \sqrt{1 - \left(\frac{ax^2 + b}{cx}\right)^2}} dx$$

$$\int \frac{\left(\frac{a}{c} - \frac{b}{cx^2}\right)}{\sqrt{1 - \left(\frac{a}{c}x + \frac{b}{cx}\right)^2}} dx$$

$$= \sin^{-1}\left(\frac{ax^2 + b}{cx}\right) + K$$

34.
$$\int \frac{dx}{\left(x + \sqrt{x(1+x)}\right)^2}$$

Ans. $2 \ln \frac{t}{2t+1} + \frac{1}{2t+1} + C$ when $t = x + \sqrt{x^2 + x}$

Sol.
$$\int \frac{dx}{\left(x + \sqrt{x(1+x)}\right)^2} = \int \frac{dx}{x^2 \left(1 + \sqrt{1 + \frac{1}{x}}\right)^2}$$

Let $1 + \frac{1}{x} = t^2$

$$= \int \frac{-2t dt}{(1+t)^2} = -2 \int \frac{t dt}{(1+t)^2}$$

$$= -2 \left[\int \frac{dt}{t+1} - \int \frac{1}{(1+t)^2} dt \right]$$

$$= -2 \ln |t+1| + \frac{2}{t+1} + C$$

$$= -2 \ln \left| \frac{\sqrt{x} + \sqrt{x+1}}{\sqrt{x}} \right| + \frac{2\sqrt{x}}{\sqrt{x+1} + \sqrt{x}} + C$$

35. $\int \frac{5x^4 + 4x^5}{(x^5 + x + 1)^2} dx$

Ans. $C - \frac{x+1}{x^5 + x + 1}$ or $C + \frac{x^5}{x^5 + x + 1}$

Sol. $\int \frac{5x^4 + 4x^5}{(x^5 + x + 1)^2} dx = \int \frac{\left(\frac{5}{x^6} + \frac{4}{x^5}\right)}{\left(1 + \frac{1}{x^4} + \frac{1}{x^5}\right)^2} dx$

Let $\frac{1}{x^4} + \frac{1}{x^5} + 1 = t$

$-\left(\frac{4}{x^5} + \frac{5}{x^6}\right) dx = dt$

$= -\int \frac{dt}{t^2} = \frac{1}{t} + C = \frac{x^5}{x^5 + x + 1} + C$

36. For any natural number m , evaluate $\int (x^{3m} + x^{2m} + x^m) (2x^{2m} + 3x^m + 6)^{\frac{1}{m}} dx$, where $x > 0$

यदि $m \in \mathbb{N}$ तो $\int (x^{3m} + x^{2m} + x^m) (2x^{2m} + 3x^m + 6)^{\frac{1}{m}} dx$, का मान ज्ञात करो जहाँ $x > 0$

Ans. $\frac{1}{6(m+1)} (2x^{3m} + 3x^{2m} + 6x^m)^{\frac{m+1}{m}} + C$

Sol. $I = \int (x^{3m} + x^{2m} + x^m)(2x^{2m} + 3x^m + 6)^{1/m} dx$

$= \int (x^{3m} + x^{2m} + x^m) \left[\frac{2x^{3m} + 3x^{2m} + 6x^m}{x^m} \right]^{1/m} dx$

$= \int \left(\frac{x^{3m} + x^{2m} + x^m}{x} \right) (2x^{3m} + 3x^{2m} + 6x^m)^{1/m} dx$

$= \int (x^{3m-1} + x^{2m-1} + x^{m-1})(2x^{3m} + 3x^{2m} + 6x^m)^{1/m} dx$

Put $2x^{3m} + 3x^{2m} + 6x^m = y$

$\therefore I = \frac{1}{6m} \int y^{1/m} dy = \frac{1}{6m} \frac{y^{(1/m)+1}}{\left(\frac{m+1}{m}\right)} + C$

$$= \frac{1}{6} \frac{y^{\frac{m+1}{m}}}{(m+1)} + C$$

$$= \frac{1}{6} \frac{(2x^{3m} + 3x^{2m} + 6x^m)^{\frac{m+1}{m}}}{m+1} + C$$

INTEGRATION OF IRRATIONAL ALGEBRAIC FUNCTIONS

37. $\int \frac{dx}{(x-\alpha)\sqrt{(x-\alpha)(x-\beta)}}$

Ans. $\frac{-2}{\alpha-\beta} \sqrt{\frac{x-\beta}{x-\alpha}} + C$

Sol. $\int \frac{dx}{(x-\alpha)\sqrt{(x-\alpha)(x-\beta)}}$

$$\text{let } x - \alpha = \frac{1}{t} \Rightarrow x = \frac{\alpha + 1}{t}$$

$$\Rightarrow dx = \frac{-1}{t^2} dt$$

$$= - \int \frac{t}{\sqrt{\frac{1}{t} \cdot \left(\alpha - \beta + \frac{1}{t} \right)}} \cdot \frac{1}{t^2} dt$$

$$= - \int \frac{1}{t \sqrt{(\alpha - \beta)t + 1}} dt$$

$$= - \int \frac{dt}{\sqrt{(\alpha - \beta)t + 1}}$$

$$= \frac{-2}{(\alpha - \beta)} \sqrt{(\alpha - \beta)t + 1} + C$$

$$= \frac{-2}{(\alpha - \beta)} \sqrt{\frac{(\alpha - \beta)}{x - \alpha} + 1} + C$$

$$= \frac{-2}{\alpha - \beta} \sqrt{\frac{x - \beta}{x - \alpha}} + C$$

MISCELLANEOUS

38. $\int \frac{x^3 + x + 1}{x^4 + x^2 + 1} dx$

Ans. $\frac{1}{2} \ln(x^2 + x + 1) - \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right) - \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right) + C$

Sol. $\int \frac{x^3 + x + 1}{x^4 + x^2 + 1} dx$

$$= \int \frac{(x^3 + 1)}{x^4 + x^2 + 1} dx + \int \frac{x}{x^4 + x^2 + 1} dx$$

$$= \frac{1}{2} \int \frac{2(x+1)(x^2 - x + 1)}{(x^2 + x + 1)(x^2 - x + 1)} + \frac{1}{2} \int \frac{2x}{\underbrace{x^4 + x^2 + 1}_{\text{Let } x^2 = t}} dx$$

$$2x dx = dt$$

$$= \frac{1}{2} \int \frac{2x+1}{x^2 + x + 1} + \frac{1}{2} \int \frac{dx}{x^2 + x + 1} \int \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} + \frac{1}{2} \int \frac{dt}{t^2 + t + 1}$$

$$= \frac{1}{2} \ln |x^2 + x + 1| + \frac{1}{2} \times \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{x + \frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + \frac{1}{2} \times \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + C$$

$$= \frac{1}{2} \ln |x^2 + x + 1| + \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right) + C$$

39. $\int \frac{(1+x^2)dx}{1-2x^2 \cos \alpha + x^4} \alpha \in (0, \pi)$

Ans. $\frac{1}{2} \left(\operatorname{cosec} \frac{\alpha}{2} \right) \cdot \tan^{-1} \left(\left(\frac{x^2 - 1}{2x} \right) \operatorname{cosec} \frac{\alpha}{2} \right)$

Sol. $\int \frac{(1+x^2)dx}{1-2x^2 \cos \alpha + x^4} \alpha \in (0, \pi)$

$$= \int \frac{(1+x^2)}{x^2 \left(x^2 - 2 \cos \alpha + \frac{1}{x^2} \right)} dx$$

$$= \int \frac{\left(1 + \frac{1}{x^2} \right) dx}{\left(x - \frac{1}{x} \right)^2 + 2 - 2 \cos \alpha}$$

$$\text{Let } x - \frac{1}{x} = t$$

$$\left(1 + \frac{1}{x^2} \right) dx = dt$$

$$= \int \frac{dt}{t^2 + 2 - 2 \cos \alpha} = \int \frac{dt}{t^2 + \left(2 \sin \frac{\alpha}{2} \right)^2}$$

$$= \frac{1}{2 \sin \frac{\alpha}{2}} \left[\tan^{-1} \left(\frac{t}{2 \sin \frac{\alpha}{2}} \right) \right] + C$$

$$= \frac{1}{2} \operatorname{cosec} \frac{\alpha}{2} \tan^{-1} \left(\left(\frac{x^2 - 1}{2x} \right) \operatorname{cosec} \frac{\alpha}{2} \right) + C$$

DEFINITE INTEGRATION WITH THE HELP OF INDEFINITE INTEGRATION

40. Evaluate: $\int_0^1 e^{\ln \tan^{-1} x} \cdot \sin^{-1}(\cos x) dx$.

ज्ञात करो: $\int_0^1 e^{\ln \tan^{-1} x} \cdot \sin^{-1}(\cos x) dx$.

Ans. $\frac{\pi^2}{8} - \frac{\pi}{4} (1 + \ln 2) + \frac{1}{2}$

Sol. $\int_0^1 e^{\ln \tan^{-1} x} \cdot \sin^{-1}(\cos x) dx$

$$= \int_0^1 \tan^{-1} x \cdot \sin^{-1}(\cos x) dx = \int_0^1 \tan^{-1} x \left(\frac{\pi}{2} - x \right) dx$$

$$= \frac{\pi}{2} \int_0^1 \tan^{-1} x dx - \int_0^1 x \tan^{-1} x dx$$

$$= \frac{\pi}{2} \int_0^1 \tan^{-1} x dx - \int_0^1 x \tan^{-1} x dx$$

Let $I = \int_0^1 \tan^{-1} x dx = \left[x \tan^{-1} x \right]_0^1 - \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} dx$

$$= \frac{\pi}{4} - \frac{1}{2} \left[\ln |1+x^2| \right]_0^1$$

$$= \frac{\pi}{4} - \frac{\ln 2}{2}$$

Now, $I' = \int_0^1 x \tan^{-1} x = \left[\frac{x^2}{2} \tan^{-1} x \right]_0^1 - \frac{1}{2} \int_0^1 \frac{x^2}{1+x^2} dx$

$$= \frac{\pi}{8} - \frac{1}{2} \left[I - \frac{\pi}{4} \right] = \frac{\pi}{4} - \frac{1}{2}$$

$$\Rightarrow \int_0^1 e^{\ln \tan^{-1} x} \cdot \sin^{-1}(\cos x) dx = \frac{\pi^2}{8} - \frac{\pi}{4} \ln 2 - \frac{\pi}{4} + \frac{1}{2} + C$$

41. Prove that :

(a) $\int_{\alpha}^{\beta} \sqrt{(x-\alpha)(\beta-x)} dx = \frac{(\beta-\alpha)^2 \pi}{8}$

(b) $\int_{\alpha}^{\beta} \sqrt{\frac{x-\alpha}{\beta-x}} dx = (\beta-\alpha) \frac{\pi}{2}$

(c) $\int_{\alpha}^{\beta} \frac{dx}{x \sqrt{(x-\alpha)(\beta-x)}} = \frac{\pi}{\sqrt{\alpha\beta}}$ where $\alpha, \beta > 0$

(d) $\int_{\alpha}^{\beta} \frac{x \cdot dx}{\sqrt{(x-\alpha)(\beta-x)}} = (\alpha+\beta) \frac{\pi}{2}$ where $\alpha < \beta$

सिद्ध कीजिए :

$$(a) \int_{\alpha}^{\beta} \sqrt{(x-\alpha)(\beta-x)} \, dx = \frac{(\beta-\alpha)^2 \pi}{8}$$

$$(b) \int_{\alpha}^{\beta} \sqrt{\frac{x-\alpha}{\beta-x}} \, dx = (\beta-\alpha) \frac{\pi}{2}$$

$$(c) \int_{\alpha}^{\beta} \frac{dx}{x\sqrt{(x-\alpha)(\beta-x)}} = \frac{\pi}{\sqrt{\alpha\beta}} \quad \text{जहाँ } \alpha, \beta > 0$$

$$(d) \int_{\alpha}^{\beta} \frac{x \cdot dx}{\sqrt{(x-\alpha)(\beta-x)}} = (\alpha+\beta) \frac{\pi}{2} \quad \text{जहाँ } \alpha < \beta$$

Ans.

Sol. (a) $\int_{\alpha}^{\beta} \sqrt{(x-\alpha)(\beta-x)} \, dx = \frac{(\beta-\alpha)^2 \pi}{8}$

Let $x = \alpha \cos^2 \theta + \beta \sin^2 \theta \Rightarrow (x - \alpha) = (\beta - \alpha) \sin^2 (\theta)$ and $(\beta - \alpha) \cos^2 (\theta) = \beta - x$

$$dx = (\beta - \alpha) \sin 2\theta d\theta$$

$$= \frac{1}{2} \int (\beta - \alpha) \sin 2\theta \cdot (\beta - \alpha) \sin 2\theta d\theta$$

$$= \frac{(\beta - \alpha)^2}{2} \int_0^{\pi/2} \sin^2 2\theta \cdot d\theta = \frac{(\beta - \alpha)^2}{2} \cdot \frac{\pi}{4} = \frac{(\beta - \alpha)^2 \pi}{8}$$

$$(b) \int_{\alpha}^{\beta} \sqrt{\frac{x-\alpha}{\beta-x}} \, dx = (\beta - \alpha) \frac{\pi}{2}$$

Let $x = \alpha \cos^2 \theta + \beta \sin^2 \theta$

$$= (\beta - \alpha) \int \sqrt{\tan^2 \theta} \cdot \sin^2 \theta d\theta$$

$$= 2(\beta - \alpha) \int \sin^2 \theta d\theta = 2(\beta - \alpha) \cdot \frac{\pi}{4} = \frac{\pi(\beta - \alpha)}{2}$$

$$(c) \int_{\alpha}^{\beta} \frac{dx}{x\sqrt{(x-\alpha)(\beta-x)}}$$

Let $x = \alpha \cos^2 \theta + \beta \sin^2 \theta$

$$I = \int_0^{\pi/2} \frac{(\beta - \alpha) \sin 2\theta}{(\alpha \cos^2 \theta + \beta \sin^2 \theta) \sqrt{(\beta - \alpha)^2 \sin^2 \theta \cos^2 \theta}} d\theta$$

$$I = 2 \int_0^{\pi/2} \frac{1}{(\alpha \cos^2 \theta + \beta \sin^2 \theta)} d\theta$$

$$I = 2 \int_0^{\pi/2} \frac{\sec^2 \theta}{\alpha + \beta \tan^2 \theta} d\theta$$

Let $\tan \theta = t$

$$\sec^2 \theta d\theta = dt$$

$$I = \frac{2}{\alpha} \int_0^{\infty} \frac{dt}{1 + \frac{\beta t^2}{\alpha}}$$

If $\alpha, \beta > 0$, then

$$I = \frac{2}{\alpha} \cdot \frac{\sqrt{\alpha}}{\sqrt{\beta}} = \left[\tan^{-1} \frac{t\sqrt{\alpha}}{\sqrt{\beta}} \right]_0^{\infty}$$

$$I = \frac{2}{\sqrt{\alpha\beta}} \cdot \frac{\pi}{2} = \frac{\pi}{\sqrt{\alpha\beta}}$$

$$(d) \int_{\alpha}^{\beta} \frac{x \cdot dx}{\sqrt{(x-\alpha)(\beta-x)}}$$

If $\beta > \alpha$ then

$$\text{Let } x = \alpha \cos^2 \theta + \beta \sin^2 \theta$$

$$= \int_0^{\pi/2} \frac{(\alpha \cos^2 \theta + \beta \sin^2 \theta)}{(\beta - \alpha) \sin \theta \cos \theta} \cdot (\beta - \alpha) \sin 2\theta d\theta$$

$$= 2 \int_0^{\pi/2} (\alpha \cos^2 \theta + \beta \sin^2 \theta) d\theta$$

$$= (\alpha + \beta) \frac{\pi}{2}$$

42. Evaluate: (a) $\int_0^{\pi/2} e^x \left\{ \cos(\sin x) \cos^2 \frac{x}{2} + \sin(\sin x) \sin^2 \frac{x}{2} \right\} dx$ (b) $\int_1^e \left\{ (1+x)e^x + (1-x)e^{-x} \right\} \cdot \ln x dx$

मान बताइये : (a) $\int_0^{\pi/2} e^x \left\{ \cos(\sin x) \cos^2 \frac{x}{2} + \sin(\sin x) \sin^2 \frac{x}{2} \right\} dx$ (b) $\int_1^e \left\{ (1+x)e^x + (1-x)e^{-x} \right\} \cdot \ln x dx$

Ans. (a) $\frac{1}{2} [e^{\pi/2} (\cos 1 + \sin 1) - 1]$; (b) $e^{1+e} + e^{1-e} + e^{-e} - e^e + e - e^{-1}$

Sol. (a) $\int_0^{\pi/2} e^x \left\{ \cos(\sin x) \cos^2 \frac{x}{2} + \sin(\sin x) \sin^2 \frac{x}{2} \right\} dx$

$$\because \cos^2 \frac{x}{2} = \frac{1 + \cos x}{2} \text{ \& } \sin^2 \frac{x}{2} = \frac{1 - \cos x}{2}$$

$$= \frac{1}{2} \int_0^{\pi/2} e^x \left(\underbrace{\cos(\sin x) + \sin(\sin x)}_{f(x)} + \underbrace{-\sin(\sin x) \cdot \cos x + \cos(\sin x) \cos x}_{f'(x)} \right) dx$$

$$= \frac{1}{2} [e^x (\cos(\sin x) + (\sin x))]_0^{\pi/2}$$

$$= \frac{1}{2} [e^{\pi/2} (\cos 1 + \sin 1) - 1]$$

(b) $I = \int_1^e \underbrace{e^x (1+x)}_{II} \underbrace{\ln x}_{I} dx + \int_1^e \underbrace{e^{-x} (1-x)}_{II} \underbrace{\ln x}_{I} dx$

$$= [\ln x e^x \cdot x]_1^e - \int_1^e e^x dx + [e^{-x} (x) \cdot \ln x]_1^e - \int_1^e e^{-x} dx$$

$$= e^{1+e} + e^{1-e} + e^{-e} - e^e + e - e^{-1}$$

43. If $P = \int_0^\infty \frac{x^2}{1+x^4} dx$; $Q = \int_0^\infty \frac{x dx}{1+x^4}$ and $R = \int_0^\infty \frac{dx}{1+x^4}$ then prove that

(a) $P = \frac{\pi}{4}$, (b) $P = R$, (c) $P - \sqrt{2} Q + R = \frac{\pi}{2\sqrt{2}}$

यदि $P = \int_0^\infty \frac{x^2}{1+x^4} dx$; $Q = \int_0^\infty \frac{x dx}{1+x^4}$ तथा $R = \int_0^\infty \frac{dx}{1+x^4}$ तो सिद्ध कीजिए।

(a) $P = \frac{\pi}{4}$, (b) $P = R$, (c) $P - \sqrt{2} Q + R = \frac{\pi}{2\sqrt{2}}$

Ans.

Sol. $P = \int_0^\infty \frac{x^2}{1+x^4} dx$

$$\text{Let } x^2 = \tan \theta \Rightarrow 2x \, dx = \sec^2 \theta \, d\theta$$

$$\Rightarrow P = \int_0^{\pi/2} \frac{\tan \theta}{1 + \tan^2 \theta} \cdot \frac{\sec^2 \theta \, d\theta}{2\sqrt{\tan \theta}}$$

$$P = \frac{1}{2} \int_0^{\pi/2} \sqrt{\tan \theta} \, d\theta$$

Apply king prop.

$$P = \frac{1}{2} \int_0^{\pi/2} \sqrt{\cot \theta} \, d\theta$$

$$\Rightarrow 2P = \frac{1}{2} \int_0^{\pi/2} (\sqrt{\tan \theta} + \sqrt{\cot \theta}) \, d\theta$$

$$\Rightarrow 2P = \frac{1}{4} \int_0^{\pi/2} \frac{(1 + \tan \theta)}{\sqrt{\tan \theta}} \, d\theta$$

$$\Rightarrow P = \frac{\pi}{4}$$

$$R = \int_0^{\infty} \frac{dx}{1+x^4}$$

$$\text{Let } x^2 = \tan \theta$$

$$2x \, dx = \sec^2 \theta \, d\theta$$

$$R = \int_0^{\pi/2} \frac{1}{\sec^2 \theta} \frac{\sec^2 \theta}{2\sqrt{\tan \theta}} \, d\theta$$

$$\Rightarrow \frac{1}{2} \int_0^{\pi/2} \sqrt{\cot \theta} \, d\theta$$

Apply king prop.

$$\Rightarrow R = \frac{1}{2} \int_0^{\pi/2} \sqrt{\tan \theta} \, d\theta$$

$$\Rightarrow R = \frac{\pi}{4}$$

$$\Rightarrow P = R$$

$$P - \sqrt{2}Q + R = \int_0^{\infty} \frac{(x^2 - \sqrt{2}x + 1)}{x^4 + 1} dx$$

$$= \int_0^{\infty} \frac{(x^2 - \sqrt{2}x + 1) dx}{(x^2 - \sqrt{2}x + 1)(x^2 + \sqrt{2}x + 1)} = \int_0^{\infty} \frac{dx}{x^2 + \sqrt{2}x + 1}$$

$$= \int_0^{\infty} \frac{dx}{\left(x + \frac{1}{\sqrt{2}}\right)^2 + \frac{1}{2}} = \sqrt{2} \left[\tan^{-1} \left(\frac{x + \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} \right) \right]_0^{\infty} = \frac{\pi}{2\sqrt{2}}$$

44. For $a \geq 2$, if the value of the definite integral $\int_0^{\infty} \frac{dx}{a^2 + (x - (1/x))^2} = \frac{\pi}{5050}$. Find the value of a .

$a \geq 2$ के लिये यदि $\int_0^{\infty} \frac{dx}{a^2 + (x - (1/x))^2}$ का मान $\frac{\pi}{5050}$ है तो a का मान बताओं।

Ans. 2525

Sol.
$$\int_0^{\infty} \frac{dx}{a^2 + (x - (1/x))^2} = \frac{\pi}{5050}$$

Let $x = \frac{1}{t}$

$$= \int_{\infty}^0 \frac{-1}{t^2} \cdot \frac{1}{\left(a^2 + \left(\frac{1}{t} - t\right)^2\right)} dt$$

$$I = \int_0^{\infty} \frac{1}{x^2 \left(a^2 + \left(x - \frac{x}{x}\right)^2\right)} dx$$

$$\Rightarrow 2I = \int_0^{\infty} \frac{\left(1 + \frac{1}{x^2}\right)}{a^2 + \left(x - \frac{1}{x}\right)^2} \text{ Let } x - \frac{1}{x} = \theta$$

$$2I = \int_0^{\infty} \frac{dx}{a^2 + \theta^2} = \frac{1}{a} \left[\tan^{-1} \left(x - \frac{1}{x} \right) \right]_0^{\infty}$$

$$\Rightarrow I = \frac{1}{2a} \left[\frac{\pi}{2} + \frac{\pi}{2} \right] = \frac{\pi}{2a}$$

$$\Rightarrow a = 2525$$

45. If $\int_0^{\pi} \sqrt{(\cos x + \cos 2x + \cos 3x)^2 + (\sin x + \sin 2x + \sin 3x)^2} dx$ has the value equal to $\left(\frac{\pi}{k} + \sqrt{w} \right)$ where k and w are positive integers find the value of $(k^2 + w^2)$.

यदि $\int_0^{\pi} \sqrt{(\cos x + \cos 2x + \cos 3x)^2 + (\sin x + \sin 2x + \sin 3x)^2} dx$ का मान $\left(\frac{\pi}{k} + \sqrt{w} \right)$ जहाँ k एवं w धनात्मक पूर्णांक है तो $(k^2 + w^2)$ का मान होगा।

Ans. 153

Sol. $\int_0^{\pi} \sqrt{(\cos x + \cos 2x + \cos 3x)^2 + (\sin x + \sin 2x + \sin 3x)^2} dx$

$$= \int_0^{\pi} \sqrt{\cos^2 2x (1 + 2 \cos x)^2 + \sin^2 2x (1 + 2 \cos x)^2} dx$$

$$= \int_0^{\pi} |1 + 2 \cos x| dx$$

$$= \int_0^{\pi/3} (1 + 2 \cos x) dx - \int_{2\pi/3}^{\pi} (1 + 2 \cos x) dx$$

$$= \frac{2\pi}{3} + 2 \cdot \frac{\sqrt{3}}{2} - \frac{\pi}{3} + 2 \cdot \frac{\sqrt{3}}{2}$$

$$= \frac{\pi}{3} + 2\sqrt{3}$$

$$\Rightarrow k = 3 \text{ \& } w = 12$$

$$\Rightarrow k^2 + w^2 = 144 + 9 = 153$$

46. $\int_0^{\ln 3} \frac{e^x + 1}{e^{2x} + 1} dx$

Ans. $\frac{1}{2} \left[\frac{\pi}{6} + \ln 3 - \ln 2 \right]$

Sol. $\int_0^{\ln 3} \frac{e^x + 1}{e^{2x} + 1} dx =$

$$= \underbrace{\int_0^{\ln 3/2} \frac{e^x}{e^{2x} + 1} dx}_{\text{let } e^x = t} + \int_0^{\ln 3/2} \frac{dx}{1 + e^{2x}}$$

$$= \int_1^{\sqrt{3}} \frac{dt}{1+t^2} + \underbrace{\int_0^{\ln 3/2} \frac{e^{-2x}}{e^{-2x} + 1} dx}_{\substack{e^{-2x} = t \\ -2e^{-2x} dx = dt}}$$

$$= \frac{\pi}{2} - \frac{\pi}{4} - \frac{1}{2} \left[\ln(1 + e^{-2x}) \right]_0^{\ln \sqrt{3}}$$

$$= \frac{1}{2} \left[\frac{\pi}{6} + \ln 3 - \ln 2 \right]$$

47. Let $I = \int_0^1 \frac{2+3x+4x^2}{2\sqrt{1+x+x^2}} dx$. Find the value of I^2 .

माना कि $I = \int_0^1 \frac{2+3x+4x^2}{2\sqrt{1+x+x^2}} dx$. तो I^2 का मान बताइये।

Ans. 3

Sol. $I = \int_0^1 \frac{2+3x+4x^2}{2\sqrt{1+x+x^2}} dx$

$$\Rightarrow I = \int_0^1 \left(\sqrt{1+x+x^2} + \frac{x(2x+1)}{x\sqrt{1+x+x^2}} \right) dx$$

$$\text{Let } f(x) = \sqrt{1+x+x^2}$$

$$f'(x) = \frac{2x+1}{2\sqrt{1+x+x^2}}$$

$$\Rightarrow \int_0^1 (f(x) + xf'(x)) dx$$

$$= [xf(x)]_0^1$$

$$= f(1) = \sqrt{3}$$

$$\Rightarrow I^2 = 3$$

48. Show that the sum of the two integrals $\int_{-4}^{-5} e^{(x+5)^2} dx + 3 \int_{1/3}^{2/3} e^{9(x-2/3)^2} dx$ is zero.

दर्शाइये कि दोनों समाकलन का योग $\int_{-4}^{-5} e^{(x+5)^2} dx + 3 \int_{1/3}^{2/3} e^{9(x-2/3)^2} dx$ शून्य है।

Ans.

Sol. $\int_{-4}^{-5} e^{(x+5)^2} dx + 3 \int_{1/3}^{2/3} e^{9(x-2/3)^2} dx$

$$\int_{-4}^{-5} e^{(x+5)^2} dx + 3 \underbrace{\int_{1/3}^{2/3} e^{(3x-2)^2} dx}_{\text{let } 2-3x=t}$$

$$= \int_{-4}^{-5} e^{(x+5)^2} dx - \int_{-1}^0 e^{t^2} dt$$

$$\text{Let } x+5 = \theta$$

$$I = \int_1^0 e^{t^2} d\theta - \int_{+1}^0 e^{t^2} dt = 0$$

49. $\int_0^{\pi/2} \tan^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right] dx$

Ans. $\frac{3\pi^2}{16}$

Sol.
$$\int_0^{\pi/2} \tan^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right] dx$$

$$= \int_0^{\pi/2} \tan^{-1} \left[\frac{\left| \sin \frac{x}{2} + \cos \frac{x}{2} \right| + \left| \sin \frac{x}{2} - \cos \frac{x}{2} \right|}{\left| \sin \frac{x}{2} + \cos \frac{x}{2} \right| - \left| \sin \frac{x}{2} - \cos \frac{x}{2} \right|} \right] dx$$

$$= \int_0^{\pi/2} \tan^{-1} \left(\cot \frac{x}{2} \right) dx$$

$$= \left[\frac{\pi}{2} x - \frac{x^2}{4} \right]_0^{\pi/2}$$

$$= \frac{\pi^2}{4} - \frac{\pi^2}{16} = \frac{3\pi^2}{16}$$

50.
$$\int_{\frac{\sqrt{3a^2+b^2}}{2}}^{\frac{\sqrt{a^2+b^2}}{2}} \frac{x \cdot dx}{\sqrt{(x^2 - a^2)(b^2 - x^2)}}$$

Ans. $\frac{\pi}{12}$

Sol.
$$\int_{\frac{\sqrt{3a^2+b^2}}{2}}^{\frac{\sqrt{a^2+b^2}}{2}} \frac{x \cdot dx}{\sqrt{(x^2 - a^2)(b^2 - x^2)}}$$

Let $x^2 - a^2 = t^2$

$\Rightarrow x dx = t dt$

$$I = \int \frac{t dt}{t \sqrt{(b^2 - a^2) - t^2}} = \left(\sin^{-1} \left(\frac{t}{\sqrt{b^2 - a^2}} \right) \right)$$

$$= \left[\sin^{-1} \left(\sqrt{\frac{x^2 - a^2}{b^2 - a^2}} \right) \right]_{\frac{\sqrt{3a^2 + b^2}}{2}}^{\frac{\sqrt{a^2 + b^2}}{2}}$$

$$= \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) - \sin^{-1} \left(\frac{1}{2} \right)$$

$$= \frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12}$$

51. (a) Show that $\int_0^{\infty} \frac{dx}{x^2 + 2x \cos \theta + 1} = 2 \int_0^1 \frac{dx}{x^2 + 2x \cos \theta + 1}$

(b) Evaluate: $f(\theta) = \int_0^{\infty} \frac{\tan^{-1} x}{x^2 + 2x \cos \theta + 1} dx, \theta \in (0, \pi)$

(a) सिद्ध करो कि $\int_0^{\infty} \frac{dx}{x^2 + 2x \cos \theta + 1} = 2 \int_0^1 \frac{dx}{x^2 + 2x \cos \theta + 1}$

(b) ज्ञात करो : $f(\theta) = \int_0^{\infty} \frac{\tan^{-1} x}{x^2 + 2x \cos \theta + 1} dx, \theta \in (0, \pi)$

Ans. (b) $\frac{\pi \theta}{4 \sin \theta}$

Sol. (a) $\int_0^{\infty} \frac{dx}{x^2 + 2x \cos \theta + 1} = \int_0^{\infty} \frac{dx}{(x + \cos \theta)^2 + 1 - \cos^2 \theta}$

Let $x + \cos \theta = t$

$$= \int \frac{dt}{t^2 + \sin^2 \theta} = \frac{1}{\sin \theta} \tan^{-1} \left[\frac{t}{\sin \theta} \right]$$

$$= \frac{1}{\sin \theta} \tan^{-1} \left[\frac{x + \cos \theta}{\sin \theta} \right]_0^{\infty}$$

$$= \frac{1}{\sin \theta} \left(\frac{\pi}{2} - \left(\frac{\pi}{2} - \theta \right) \right) = \frac{\theta}{\sin \theta}$$

Now, $2 \int_0^1 \frac{dx}{(x + \cos \theta)^2 + \sin^2 \theta}$ Let $x + \cos \theta = t$

$$= 2 \int_{\cos \theta}^{1+\cos \theta} \frac{dt}{t^2 + \sin^2 \theta} = \frac{2}{\sin \theta} \tan^{-1} \left[\frac{t}{\sin \theta} \right]_{\cos \theta}^{1+\cos \theta}$$

$$= \frac{2}{\sin \theta} \left[\tan^{-1} [\cos \sec \theta + \cot \theta] - \tan^{-1} (\cot \theta) \right]$$

$$= \frac{2}{\sin \theta} \left[\frac{\pi}{2} - \frac{\theta}{2} - \left(\frac{\pi}{2} - \theta \right) \right] = \frac{\theta}{\sin \theta}$$

$$\Rightarrow \int_0^{\infty} \frac{dx}{x^2 + 2x \cos \theta + 1} = 2 \int_0^1 \frac{dx}{x^2 + 2x \cos \theta + 1}$$

$$(b) f(\theta) = \int_0^{\infty} \frac{\tan^{-1} x}{x^2 + 2x \cos \theta + 1} dx, \quad \theta \in (0, \pi)$$

$$\text{Let } x = \frac{1}{t}$$

$$dx = \frac{-1}{t^2} dt$$

$$f(\theta) = \int_0^{\infty} \frac{\cot^{-1} t}{t^2 + 2t \cos \theta + 1} dt$$

$$\Rightarrow 2f(\theta) = \int_0^{\infty} \frac{(\tan^{-1} t + \cot^{-1} t)}{t^2 + 2t \cos \theta + 1} dt$$

$$= \frac{\pi}{2} \times \frac{\theta}{\sin \theta}$$

$$\Rightarrow f(\theta) = \frac{\pi \theta}{4 \sin \theta}$$

52. Let $I = \int_0^{\pi/2} \frac{\cos x + 4}{3 \sin x + 4 \cos x + 25} dx$ and $J = \int_0^{\pi/2} \frac{\sin x + 3}{3 \sin x + 4 \cos x + 25} dx$. If $25I = a\pi + b \ln \frac{c}{d}$ where a, b, c and

$d \in \mathbb{N}$ and $\frac{c}{d}$ is not a perfect square of a rational then find the value of $(a + b + c + d)$.

माना कि $I = \int_0^{\pi/2} \frac{\cos x + 4}{3 \sin x + 4 \cos x + 25} dx$ और $J = \int_0^{\pi/2} \frac{\sin x + 3}{3 \sin x + 4 \cos x + 25} dx$ यदि $25I = a\pi + b \ln \frac{c}{d}$ जहाँ a, b, c

तथा $d \in \mathbb{N}$ तथा $\frac{c}{d}$ किसी परिमेय संख्या का पूर्णवर्ग नहीं है तो $(a + b + c + d)$ का मान बताइये।

Ans. 62

Sol. $I = \int_0^{\pi/2} \frac{\cos x + 4}{3 \sin x + 4 \cos x + 25} dx$

$$J = \int_0^{\pi/2} \frac{\sin x + 3}{3 \sin x + 4 \cos x + 25} dx$$

$$\Rightarrow 3I - 4J = \int_0^{\pi/2} \frac{(3 \cos x - 4 \sin x)}{(3 \sin x + 4 \cos x + 25)} dx$$

$$\Rightarrow 3I - 4J = \left[\ln |3 \sin x + 4 \cos x + 25| \right]_0^{\pi/2}$$

$$3I - 4J = \ln \left(\frac{28}{29} \right) \quad \dots\dots\dots(i)$$

$$4I + 3J = \int_0^{\pi/2} \frac{4 \cos x + 3 \sin x + 25}{3 \sin x + 4 \cos x + 25} dx$$

$$4I + 3J = \frac{\pi}{2} \quad \dots\dots\dots(ii)$$

$\Rightarrow$ From (i) & (ii)

$$25I = 3 \ln \left(\frac{28}{29} \right) + 2\pi$$

$$\Rightarrow a = 2, b = 3, c = 28, d = 29$$

$$\Rightarrow a + b + c + d = 62$$

PROPERTY FOR DISCONTINUOUS/NON UNIFORM FUNCTION

53. Let $\beta(n) = \int_0^{n\pi} \sqrt{1 - \sin t} dt$. Find the value of $\beta(2) - \beta(1)$.

माना $\beta(n) = \int_0^{n\pi} \sqrt{1 - \sin t} \, dt$ हो तो $\beta(2) - \beta(1)$ का मान ज्ञात कीजिये।

Ans. 4

Sol. $\beta(n) = \int_0^{n\pi} \sqrt{1 - \sin t} \, dt$

$$\beta(n) = \int_0^{n\pi} \left| \sin \frac{t}{2} - \cos \frac{t}{2} \right| dt$$

$$\beta(2) - \beta(1) = \int_0^{2\pi} \left| \sin \frac{t}{2} - \cos \frac{t}{2} \right| dt - \int_0^{\pi} \left| \sin \frac{t}{2} - \cos \frac{t}{2} \right| dt$$

$$= \int_{\pi}^{2\pi} \left| \sin \frac{t}{2} - \cos \frac{t}{2} \right| dt = 2 \int_{\pi/2}^{\pi} |\sin \theta - \cos \theta| d\theta \quad \text{If } \frac{t}{2} = \theta$$

$$= 2 \int_{\pi/2}^{\pi} (\sin \theta - \cos \theta) d\theta$$

$$= 2[-\cos \theta - \sin \theta]_{\pi/2}^{\pi} = 2[1 + 1] = 4$$

54. $\int_0^{\pi/2} \sqrt{\frac{1 - \sin 2x}{1 + \sin 2x}} dx$

Ans. $\ln 2$

Sol. $\int_0^{\pi/2} \sqrt{\frac{1 - \sin 2x}{1 + \sin 2x}} dx$

$$= \int_0^{\pi/2} \frac{|\sin x - \cos x|}{|\sin x + \cos x|} dx$$

$$= - \int_0^{\pi/4} \frac{(\sin x - \cos x)}{\sin x + \cos x} dx + \int_{\pi/4}^{\pi/2} \frac{(\sin x - \cos x)}{\sin x + \cos x} dx$$

$$= [\ln |\sin x + \cos x|]_0^{\pi/4} - [\ln (\sin x + \cos x)]_{\pi/4}^{\pi/2}$$

$$= \ln \sqrt{2} + \ln \sqrt{2}$$

$$= \ln 2$$

55. Find the value of the definite integral $\int_0^{\pi} |\sqrt{2} \sin x + 2 \cos x| dx$.

निश्चित समाकलन $\int_0^{\pi} |\sqrt{2} \sin x + 2 \cos x| dx$ का मान ज्ञात कीजिये।

Ans. $2\sqrt{6}$

Sol. $I = \int_0^{\pi} |\sqrt{2} \sin x + 2 \cos x| dx$

$$I = \sqrt{2} \int_0^{\pi} |\sin x + \sqrt{2} \cos x| dx$$

$$\text{let } \cos \theta = \frac{1}{\sqrt{2}}$$

$$= \sqrt{6} \int_0^{\pi} |\sin(x + \theta)| dx$$

$$\text{let } x + \theta = t$$

$$= \sqrt{6} \int_{\theta}^{\pi+\theta} |\sin t| dt$$

$$= \sqrt{6} \int_{\theta}^{\pi+\theta} |\sin t| dt$$

$$= \sqrt{6} \int_{\theta}^{\pi+\theta} |\sin t| dt$$

$$= \sqrt{6} \left[\int_{\theta}^{\pi} \sin t dt + \int_{\pi}^{\pi+\theta} -\sin t dt \right]$$

$$= \sqrt{6} \left[(-\cos t)_{\theta}^{\pi} + (\cos t)_{\pi}^{\pi+\theta} \right]$$

$$= \sqrt{6} [1 + \cos \theta - \cos \theta + 1]$$

$$= 2\sqrt{6}$$

PROPERTY FOR ODD AND EVEN FUNCTIONS

56.
$$\int_{-\sqrt{2}}^{\sqrt{2}} \frac{2x^7 + 3x^6 - 10x^5 - 7x^3 - 12x^2 + x + 1}{x^2 + 2} dx$$

Ans.
$$\frac{\pi}{2\sqrt{2}} - \frac{16\sqrt{2}}{5}$$

Sol.
$$I = \int_{-\sqrt{2}}^{\sqrt{2}} \frac{2x^7 + 3x^6 - 10x^5 - 7x^3 - 12x^2 + x + 1}{x^2 + 2} dx$$

Apply king

$$I = \int_{-\sqrt{2}}^{\sqrt{2}} \frac{-2x^7 + 3x^6 + 10x^5 + 7x^3 - 12x^2 - x + 1}{x^2 + 2} dx$$

$$\Rightarrow 2I = \int_{-\sqrt{2}}^{\sqrt{2}} \frac{6x^6 - 24x^2 + 2}{x^2 + 2} dx$$

$$\Rightarrow I = \int_{-\sqrt{2}}^{\sqrt{2}} \frac{3x^6 - 12x^2 + 1}{x^2 + 2} dx$$

$$\Rightarrow I = 2 \int_0^{\sqrt{2}} \frac{3x^6 - 12x^2 + 1}{x^2 + 2} dx$$

$$= 2 \left[\int_0^{\sqrt{2}} \frac{3(x^4 - 4)x^2}{x^2 + 2} dx + \int_0^{\sqrt{2}} \frac{dx}{x^2 + 2} \right]$$

$$= 2 \left[3 \int_0^{\sqrt{2}} x^2 (x^2 - 2) dx + \frac{1}{\sqrt{2}} \left[\tan^{-1} \left(\frac{x}{\sqrt{2}} \right) \right]_0^{\sqrt{2}} \right]$$

$$= 2 \left[3 \left[\frac{x^5}{5} - \frac{2x^3}{3} \right]_0^{\sqrt{2}} + \frac{1}{\sqrt{2}} \frac{\pi}{4} \right]$$

$$= \frac{\pi}{2\sqrt{2}} + \left[\frac{2\sqrt{2}}{5} - 4\sqrt{2} \right] 2 = \frac{\pi}{2\sqrt{2}} - \frac{16\sqrt{2}}{5}$$

57. $\int_{-2}^2 \frac{x^2 - x}{\sqrt{x^2 + 4}} dx$

Ans. $4\sqrt{2} - 4\ln(\sqrt{2} + 1)$

Sol. $\int_{-2}^2 \frac{x^2 - x}{\sqrt{x^2 + 4}} dx = \underbrace{\int_{-2}^2 \frac{x^2}{\sqrt{x^2 + 4}} dx}_{\text{even}} - \underbrace{\int_{-2}^2 \frac{x}{\sqrt{x^2 + 4}} dx}_{\text{odd}}$

$$= 2 \int_0^2 \frac{x^2}{\sqrt{x^2 + 4}} dx$$

$$= 2 \int_0^2 \left(\sqrt{x^2 + 4} - \frac{4}{\sqrt{x^2 + 4}} \right) dx$$

$$= 2 \left[\frac{x}{2} \sqrt{x^2 + 4} + 2 \ln |x + \sqrt{x^2 + 4}| - 4 \ln |x + \sqrt{x^2 + 4}| \right]_0^2$$

$$= 4\sqrt{2} - 4\ln(\sqrt{2} + 1)$$

58. Evaluate : $\int_{-1}^1 \frac{(2x^{332} + x^{998} + 4x^{1668} \cdot \sin x^{691})}{1 + x^{666}} dx$

ज्ञात बताइये : $\int_{-1}^1 \frac{(2x^{332} + x^{998} + 4x^{1668} \cdot \sin x^{691})}{1 + x^{666}} dx$

Ans. $\frac{\pi + 4}{666}$

Sol. $\int_{-1}^1 \frac{(2x^{332} + x^{998} + 4x^{1668} \cdot \sin x^{691})}{1 + x^{666}} dx$

$$= \underbrace{\int_{-1}^1 \left(\frac{2x^{332} + x^{998}}{1 + x^{666}} \right) dx}_{\text{even}} + \underbrace{\int_{-1}^1 \frac{4x^{1668} \cdot \sin x^{691}}{1 + x^{666}} dx}_{\text{odd}}$$

$$= 2 \int_0^1 \frac{2x^{332} + x^{998}}{1 + x^{666}} dx + 0$$

$$\begin{aligned}
&= 2 \int_0^1 \frac{x^{332} + x^{332} (1 + x^{666})}{1 + x^{666}} dx \\
&= 2 \left[\int_0^1 \frac{x^{332}}{1 + (x^{333})^2} dx + \int_0^1 x^{332} dx \right] \\
&= 2 \left[\frac{\tan^{-1}(x^{33})}{333} \right]_0^1 + \frac{2}{333} \\
&= \frac{\pi}{666} + \frac{2}{333} = \frac{\pi + 4}{666}
\end{aligned}$$

59. Evaluate $\int_{-\pi/3}^{\pi/3} \frac{\pi + 4x^3}{2 - \cos\left(x + \frac{\pi}{3}\right)} dx$.

ज्ञात बताइये $\int_{-\pi/3}^{\pi/3} \frac{\pi + 4x^3}{2 - \cos\left(x + \frac{\pi}{3}\right)} dx$.

Ans. $\frac{4\pi}{\sqrt{3}} \tan^{-1}\left(\frac{1}{2}\right)$

Sol. $\int_{-\pi/3}^{\pi/3} \frac{\pi + 4x^3}{2 - \cos\left(x + \frac{\pi}{3}\right)} dx = 2 \int_0^{\frac{\pi}{3}} \frac{\pi}{2 - \cos\left(x + \frac{\pi}{3}\right)} dx$

Let $x + \frac{\pi}{3} = t$

$$= 2\pi \int_{\pi/3}^{2\pi/3} \frac{\sec^2 \frac{t}{2}}{\sec^2 \frac{t}{2} + 2 \tan^2 \frac{t}{2}} dt$$

$$= \frac{4\pi}{\sqrt{3}} \tan^{-1}\left(\frac{1}{2}\right)$$

KING & QUEEN PROPERTY

60. Let $u = \int_0^{\pi/4} \left(\frac{\cos x}{\sin x + \cos x} \right)^2 dx$ and $v = \int_0^{\pi/4} \left(\frac{\sin x + \cos x}{\cos x} \right)^2 dx$. Find the value of $\frac{v}{u}$.

माना कि $u = \int_0^{\pi/4} \left(\frac{\cos x}{\sin x + \cos x} \right)^2 dx$ तथा $v = \int_0^{\pi/4} \left(\frac{\sin x + \cos x}{\cos x} \right)^2 dx$. तो $\frac{v}{u}$ का मान बताओं।

Ans. 4

Sol. $u = \int_0^{\pi/4} \left(\frac{\cos x}{\sin x + \cos x} \right)^2 dx$

$$v = \int_0^{\pi/4} \left(\frac{\sin x + \cos x}{\cos x} \right)^2 dx$$

$$\Rightarrow u = \int_0^{\pi/4} \left[\frac{\cos\left(\frac{\pi}{4} - x\right)}{\sin\left(\frac{\pi}{4} - x\right) + \cos\left(\frac{\pi}{4} - x\right)} \right]^2 dx$$

$$= \int_0^{\pi/4} \left(\frac{\cos x + \sin x}{\sqrt{2}(\sqrt{2} \cos x)} \right)^2 dx$$

$$= \frac{1}{4} \int_0^{\pi/4} \left(\frac{\cos x + \sin x}{\cos x} \right)^2 dx$$

$$\Rightarrow 4u = v$$

$$\Rightarrow \frac{v}{u} = 4$$

61. $\int_0^{\pi/4} \frac{x dx}{\cos x (\cos x + \sin x)}$

Ans. $\frac{\pi}{8} \ln 2$

Sol. $\int_0^{\pi/4} \frac{x dx}{\cos x (\cos x + \sin x)} dx \dots\dots\dots(i)$

Apply king

$$I = \int_0^{\pi/4} \frac{\left(\frac{\pi}{4} - x\right)}{\cos\left(\frac{\pi}{4} - x\right) \left(\cos\left(\frac{\pi}{4} - x\right) + \sin\left(\frac{\pi}{4} - x\right)\right)} dx$$

$$\Rightarrow I = \int_0^{\pi/4} \frac{\sqrt{2} \left(\frac{\pi}{4} - x\right)}{(\cos x + \sin x) \sqrt{2} \cos x} dx \quad \dots\dots\dots(ii)$$

By (i) & (ii)

$$2I = \frac{\pi}{4} \int_0^{\pi/4} \frac{1}{\cos x (\cos x + \sin x)} dx$$

$$= \frac{\pi}{4} \int_0^{\pi/4} \frac{\sec^2 x dx}{(1 + \tan x)} x$$

$$= \frac{\pi}{4} [\ln |1 + \tan x|]_0^{\pi/4}$$

$$2I = \frac{\pi}{4} \ln 2$$

$$\Rightarrow I = \frac{\pi}{8} \ln 2$$

62. $\int_0^1 \frac{\sin^{-1} \sqrt{x}}{x^2 - x + 1} dx$

Ans. $\frac{\pi^2}{6\sqrt{3}}$

Sol. $I = \int_0^1 \frac{\sin^{-1} \sqrt{x}}{x^2 - x + 1} dx \quad \dots\dots\dots(i)$

Apply king property

$$= \int_0^1 \frac{\sin^{-1} \sqrt{1-x}}{(1-x)^2 - (1-x) + 1} dx = \int_0^1 \frac{\sin^{-1} \sqrt{1-x}}{x^2 - x + 1} dx \quad \dots\dots\dots(ii)$$

Add. (i) and (ii)

$$2I = \int_0^1 \frac{\sin^{-1} \sqrt{x} + \sin^{-1} \sqrt{1-x}}{x^2 - x + 1} dx$$

$$\Rightarrow 2I = \frac{\pi}{2} \int_0^1 \frac{dx}{(x-1/2)^2 + (\sqrt{3}/2)^2}$$

$$\Rightarrow I = \frac{\pi}{4} \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{2x-1}{\sqrt{3}} \right]_0^1 = \frac{\pi^2}{6\sqrt{3}}$$

63. $\int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin\left(\frac{\pi}{4} + x\right)} dx$

Ans. $\frac{\pi(a+b)}{2\sqrt{2}}$

Sol. $I = \int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin\left(\frac{\pi}{4} + x\right)} dx$

$$I = \sqrt{2} \int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin x + \cos x} dx \quad \dots\dots\dots(1)$$

Apply king property

$$I = \sqrt{2} \int_0^{\pi/2} \frac{a \cos x + b \sin x}{\cos x + \sin x} dx \quad \dots\dots\dots(2)$$

add.(1) and (2)

$$2I = \sqrt{2}(a+b) \int_0^{\pi/2} dx \Rightarrow I = \frac{(a+b)\pi}{2\sqrt{2}}$$

64. $\int_0^{\pi} \frac{(ax+b) \sec x \tan x}{4 + \tan^2 x} dx \quad (a, b > 0)$

Ans. $\frac{(a\pi + 2b)\pi}{3\sqrt{3}}$

Sol. $\int_0^{\pi} \frac{(ax+b) \sec x \tan x}{4 + \tan^2 x} dx \quad (a, b > 0)$

Apply king property

$$I = \int_0^{\pi} \frac{(a\pi + b - ax) \sec x \tan x}{4 + \tan^2 x} dx$$

$$\Rightarrow 2I = \int_0^{\pi} \frac{(a\pi + 2b) \sec x \tan x}{4 + \tan^2 x} dx$$

$$\Rightarrow 2I = (a\pi + 2b) \int_0^{\pi} \frac{\sec x \tan x}{3 + \sec^2 x} dx$$

Let $\sec x = t$

$$\sec x \tan x dx = dt$$

$$\Rightarrow 2I = (a\pi + 2b) \int_1^{-1} \frac{dt}{3 + t^2}$$

$$\Rightarrow 2I = -(a\pi + 2b) \left[\frac{1}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \right]_{-1}^1$$

$$\Rightarrow 2I = \frac{-(a\pi + 2b)}{\sqrt{3}} \left[\frac{2\pi}{6} - \pi \right] \Rightarrow I = \frac{(a\pi + 2b)\pi}{3\sqrt{3}}$$

65. $\int_0^{\pi} \frac{(2x + 3) \sin x}{(1 + \cos^2 x)} dx$

Ans. $\frac{\pi(\pi + 3)}{2}$

Sol. $\int_0^{\pi} \frac{(2x + 3) \sin x}{(1 + \cos^2 x)} dx$

Apply king property

$$I = \int_0^{\pi} \frac{(2\pi - 2x + 3) \sin x}{1 + \cos^2 x} dx$$

$$\Rightarrow 2I = \int_0^{\pi} \frac{(2\pi + 6) \sin x}{1 + \cos^2 x} dx$$

$$\Rightarrow I = (\pi + 3) \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

Let $\cos x = t$

$$\sin x dx = dt$$

$$\Rightarrow I = -(\pi + 3) \int_1^{-1} \frac{dt}{1 + t^2}$$

$$\Rightarrow I = (\pi + 3) \int_{-1}^1 \frac{dt}{1 + t^2}$$

$$\Rightarrow I = (\pi + 3) [\tan^{-1} t]_{-1}^1$$

$$I = \frac{\pi(\pi + 3)}{2}$$

66. Let $I_n = \int_{-n}^n \left(\{x+1\} \cdot \{x^2+2\} + \{x^2+2\} \{x^3+4\} \right) dx$ where $\{ \}$ denotes the fractional part of x . Find I_1 .

माना $I_n = \int_{-n}^n \left(\{x+1\} \cdot \{x^2+2\} + \{x^2+2\} \{x^3+4\} \right) dx$ जहाँ $\{ \}$ भिन्नात्मक फलन है तो I_1 का मान बताइये।

Ans. $2/3$

Sol. $I_n = \int_{-n}^n \left(\{x+1\} \cdot \{x^2+2\} + \{x^2+2\} \{x^3+4\} \right) dx$

$$\Rightarrow I_n = \int_{-n}^n \left(\{x\} \{x^2\} + \{x^2\} \{x^3\} \right) dx$$

$$\Rightarrow I_1 = \int_{-1}^1 \left(\{x\} + \{x^3\} \right) \{x^2\} dx$$

Apply king prop.

$$\Rightarrow I_1 = \int_{-1}^1 \left(\{-x\} + \{-x^3\} \right) \{x^2\} dx$$

$$\Rightarrow 2I_1 = \int_{-1}^1 \left(\{x\} + \{-x\} + \{x^3\} + \{-x^3\} \right) \{x^2\} dx$$

$$\Rightarrow 2I_1 = \int_{-1}^1 x^2 (1+1) dx = 2 \left[\frac{x^3}{3} \right]_{-1}^1$$

$$= \frac{4}{3}$$

$$\Rightarrow I_1 = \frac{2}{3}$$

67. $\int_0^{\pi} \frac{x^2 \sin 2x \sin\left(\frac{\pi}{2} \cos x\right)}{2x - \pi} dx.$

Ans. $\frac{8}{\pi}$

Sol. $I = \int_0^{\pi} \frac{x \sin 2x \sin\left(\frac{\pi}{2} \cos x\right)}{2x - \pi} dx$ _____ (1)

Apply king property

$$I = \int_0^{\pi} \frac{(\pi - x) \sin 2(\pi - x) \sin\left(\frac{\pi}{2} \cos(\pi - x)\right)}{2(\pi - x) - \pi} dx$$

$$= \int_0^{\pi} \frac{(\pi - x) \sin 2x \sin\left(\frac{\pi}{2} \cos x\right)}{\pi - 2x} dx$$

$$= \int_0^{\pi} \frac{(x - \pi) \sin 2x \sin\left(\frac{\pi}{2} \cos x\right)}{2x - \pi} dx$$
 _____ (2)

add equation (1) & (2)

$$2I = \int_0^{\pi} \sin 2x \sin\left(\frac{\pi}{2} \cos x\right) dx$$

$$\therefore I = \int_0^{\pi} \sin x \cos x \sin\left(\frac{\pi}{2} \cos x\right) dx$$

$$\text{Put } \frac{\pi}{2} \cos x = t \Rightarrow \sin x dx = -\frac{2}{\pi} dt$$

$$\therefore I = -\frac{2}{\pi} \int_{\pi/2}^{-\pi/2} \frac{2t}{\pi} \sin t dt = \frac{4}{\pi^2} \int_{-\pi/2}^{\pi/2} t \sin t dt$$

$$\Rightarrow I = \frac{4}{\pi^2} \int_{-\pi/2}^{\pi/2} t \sin t dt = \frac{4}{\pi^2} [-t \cos t + \sin t]_{-\pi/2}^{\pi/2}$$

$$= \frac{4}{\pi^2} \times 2 = \frac{8}{\pi^2}$$

68. $\int_0^{2\pi} \frac{dx}{2 + \sin 2x}$.

Ans. $\frac{2\pi}{\sqrt{3}}$

Sol. $\int_0^{2\pi} \frac{dx}{2 + \sin 2x} = \int_0^{2\pi} \frac{dx}{2 + \frac{2 \tan x}{1 + \tan^2 x}}$

$$= \frac{1}{2} \int_0^{2\pi} \frac{1 + \tan^2 x}{1 + \tan^2 x + \tan x} dx$$

Apply queen property

$$= \frac{4}{2} \int_0^{\pi/2} \frac{\sec^2 x dx}{\left(\tan x + \frac{1}{2}\right) + \frac{3}{4}}$$

Let $\tan x = t$

$$\Rightarrow \sec^2 x dx = dt$$

$$= 2 \int_0^{\infty} \frac{dt}{\left(t + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = \frac{4}{\sqrt{3}} \left[\tan^{-1} \left(\frac{2t+1}{\sqrt{3}} \right) \right]_0^{\infty}$$

$$= \frac{4}{\sqrt{3}} \left[\frac{\pi}{2} \right]$$

69. If f is an even function then prove that $\int_0^{\pi/2} f(\cos 2x) \cos x dx = \sqrt{2} \int_0^{\pi/4} f(\sin 2x) \cos x dx$.

यदि f एक सम फलन है तो सिद्ध कीजिए $\int_0^{\pi/2} f(\cos 2x) \cos x \, dx = \sqrt{2} \int_0^{\pi/4} f(\sin 2x) \cos x \, dx$

Ans.

Sol. $I = \int_0^{\pi/2} f(\cos 2x) \cos x \, dx$ _____(1)

Given f is an even function.

$$I = \int_0^{\pi/2} f(\cos(\pi - 2x)) \cos\left(\frac{\pi}{2} - x\right) dx$$

$$\Rightarrow I = \int_0^{\pi/2} f(\cos 2x) \sin x \, dx$$

By (1) + (2)

$$\Rightarrow 2I = \int_0^{\pi/2} f(\cos 2x)(\sin x + \cos x) \, dx$$

$$2I = \sqrt{2} \int_0^{\pi/2} \cos\left(x - \frac{\pi}{4}\right) f(\cos 2x) \, dx$$

$$\text{let } x - \frac{\pi}{4} = t$$

$$I = \frac{1}{\sqrt{2}} \int_{-\pi/4}^{\pi/4} \cos t f(\sin 2t) \, dt$$

$$I = \sqrt{2} \int_0^{\pi/4} \cos t f(\sin 2t) \, dt$$

$$I = \sqrt{2} \int_0^{\pi/4} \cos x f(\sin 2x) \, dx$$

PROPERTY FOR PERIODIC FUCTIONS

70. Show that $\int_0^{p+q\pi} |\cos x| dx = 2q + \sin p$ where $q \in \mathbb{N}$ & $-\frac{\pi}{2} < p < \frac{\pi}{2}$.

दर्शाइये कि $\int_0^{p+q\pi} |\cos x| dx = 2q + \sin p$ जहाँ $q \in \mathbb{N}$ तथा $-\frac{\pi}{2} < p < \frac{\pi}{2}$

Ans.

Sol.
$$\int_0^{p+q\pi} |\cos x| dx = \int_0^p |\cos x| dx + \int_p^{p+q\pi} |\cos x| dx$$

$$= \int_0^p |\cos x| dx + \int_0^{q\pi} |\cos x| dx$$

$$= \int_0^p \cos x dx + q \int_0^\pi |\cos x| dx$$

$\{\because q \in \mathbb{N}\}$ $\{ \& (|\cos x|) \text{ is periodic function with period } \pi \}$

$$= \sin p + q \left(\int_0^{\pi/2} \cos x dx - \int_{\pi/2}^\pi \cos x dx \right)$$

$$= \sin p + 2q$$

71. For any real number x , let $[x]$ denote the largest integer less than or equal to x . Let f be a real valued function defined on the interval $[-10, 10]$ by $f(x) = \begin{cases} x - [x] & \text{if } [x] \text{ is odd} \\ 1 + [x] - x & \text{if } [x] \text{ is even} \end{cases}$. Then the value of $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x dx$

is.

[JEE-2010]

यदि x एक वास्तविक संख्या है, माना $[x]$, x के महत्तम पूर्णांक मान जो x से छोटा अथवा बराबर है को दर्शाता है। यदि f एक वास्तविक

मान का फलन अन्तराल $[-10, 10]$ में $f(x) = \begin{cases} x - [x] & \text{यदि } [x] \text{ विषम है} \\ 1 + [x] - x & \text{यदि } [x] \text{ सम है} \end{cases}$ द्वारा परिभाषित है। तो $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x dx$ का

मान है।

[JEE-2010]

Ans. 4

Sol.
$$f(x) = \begin{cases} \{x\} & \text{when } -9 \leq x < -8; -7 \leq x < -6, \dots\dots\dots \\ 1 - \{x\} & \text{when } -10 \leq x \leq -9; -8 \leq x < -7, \dots\dots \end{cases}$$

Since $f(x)$ & $\cos \pi x$ both are periodic functions having period 2.

$$\begin{aligned}
I &= \frac{10 \times \pi^2}{10} \left(\int_0^1 (1 - \{x\}) \cos \pi x dx + \int_1^2 \{x\} \cos \pi x dx \right) \\
&= \pi^2 \left(\int_0^1 (1 - x) \cos \pi x dx + \int_1^2 (x - 1) \cos \pi x dx \right) \\
&= \pi^2 \left(\int_0^1 \cos \pi x dx - \int_1^2 \cos \pi x dx + \int_1^2 x \cos \pi x dx - \int_0^1 x \cos \pi x dx \right) \\
\Rightarrow I &= 4
\end{aligned}$$

DIFFERENTIATION OF INTEGRATION BY LEIBNITZ

72. $\lim_{n \rightarrow \infty} n^2 \int_{-1/n}^{1/n} (2010 \sin x + 2012 \cos x) |x| dx.$

Ans. 2012

Sol. $\lim_{n \rightarrow \infty} n^2 \int_{-1/n}^{1/n} (2010 \sin x + 2012 \cos x) |x| dx$

Let $g(x) = \int_{-\frac{1}{n}}^{1/n} \underbrace{(2010 \sin x + 2012 \cos x) |x| dx}_{\text{let } f(x)}$

$$= \int_0^{1/n} (f(x) + f(-x)) dx$$

$$= \int_0^{1/n} 1024 \cos x |x| dx$$

$$= 1024 \int_0^{1/n} \cos x \cdot |x| dx$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1024 \int_0^{1/n} \cos x \cdot x dx}{\frac{1}{\pi^2}} \left(\frac{0}{0} \text{ form} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1024 \cdot \cos\left(\frac{1}{n}\right) \cdot \frac{1}{n} \cdot -\frac{1}{n^2}}{\frac{-2}{n^3}} = 2012$$

73. If $y(x) = \int_{\pi^2/16}^{x^2} \frac{\cos x \cdot \cos \sqrt{\theta}}{1 + \sin^2 \sqrt{\theta}} \cdot d\theta$ then find $\frac{dy}{dx}$ at $x = \pi$.

यदि $y(x) = \int_{\pi^2/16}^{x^2} \frac{\cos x \cdot \cos \sqrt{\theta}}{1 + \sin^2 \sqrt{\theta}} \cdot d\theta$ तो $\frac{dy}{dx}$ का $x = \pi$ पर मान बताइये।

Ans. 2π

Sol. $y(x) = \int_{\pi^2/16}^{x^2} \frac{\cos x \cos \sqrt{\theta}}{1 + \sin^2 \sqrt{\theta}} d\theta$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos^2 x}{1 + \sin^2 x} \cdot 2x$$

$$\Rightarrow \left. \frac{dy}{dx} \right|_{\text{at } x=\pi} = \frac{2\pi}{1} = 2\pi$$

DETERMINATION OF FUNCTION

74. A continuous real function f satisfies $f(2x) = 3f(x) \forall x \in \mathbb{R}$. If $\int_0^1 f(x) dx = 1$, then compute the value of definite

integral $\int_1^2 f(x) dx$.

एक सतत वास्तविक फलन f सम्बन्ध $f(2x) = 3f(x) \forall x \in \mathbb{R}$ को संतुष्ट कर रहा है। यदि $\int_0^1 f(x) dx = 1$, तो $\int_1^2 f(x) dx$ का मान बताइये।

Ans. 5

Sol. $f(2x) = 3f(x) \forall x \in \mathbb{R}$

$$\Rightarrow \int_0^1 \frac{1}{3} f(2x) dx = \int_0^1 f(x) dx$$

$$= \text{let } 2x = t$$

$$\Rightarrow dx = \frac{dt}{2}$$

$$\Rightarrow \frac{1}{6} \int_0^2 f(t) dt = 1$$

$$\Rightarrow \int_0^1 f(t) dt + \int_1^2 f(t) dt = 6$$

$$\Rightarrow \int_1^2 f(x) dx = 5$$

75. Let $f(x)$ be a function defined on R such that $f'(x) = f'(3-x) \forall x \in [0, 3]$ with $f(0) = -32$ and $f(3) = 46$. Then find the value of $\int_0^3 f(x) dx$.

माना $f(x)$, R पर परिभाषित फलन निम्न प्रकार से है $f'(x) = f'(3-x) \forall x \in [0, 3]$ तथा $f(0) = -32$, $f(3) = 46$ तो $\int_0^3 f(x) dx$ का मान ज्ञात कीजिये।

Ans. 21

Sol. $f'(x)f'(3-x) \forall x \in [0, 3]$, $f(0) = -32$
 $f(3) = 46$

$$\text{let } I = \int_0^3 f(x) dx$$

$$\Rightarrow I = [xf(x)]_0^3 - \int_0^3 xf'(x) dx$$

$$\Rightarrow I = 3f(3) - \int_0^3 xf'(x) dx$$

$$\text{let } I' = \int_0^3 xf'(x) dx$$

Apply king prop.

$$I' = \int_0^3 (3-x)f'(3-x) dx = \int_0^3 f'(x).(3-x) dx$$

$$\Rightarrow I' = 3I[f(x)]_0^3 - I'$$

$$\Rightarrow 2I' = 3[f(3) - f(0)] = 3.[46 + 32]$$

$$= 3.[78] = 234$$

$$\Rightarrow I' = 117$$

$$\Rightarrow I = 138 - 117 = 21$$

76. Let $f: R \rightarrow R$ be a continuous function which satisfies $f(x) = \int_0^x f(t) dt$. Then the value of $f(\ln 5)$ is.

[JEE-2009]

यदि $f: R \rightarrow R$ एक सतत् फलन है जो कि $f(x) = \int_0^x f(t) dt$ को संतुष्ट करता है तो $f(\ln 5)$ का मान बताइये। [JEE 2009]

Ans. 0

Sol. $f(x) = \int_0^x f(t)dt \Rightarrow f(0) = 0$

$$\Rightarrow f'(x) = f(x)$$

$$\Rightarrow \int \frac{f'(x)}{f(x)} dx = \int dx$$

$$\Rightarrow \ln |f(x)| = (x) + \ln c$$

$$\Rightarrow f(x) = ce^x$$

$$\Rightarrow \because f(0) = 0$$

$$\Rightarrow c = 0$$

$$\Rightarrow f(x) = 0$$

$$\Rightarrow f(\ln 5) = 0$$

DEFINITE INTEGRAL AS A LIMIT OF A SUM

77. Evaluate: $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=0}^{n-1} \left[k \int_k^{k+1} \sqrt{(x-k)(k+1-x)} dx \right]$

ज्ञात कीजिए : $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=0}^{n-1} \left[k \int_k^{k+1} \sqrt{(x-k)(k+1-x)} dx \right]$

Ans. $\frac{\pi}{16}$

Sol. $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=0}^{n-1} \left[k \int_k^{k+1} \sqrt{(x-k)(k+1-x)} dx \right]$

for $\int_k^{k+1} \sqrt{(x-k)(k+1-x)} dx$

let $x = k \cos^2 \theta + (k+1) \sin^2 \theta$

$$\Rightarrow dx = \sin^2 \theta d\theta$$

$$\Rightarrow x - k = \sin^2 \theta$$

$$(k+1-x) = \cos^2 \theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \sin^2 2\theta d\theta = \frac{\pi}{8}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n^2} \left[\sum_{k=0}^{n-1} \frac{\pi k}{8} \right] = \frac{\pi}{8} \left(\lim_{n \rightarrow \infty} \frac{n(n-1)}{2n^2} \right) = \frac{\pi}{16}$$

78. Evaluate: (a) $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \left(1 + \frac{3^2}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right) \right]^{1/n}$; (b) $\lim_{n \rightarrow \infty} \frac{1}{n} \left[\frac{1}{n+1} + \frac{2}{n+2} + \dots + \frac{3n}{4n} \right]$

ज्ञात कीजिए : (a) $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \left(1 + \frac{3^2}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right) \right]^{1/n}$; (b) $\lim_{n \rightarrow \infty} \frac{1}{n} \left[\frac{1}{n+1} + \frac{2}{n+2} + \dots + \frac{3n}{4n} \right]$

Ans. (a) $2e^{\frac{\pi-4}{2}}$; (b) $3 - \ln 4$

Sol. (A) $\left(\sum_{r=1}^n \left(1 + \frac{r^2}{n^2} \right) \right)^{1/n} = \{A(\text{let})\}$

$$\Rightarrow \sum_{r=1}^n \frac{1}{n} \left(\ln \left(1 + \frac{r}{n} \right)^2 \right) = \ln(A)$$

$$\Rightarrow \lim_{n \rightarrow \infty} \ln A = \lim_{n \rightarrow \infty} \sum_{r=1}^n \ln \left(1 + \left(\frac{r}{n} \right)^2 \right) \cdot \frac{1}{n}$$

$$= \int_0^1 \ln(1+x^2) dx$$

$$\Rightarrow \ln A = \left[x \ln(1+x^2) \right]_0^1 - \int_0^1 \frac{2x^2}{1+x^2} dx$$

$$= \ln 2 - 2 \int_0^1 \frac{(x^2+1-1)}{x^2+1} dx$$

$$= \ln 2 - 2 + 2 \left[\tan^{-1} x \right]_0^1$$

$$= \ln 2 - 2 + 2 \cdot \frac{\pi}{4}$$

$$\ln A = \frac{\pi-4}{2} + \ln 2$$

$$\Rightarrow A = 2 \cdot e^{\frac{\pi-4}{2}}$$

$$(B) \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{3n} \frac{r}{n+r}$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^{3n} \left(\frac{\frac{r}{n}}{1 + \frac{r}{n}} \right) \cdot \frac{1}{n}$$

$$= \int_0^3 \frac{x}{1+x} dx = \int_0^3 \left(1 - \frac{1}{1+x} \right) dx$$

$$= 3 - \ln 4$$

79. Consider a function $f(n) = \frac{1}{1+n^2}$. Let $\alpha_n = \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right)$ and $\beta_n = \frac{1}{n} \sum_{r=0}^{n-1} f\left(\frac{r}{n}\right)$ for $n = 1, 2, 3, \dots$

Also $\alpha = \lim_{n \rightarrow \infty} \alpha_n$ & $\beta = \lim_{n \rightarrow \infty} \beta_n$. Then prove (a) $\alpha_n < \beta_n$ (b) $\alpha = \beta$ (c) $\alpha_n < \frac{\pi}{4} < \beta_n$

दिया है $f(n) = \frac{1}{1+n^2}$. माना कि $\alpha_n = \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right)$ एवं $\beta_n = \frac{1}{n} \sum_{r=0}^{n-1} f\left(\frac{r}{n}\right)$ जहाँ $n = 1, 2, 3, \dots$ तथा $\alpha = \lim_{n \rightarrow \infty} \alpha_n$ और

$\beta = \lim_{n \rightarrow \infty} \beta_n$. तो सिद्ध कीजिए कि (a) $\alpha_n < \beta_n$ (b) $\alpha = \beta$ (c) $\alpha_n < \frac{\pi}{4} < \beta_n$

Ans.

Sol. (a) $\alpha_n = \frac{1}{n} \left[\sum_{r=1}^n f\left(\frac{r}{n}\right) \right]$

$$= \frac{1}{n} \left[f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots + f\left(\frac{n-1}{n}\right) + f(1) \right]$$

$$\beta_n = \frac{1}{n} \left[\sum_{r=0}^{n-1} f\left(\frac{r}{n}\right) \right]$$

$$= \frac{1}{n} \left[f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots + f\left(\frac{n-1}{n}\right) \right]$$

$$\Rightarrow \alpha_n = \beta_n + \frac{1}{2n} - \frac{f(0)}{n}$$

$$\Rightarrow \alpha_n = \beta_n - \frac{1}{2n}$$

$$\therefore \beta_n > \alpha_n$$

(b) $f(n) = \frac{1}{1+n^2}$

$$\lim_{n \rightarrow \infty} \alpha_n = \lim_{n \rightarrow \infty} \left(\sum_{r=1}^n f\left(\frac{r}{n}\right) \right) \frac{1}{n} = \int_0^1 f(x) dx$$

$$= \int_0^1 \frac{1}{1+x^2} dx = \left[\tan^{-1} x \right]_0^1 = \frac{\pi}{4}$$

$$\Rightarrow \alpha = \frac{\pi}{4}$$

$$\lim_{n \rightarrow \infty} \beta_n = \left(\sum_{r=1}^{n-1} f\left(\frac{r}{n}\right) \right) \frac{1}{n} = \int_0^1 f(x) dx = \frac{\pi}{4}$$

$$\Rightarrow \beta = \frac{\pi}{4}$$

$$\Rightarrow \alpha = \beta$$

$$(c) \alpha_n < \frac{\pi}{4} < \beta_n$$

DEFINITE INTEGRATION BY ESTIMATION

80. Prove the inequalities :

सिद्ध कीजिए :

$$(a) \frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi\sqrt{2}}{8}$$

$$(b) 2e^{-1/4} < \int_0^2 e^{x^2-x} dx < 2e^2.$$

$$(c) \frac{1}{3} < \int_0^1 x^{(\sin x + \cos x)^2} dx < \frac{1}{2}$$

$$(d) \frac{1}{2} \leq \int_0^2 \frac{dx}{2+x^2} \leq \frac{5}{6}$$

Ans.

Sol. (a) $\int_0^1 \frac{dx}{\sqrt{4-x^2}} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^2}}$

$$\Rightarrow \frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi\sqrt{2}}{8}$$

(b) If $x \in [0, 2]$

$$\text{then } y = x^2 - x \in \left[\frac{-1}{4}, 2 \right]$$

$$2e^{-1/4} < \int_0^2 e^{x^2-x} dx < 2e^2.$$

(c) If $x \in [0, 1]$

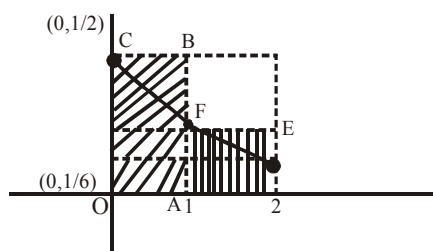
then $1 + \sin 2x \in [1, 2]$

$$\Rightarrow \int_0^1 x^2 dx < \int_0^1 x^{1+\sin 2x} dx < \int_0^1 x dx$$

$$\Rightarrow \frac{1}{3} < \int_0^1 x^{1+\sin 2x} dx < \frac{1}{2}$$

(d) If $x \in [0, 2]$

then $x^2 + 2 \in [2, 6]$



$$\Rightarrow \frac{1}{2} \leq \int_0^2 \frac{dx}{2+x^2} \leq \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

$\uparrow$ $\uparrow$
 Area of Area of + Area of
 OABC OABC ADEF

WALLI'S THEOREM

81. If a_1 , a_2 and a_3 are the three values of a which satisfy the equation

$$\int_0^{\pi/2} (\sin x + a \cos x)^3 dx - \frac{4a}{\pi-2} \int_0^{\pi/2} x \cos x dx = 2 \text{ then find the value of } 1000(a_1^2 + a_2^2 + a_3^2)$$

यदि a के तीन मान a_1 , a_2 एवं a_3 हैं तथा समीकरण $\int_0^{\pi/2} (\sin x + a \cos x)^3 dx - \frac{4a}{\pi-2} \int_0^{\pi/2} x \cos x dx = 2$ को संतुष्ट करते हैं तो

$1000(a_1^2 + a_2^2 + a_3^2)$ का मान बताइये।

Ans. 5250

Sol. $\int_0^{\pi/2} (\sin x + a \cos x)^3 dx - \frac{4a}{\pi-2} \int_0^{\pi/2} x \cos x dx = 2$

$$\int_0^{\pi/2} \sin^3 x dx + a^3 \int_0^{\pi/2} \cos^3 x dx + 3a \int_0^{\pi/2} \sin^2 x \cos x dx + 3a^2 \int_0^{\pi/2} \cos^2 x \sin x dx + \frac{4a}{\pi-2} [x \sin x + \cos x]_0^{\pi/2} = 2$$

$$\Rightarrow \frac{2}{3} + \frac{2}{3}a^3 + a + a^2 - \frac{4a}{\pi-2} \left[\frac{\pi-2}{2} \right] = 2$$

$$\Rightarrow \frac{2a^3}{3} + a^2 + a + \frac{2}{3} - 2a = 2$$

$$\Rightarrow 2a^3 + 3a^2 - 3a - 4 = 0 \begin{cases} a_1 \\ a_2 \\ a_3 \end{cases}$$

$$(a_1 + a_2 + a_3)^2 = a_1^2 + a_2^2 + a_3^2 + \left(\frac{-3}{2}\right) \cdot 2$$

$$\Rightarrow a_1^2 + a_2^2 + a_3^2 = \frac{9}{4} + 3 = \frac{21}{4}$$

$$\Rightarrow 1000(a_1^2 + a_2^2 + a_3^2) = 5250$$

INTEGRATION BY REDUCTION

82. Determine a positive integer $n \leq 5$, such that $\int_0^1 e^x (x-1)^n dx = 16 - 6e$.

धनात्मक पूर्णांक $n \leq 5$ ज्ञात कीजिए जहाँ $\int_0^1 e^x (x-1)^n dx = 16 - 6e$.

Ans. $n = 3$

Sol. $I_n = \int_0^1 e^x \cdot (x-1)^n dx = 16 - 6e; n \leq 5$

$$I_n = e^x \cdot (x-1)^n \Big|_0^1 - n \int_0^1 e^x (x-1)^{n-1} dx$$

$$I_n = -(-1)^n - n I_{n-1} = (-1)^{n+1} - n I_{n-1}$$

$$n = 1, I = \int_0^1 e^x (x-1) dx$$

$$= (x-1)e^x \Big|_0^1 - \int_0^1 e^x dx = 2 - e$$

$$I_2 = -1 - 2(2 - e) = 2e - 5$$

$$I_3 = 1 - 3(2e - 5) = 16 - 6e$$

so $n = 3$

83. Let $U_{10} = \int_0^{\frac{\pi}{2}} x \sin^{10} x dx$, then find the value of $\left(\frac{100U_{10} - 1}{U_8} \right)$

माना $U_{10} = \int_0^{\frac{\pi}{2}} x \sin^{10} x dx$, हो तो $\left(\frac{100U_{10} - 1}{U_8} \right)$ का मान ज्ञात कीजिये।

Ans. 90

Sol. Use reduction formula

$$10U_{10} = \frac{1}{10} + 9U_8$$

$$100U_{10} - 1 = 90U_8$$

84. Find the value of $\frac{5050 \int_0^1 (1-x^{50})^{100} dx}{\int_0^1 (1-x^{50})^{101} dx}$

$$\frac{5050 \int_0^1 (1-x^{50})^{100} dx}{\int_0^1 (1-x^{50})^{101} dx} \text{ ज्ञात कीजिए।}$$

Ans. 5051

Sol. Let $I = \int_0^1 (1-x^{50})^{100} dx$ and $I' = \int_0^1 (1-x^{50})^{101} dx$

$$\text{Then, } I' = \int_0^1 1 \cdot (1-x^{50})^{101} dx = (x(1-x^{50})^{101})_0^1$$

$$+ 101 \int_0^1 50x^{50} (1-x^{50})^{100} dx$$

$$= 5050 \int_0^1 x^{50} (1-x^{50})^{100} dx$$

$$-I' = 5050 \int_0^1 -x^{50} (1-x^{50})^{100} dx$$

$$\Rightarrow 5050I - I' = 5050 \int_0^1 (1-x^{50})^{100} dx$$

$$+ 5050 \int_0^1 -x^{50} (1-x^{50})^{100} dx$$

$$\Rightarrow 5050 \int_0^1 (1-x^{50})^{101} dx = 5050 I'$$

$$\Rightarrow 5050 I = 5051 I' \Rightarrow 5050 \frac{I}{I'} = 5051$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 85 to 87

Let the function f satisfies $f(x) \cdot f'(-x) = f(-x) \cdot f'(x)$ for all x and $f(0) = 3$.

माना फलन f प्रतिबन्ध $f(x) \cdot f'(-x) = f(-x) \cdot f'(x)$ को x के सभी मानों के लिए सन्तुष्ट करता है तथा $f(0) = 3$ है।

85. The value of $f(x) \cdot f(-x)$ for all x , is :

$f(x) \cdot f(-x)$ का x के सभी मानों के लिए मान होगा :

- (A) 4 (B) 9 (C) 12 (D) 16

Ans. B

Sol. $f(x) \cdot f'(-x) = f(-x) \cdot f'(x) \forall x$

$$f(0) = 3$$

$$\Rightarrow \int \frac{f'(x)}{f(x)} dx = \int \frac{f'(-x)}{f(-x)} dx$$

$$= \log |f(x)| = -\log |f(-x)| + \log c$$

$$\Rightarrow f(x) \cdot f(-x) = c$$

$$\text{at } x = 0$$

$$(f(0))^2 = C \Rightarrow C = 9$$

$$\Rightarrow f(x) \cdot f(-x) = 9$$

86. $\int_{-51}^{51} \frac{dx}{3+f(x)}$ has the value equal to :

$$\int_{-51}^{51} \frac{dx}{3+f(x)} \text{ का मान है :}$$

- (A) 17 (B) 34 (C) 102 (D) 0

Ans. A

Sol. $I = \int_{-51}^{51} \frac{dx}{3+f(x)} \quad \underline{\hspace{2cm}} (1)$

Apply king prop.

$$\Rightarrow I = \int_{-51}^{51} \frac{1}{3+f(-x)} dx = \int_{-51}^{51} \frac{f(x)}{3f(x)+9} dx \quad \underline{\hspace{2cm}} (2)$$

by (1) + (2)

$$\Rightarrow 2I = \int_{-51}^{51} \left(1 + \frac{f(x)}{3}\right) \cdot \frac{1}{3 + f(x)} dx = \frac{1}{3} \cdot 102$$

$$\Rightarrow I = 17$$

87. Number of roots of $f(x) = 0$ in $[-2, 2]$ is :

समीकरण $f(x) = 0$ का अन्तराल $[-2, 2]$ में हलों की संख्या होगी :

(A) 0 (B) 1 (C) 2 (D) 4

Ans. A

Sol. Let $x = \alpha$ be the root of $f(x) = 0$

$$\therefore f(\alpha) = 0$$

$$f(x) \cdot f(-x) = 9$$

$$\text{put } x = \alpha \Rightarrow 0 = 9 \text{ impossible}$$

hence $f(x)$ has no root

$$\text{but } f(0) = 3$$

hence $f(x) > 0 \forall x \in \mathbb{R}$ as f is continuous

possible function $f(x) = 3e^{-x}$

Paragraph for Question nos. 88 to 90

$$\text{Consider the integral } I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1 + e^{2\sin^3 4x}} dx$$

$$\text{माना समाकलन } I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1 + e^{2\sin^3 4x}} dx \text{ है।}$$

88. If $I = k \int_0^{\pi/2} \cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x dx$ then 'k' is equal to :

$$\text{यदि } I = k \int_0^{\pi/2} \cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x dx \text{ हो, तो } k \text{ का मान है :}$$

(A) 1 (B) 5 (C) 10 (D) 20

Ans. C

89. If $I = c \cdot \int_0^{\pi/4} \cos 6x \cdot \cos 8x \cdot \cos 2x dx$ then 'c' equals :

$$\text{यदि } I = c \cdot \int_0^{\pi/4} \cos 6x \cdot \cos 8x \cdot \cos 2x dx \text{ हो, तो } c \text{ का मान होगा :}$$

- (A) 5 (B) 10 (C) 20 (D) $\frac{5}{2}$

Ans. B

90. The value of 'I' equals :

I का मान होगा :

- (A) $\frac{5\pi}{4}$ (B) $\frac{5\pi}{16}$ (C) $\frac{5\pi}{32}$ (D) $\frac{5\pi}{8}$

Ans. D

Sol. (Que. 88, 89, 90)

$$I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1 + e^{2\sin^3 4x}} dx \quad \dots(1)$$

$$I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1 + e^{-2\sin^3 4x}} dx \quad (\text{using King})$$

$$I = \int_0^{10\pi} \frac{e^{2\sin^3 4x} (\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x)}{1 + e^{2\sin^3 4x}} dx \quad (\text{using King}) \quad \dots(2)$$

add (1) + (2)

$$2I = \int_0^{10\pi} \underbrace{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}_{f(x)} dx$$

f(x) repeats after 2π

$$2I = 5 \int_0^{2\pi} f(x) dx ; \quad \text{again } f(2\pi - x) = f(x)$$

$$2I = 10 \int_0^{\pi} f(x) dx ; \quad \text{again } f(\pi - x) = f(x)$$

$$2I = 20 \int_0^{\pi/2} f(x) dx \Rightarrow I = 10 \int_0^{\pi/2} f(x) dx$$

$$\therefore I = 10 \int_0^{\pi/2} \cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x dx \quad \dots(3)$$

$$I = 10 \int_0^{\pi/2} \cos(3\pi - 6x) \cos\left(\frac{7\pi}{2} - 7x\right) \cos(4\pi - 8x) \cos\left(\frac{9\pi}{2} - 9x\right) dx$$

$$I = 10 \int_0^{\pi/2} \cos 6x \cdot \sin 7x \cdot \cos 8x \cdot \sin 9x \, dx \quad \dots(4)$$

(3) + (4)

$$2I = 10 \int_0^{\pi/2} \cos 6x \cdot \cos 8x [\cos 7x \cdot \cos 9x + \sin 7x \cdot \sin 9x] \, dx$$

$$2I = 10 \int_0^{\pi/2} \cos 6x \cdot \cos 8x \cdot \cos 2x \, dx$$

$$I = 5 \int_0^{\pi/2} \cos 6x \cdot \cos 8x \cdot \cos 2x \, dx = 10 \int_0^{\pi/4} \cos 6x \cdot \cos 8x \cdot \cos 2x \, dx \Rightarrow (B)$$

$$I = 5 \int_0^{\pi/2} \frac{\cos 8x}{2} (\cos 8x + \cos 4x) \, dx = \frac{5}{2} \int_0^{\pi/2} \cos 8x (\cos 8x + \cos 4x) \, dx$$

$$= \frac{5}{2} \int_0^{\pi/2} \cos^2 8x \, dx + \underbrace{\frac{5}{2} \int_0^{\pi/2} \cos 8x \cos 4x \, dx}_{\text{zero (using Queen)}}$$

$$\left(\int_0^{\pi/2} (\cos 12x + \cos 4x) = \frac{\sin 12x}{12} + \frac{1}{4} \sin 4x \right) \Bigg|_0^{\pi/2} \rightarrow 0$$

$$= \frac{1}{8} \cdot \frac{5}{2} \int_0^{4\pi} \cos^2 t \, dt \quad (8x = t)$$

$$= \frac{5}{4} \int_0^{\pi} \cos^2 t \, dt \quad (\text{using Queen})$$

$$I = \frac{5}{2} \int_0^{\pi/2} \cos^2 t \, dt = \frac{5}{2} \int_0^{\pi/2} \sin^2 t \, dt$$

$$2I = \frac{5}{2} \int_0^{\pi/2} dt \Rightarrow I = \frac{5}{4} \cdot \frac{\pi}{2} \Rightarrow I = \frac{5\pi}{8}$$

Paragraph for Question nos. 91 to 93

Let $y = f(x)$ be a twice differentiable, non-negative function defined on $[a, b]$. The area $\int_a^b f(x)dx$, $b > a$ bounded

by $y = f(x)$, the x -axis and the ordinates at $x = a$ and $x = b$ can be approximated as $\int_a^b f(x)dx \cong \frac{(b-a)}{2} \{f(a) + f(b)\}$.

Since $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$, $c \in (a, b)$, a better approximation to $\int_a^b f(x)dx$ can be written as

$$\int_a^b f(x)dx \cong \frac{(c-a)}{2} \{f(a) + f(c)\} + \frac{(b-c)}{2} \{f(c) + f(b)\} \equiv F(c). \text{ If } c = \frac{a+b}{2}, \text{ then this gives : } \int_a^b f(x)dx \cong \frac{b-a}{4} \{f(a) + 2f(c) + f(b)\}, \dots\dots (1)$$

[JEE-2006]

यदि $y = f(x)$ अन्तराल $[a, b]$ में ऋणात्मक नहीं है तथा दो बार अवकलनीय है तो क्षेत्रफल $\int_a^b f(x)dx$, $b > a$ जो कि $y = f(x)$, x -अक्ष

तथा रेखा $x = a$ एवं $x = b$ से परिबद्ध है को $\int_a^b f(x)dx \cong \frac{(b-a)}{2} \{f(a) + f(b)\}$ के लगभग लिखा जा सकता है। क्योंकि

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx, c \in (a, b), \text{ तो उपरोक्त का एक और लगभग मान } \int_a^b f(x)dx \text{ यह भी होगा}$$

$$\int_a^b f(x)dx \cong \frac{(c-a)}{2} \{f(a) + f(c)\} + \frac{(b-c)}{2} \{f(c) + f(b)\} \equiv F(c). \text{ यदि } c = \frac{a+b}{2}, \text{ तो दिया गया है : } \int_a^b f(x)dx \cong \frac{b-a}{4} \{f(a) + 2f(c) + f(b)\}, \dots\dots (1)$$

[JEE-2006]

91. The approximate value of $\int_0^{\pi/2} \sin x dx$ using rule (1) given above is :

नियम 1 से $\int_0^{\pi/2} \sin x dx$ का लगभग मान होगा :

- (A) $\frac{\pi}{8\sqrt{2}}(1+\sqrt{2})$ (B) $\frac{\pi}{4\sqrt{2}}(1+\sqrt{2})$ (C) $\frac{\pi}{8}(1+\sqrt{2})$ (D) $\frac{\pi}{4}(1+\sqrt{2})$

Ans. C

Sol. $\int_0^{\pi/2} \sin x dx = \frac{\left(\frac{\pi}{2} - 0\right)}{4} \left(\sin 0 + \sin \frac{\pi}{2} + 2 \sin \frac{\pi}{4} \right)$

$$= \frac{\pi}{8}(1 + \sqrt{2})$$

92. If $\lim_{t \rightarrow a} \left\{ \frac{\int_a^t f(x) dx - \frac{(t-a)}{2}(f(t) + f(a))}{(t-a)^3} \right\} = 0$, for each fixed a , then $f(x)$ is a polynomial of degree utmost :

यदि $\lim_{t \rightarrow a} \left\{ \frac{\int_a^t f(x) dx - \frac{(t-a)}{2}(f(t) + f(a))}{(t-a)^3} \right\} = 0$, जहाँ a एक निश्चित संख्या है तो, $f(x)$ किस अधिकतम घात का बहुपदीय फलन

होगा :

(A) 4

(B) 3

(C) 2

(D) 1

Ans. D

Sol. $\lim_{x \rightarrow a} \frac{\int_a^x f(x) dx - \left(\frac{x-a}{2} \right) (f(x) + f(a))}{(x-a)^3} = 0$

$$\lim_{h \rightarrow 0} \frac{\int_a^{a+h} f(x) dx - \frac{h}{2} (f(a+h) + f(a))}{h^3} = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(a+h) - \frac{1}{2}[f(a) + f(a+h)] - \frac{h}{2}(f'(a+h))}{3h^2} = 0$$

[Using L'Hospital rule]

$$\Rightarrow \lim_{h \rightarrow 0} \frac{\frac{1}{2}f(a+h) - \frac{1}{2}f(a) - \frac{h}{2}f'(a+h)}{3h^2} = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{\frac{1}{2}f'(a+h) - \frac{1}{2}f'(a) - \frac{h}{2}f''(a+h)}{6h} = 0$$

[Using L'Hospital rule]

$$\Rightarrow \lim_{h \rightarrow 0} \frac{-f''(a+h)}{12} = 0 \Rightarrow f''(x) = 0, \forall a \in \mathbb{R}$$

$\Rightarrow f(x)$ must be of max. degree 1

93. If $f''(x) < 0$, $x \in (a, b)$, then at the point $C(c, f(c))$ on $y = f(x)$ for which $F(c)$ is a maximum, $f'(c)$ is given by :

यदि $f''(x) < 0$, जहाँ $x \in (a, b)$ तो वक्र $y = f(x)$ का बिन्दु $C(c, f(c))$ पर वह मान जिस पर $F(c)$ अधिकतम है तो $f'(c)$ का मान होगा :

$$(A) f'(c) = \frac{f(b) - f(a)}{b - a} \quad (B) f'(c) = \frac{f(b) - f(a)}{a - b}$$

$$(C) f'(c) = \frac{2(f(b) - f(a))}{b - a} \quad (D) f'(c) = 0$$

Ans. A

Sol. $f''(x) < 0$, $\forall x \in (a, b)$, for $c \in (a, b)$

$$F(c) = \frac{c-a}{2}(f(a) + f(c)) + \frac{b-c}{2}(f(b) + f(c))$$

$$= \frac{b-a}{2}f(c) + \frac{c-a}{2}f(a) + \frac{b-c}{2}f(b)$$

$$\Rightarrow F'(c) = \frac{b-a}{2}f'(c) + \frac{1}{2}f(a) - \frac{1}{2}f(b)$$

$$= \frac{1}{2}[(b-a)f'(c) + f(a) - f(b)]$$

$$F''(c) = \frac{1}{2}(b-a)f''(c) < 0$$

$$[\because f''(x) < 0, \forall x \in (a, b) \text{ and } b > a]$$

$\therefore F(c)$ is max. at the point $(c, f(c))$ where

$$F'(c) = 0 \Rightarrow f'(c) = \left(\frac{f(b) - f(a)}{b - a} \right)$$

Paragraph for Questions 94 to 96

Consider the function $f(x)$ and $g(x)$, both defined from $\mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{x^3}{2} + 1 - x \int_0^x g(t) dt$ and

$$g(x) = x - \int_0^1 f(t) dt, \text{ then}$$

फलन $f(x)$ तथा $g(x)$, दोनों $R \rightarrow R$ में परिभाषित है। $f(x) = \frac{x^3}{2} + 1 - x \int_0^x g(t) dt$ तथा $g(x) = x - \int_0^1 f(t) dt$ है, तो

94. Minimum value of $f(x)$ is :

- (A) 0 (B) 1 (C) $\frac{3}{2}$ (D) Does not exist

$f(x)$ का न्यूनतम मान होगा :

- (A) 0 (B) 1 (C) $\frac{3}{2}$ (D) विद्यमान नहीं है

Ans. B

95. The number of points of intersection of $f(x)$ and $g(x)$ is/are :

$f(x)$ तथा $g(x)$ के प्रतिच्छेद बिन्दुओं की संख्या है :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. A

96. The area bounded by $g(x)$ with co-ordinate axes is (in square units) :

- (A) $\frac{9}{4}$ (B) $\frac{9}{2}$ (C) $\frac{9}{8}$ (D) None of these

$g(x)$ तथा निर्देशांकी अक्षों के मध्य का क्षेत्रफल होगा (वर्ग इकाई में) :

- (A) $\frac{9}{4}$ (B) $\frac{9}{2}$ (C) $\frac{9}{8}$ (D) इनमें से कोई नहीं

Ans. C

Sol. (Que. 94, 95, 96)

$$f(x) = \frac{x^3}{2} + 1 - x \int_0^x g(t) dt$$

$$\text{and } g(x) = x - \int_0^1 f(t) dt = x - k \quad \text{where } k = \int_0^1 f(t) dt$$

$$\Rightarrow f(x) = \frac{x^3}{2} + 1 - x \int_0^x (t - k) dt$$

$$\Rightarrow f'(x) = \frac{3x^2}{2} - \int_0^x (t - k) dt - x(x - k)$$

$$= \frac{3x^2}{2} - \left[\frac{t^2}{2} - kt \right]_0^x - x^2 + kx$$

$$= \frac{3x^2}{2} - \frac{x^2}{2} + kx - x^2 + kx$$

$$f'(x) = 2kx$$

$$\Rightarrow \int f'(x) dx = 2k \int x dx$$

$$\Rightarrow f(x) = kx^2 + C$$

$$\therefore f(0) = 1$$

$$\Rightarrow C = 1$$

$$\Rightarrow f(x) = kx^2 + 1$$

$$\text{Now } k = \int_0^1 f(t) dt$$

$$\Rightarrow k = \int_0^1 (kt^2 + 1) dt = \frac{k}{3} + 1$$

$$\Rightarrow k = \frac{3}{2}$$

$$\Rightarrow f(x) = \frac{3}{2}x^2 + 1 \text{ and } g(x) = x - \frac{3}{2}$$

$\therefore$ minimum value of $f(x)$ is 1

intersection point of $f(x)$ and $g(x)$ are

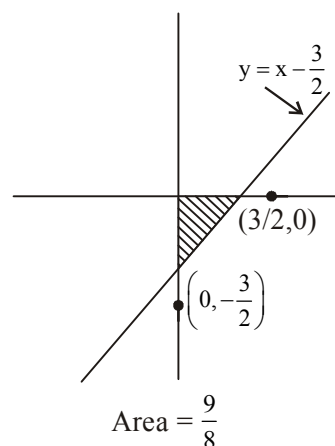
$$\frac{3}{2}x^2 + 1 = x - \frac{3}{2}$$

$$\Rightarrow 3x^2 + 2 = 2x - 3$$

$$\Rightarrow 3x^2 - 2x + 5 = 0$$

Here $a = 3$ and $D < 0$

$\Rightarrow$ Intersection point of $f(x)$ and $g(x)$ are zero.



Paragraph for Questions 97 to 98

Let $f: \mathbb{R} \rightarrow \left[\frac{3}{4}, \infty\right)$ be a surjective quadratic function with line of symmetry $2x - 1 = 0$ and $f(1) = 1$.

माना $f: \mathbb{R} \rightarrow \left[\frac{3}{4}, \infty\right)$ एक आच्छादक द्विघात फलन जो कि रेखा $2x - 1 = 0$ के सापेक्ष सम्मित है तथा $f(1) = 1$.

97. If $g(x) = \frac{f(x) + f(-x)}{2}$ then $\int \frac{dx}{\sqrt{g(e^x) - 2}}$ is equal to : (Where C is constant of integration)

- (A) $\sec^{-1}(e^{-x}) + C$ (B) $\sec^{-1}(e^x) + C$ (C) $\sin^{-1}(e^{-x}) + C$ (D) $\sin^{-1}(e^x) + C$

यदि $g(x) = \frac{f(x) + f(-x)}{2}$ है, तो $\int \frac{dx}{\sqrt{g(e^x) - 2}}$ बराबर होगा : (जहाँ C एक समाकलन नियतांक है)

- (A) $\sec^{-1}(e^{-x}) + C$ (B) $\sec^{-1}(e^x) + C$ (C) $\sin^{-1}(e^{-x}) + C$ (D) $\sin^{-1}(e^x) + C$

Ans. B

98. $\int \frac{e^x}{f(e^x)} dx :$

- (A) $\cot^{-1}\left(\frac{2e^x - 1}{\sqrt{3}}\right) + C$ (B) $\frac{2}{\sqrt{3}} \cot^{-1}\left(\frac{2e^x + 1}{\sqrt{3}}\right) + C$
 (C) $\tan^{-1}\left(\frac{2e^x + 1}{\sqrt{3}}\right) + C$ (D) $\frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2e^x - 1}{\sqrt{3}}\right) + C$

Ans. D

Sol. (Que. 97, 98)

$$f(x) = a\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$\therefore f(1) = 1$$

$$\Rightarrow a = 1$$

$$\Rightarrow g(x) = \frac{f(x) + f(-x)}{2} = \frac{\left(x - \frac{1}{2}\right)^2 + \left(x + \frac{1}{2}\right)^2 + \frac{3}{2}}{2}$$

$$= x^2 + 1$$

$$\Rightarrow \int \frac{dx}{\sqrt{g(e^x) - 2}} = \int \frac{dx}{\sqrt{e^{2x} + 1 - 2}} = \int \frac{dx}{\sqrt{(e^x)^2 - 1}}$$

$$= \int \frac{e^x dx}{e^x \sqrt{e^{2x} - 1}} = \sec^{-1}(e^x) + C$$

$$98. \int \frac{e^x}{f(e^x)} dx$$

$$= \int \frac{e^x}{\left(e^x - \frac{1}{2}\right)^2 + \frac{3}{4}} dx$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2e^x - 1}{\sqrt{3}} \right) + C$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

99. Suppose $J = \int \frac{\sin^2 x + \sin x}{1 + \sin x + \cos x} dx$ and $K = \int \frac{\cos^2 x + \cos x}{1 + \sin x + \cos x} dx$. If C is an arbitrary constant of integration then which of the following is/are correct :

माना $J = \int \frac{\sin^2 x + \sin x}{1 + \sin x + \cos x} dx$ और $K = \int \frac{\cos^2 x + \cos x}{1 + \sin x + \cos x} dx$. यदि C स्वेच्छ समाकलन अचर है। तब निम्न में से कौन कौन से सत्य है :

(A) $J = \frac{1}{2} (x - (\sin x + \cos x)) + C$

(B) $J = K - (\sin x + \cos x) + C$

(C) $J = x - K + C$

(D) $K = \frac{1}{2} (x - \sin x + \cos x) + C$

Ans. ABC

Sol. $J + K = \int \frac{1 + \sin x + \cos x}{1 + \sin x + \cos x} dx$

$J + K = x + C \dots(1) \Rightarrow$ (C)

again $J - K = \int \frac{(\sin^2 x - \cos^2 x) + \sin x - \cos x}{1 + \sin x + \cos x} dx = \int \frac{(\sin x - \cos x)(\sin x + \cos x + 1)}{1 + \sin x + \cos x} dx$

$J - K = -\cos x - \sin x + C \dots(2)$

hence $J = K - (\sin x + \cos x) + C \Rightarrow$ (B)

Also (1) + (2)

$2J = x - (\cos x + \sin x) + C$

$J = \frac{1}{2} [x - \sin x - \cos x] + C$

and (1) - (2)

$2K = x + (\sin x + \cos x) + C$

$K = \frac{1}{2} (x + \sin x + \cos x) + C$

from (1) $J = x - K + C \Rightarrow$ (C)

100. Let $u = \int_0^\infty \frac{dx}{x^4 + 7x^2 + 1}$ & $v = \int_0^\infty \frac{x^2 dx}{x^4 + 7x^2 + 1}$ then :

माना $u = \int_0^\infty \frac{dx}{x^4 + 7x^2 + 1}$ और $v = \int_0^\infty \frac{x^2 dx}{x^4 + 7x^2 + 1}$ तो :

(A) $v > u$ (B) $6v = \pi$ (C) $3u + 2v = 5\pi/6$ (D) $u + v = \pi/3$

Ans. BCD

Sol. $u = \int_0^\infty \frac{dx}{x^4 + 7x^2 + 1}$ $v = \int_0^\infty \frac{x^2 dx}{x^4 + 7x^2 + 1}$

$\underbrace{\hspace{10em}}_{\text{let } x = \frac{1}{t}}$

$$\Rightarrow u = \int_\infty^0 \frac{t^4 - \frac{1}{t^2} dt}{t^4 + 7t^2 + 1}$$

$$= \int_0^\infty \frac{t^4}{t^2(t^4 + 7t^2 + 1)} dt = \int_0^\infty \frac{t^2}{t^4 + 7t^2 + 1} dt$$

$$\Rightarrow u = v$$

$$u = \int_0^\infty \frac{dx}{x^4 + 7x^2 + 1} = \int \frac{\frac{1}{x^2} dx}{\left(x^2 + \frac{1}{x^2} + 7\right)}$$

$$u = \int_0^\infty \frac{\frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 9} dx \text{ and } v = \int_0^\infty \frac{1}{\left(x - \frac{1}{x}\right)^2 + 9} .dx$$

$$\Rightarrow u + v = \int_0^\infty \frac{\left(1 + \frac{1}{x^2}\right) dx}{\left(x - \frac{1}{x}\right)^2 + 9}$$

$$= \int \frac{dt}{t^2 + 9} = \frac{1}{3} \tan^{-1} \left(\frac{t}{3} \right)$$

$$= \left[\frac{1}{3} \tan^{-1} \left[\frac{x - \frac{1}{x}}{3} \right] \right]_0^\infty = \frac{\pi}{6} + \frac{\pi}{6}$$

$$\Rightarrow u + v = \frac{\pi}{3}$$

$$\Rightarrow u = \frac{\pi}{6} = v$$

$$\Rightarrow 3u + 2v = 54 = \frac{5\pi}{6}$$

101. Let $S_n = \sum_{k=1}^n \frac{k^2 + n^2}{n^3}$ and $T_n = \sum_{k=0}^{n-1} \frac{k^2 + n^2}{n^3}$ for $n = 1, 2, 3, \dots$. Then :

माना $S_n = \sum_{k=1}^n \frac{k^2 + n^2}{n^3}$ और $T_n = \sum_{k=0}^{n-1} \frac{k^2 + n^2}{n^3}$, जहाँ $n = 1, 2, 3, \dots$ तो :

(A) $S_n < \frac{4}{3}$ (B) $T_n > \frac{4}{3}$ (C) $S_n > \frac{4}{3}$ (D) $T_n < \frac{4}{3}$

Ans. CD

Sol. We have $S_n = \frac{1}{n} \sum_{k=1}^n \left(\frac{k^2}{n^2} + 1 \right) = \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right)$

$$\Rightarrow S_n = \frac{1}{n} \left[f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots + f\left(\frac{n}{n}\right) \right]$$

We have $f(x) = x^2 + 1$, which is increasing in $[0, 1]$

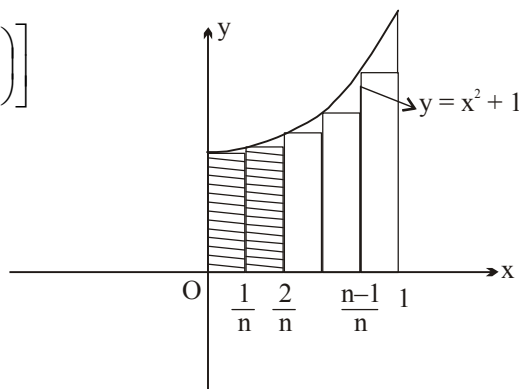
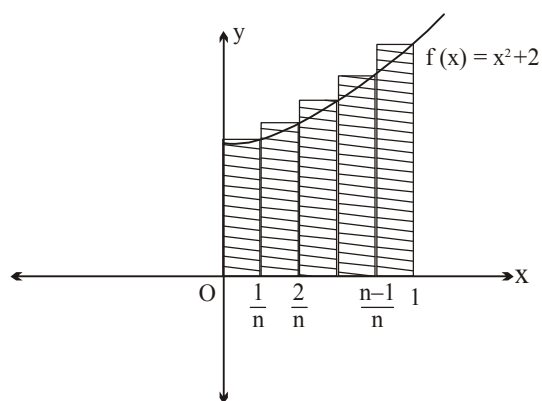
$$\text{Clearly } S_n > \int_0^1 (x^2 + 1) dx \Rightarrow S_n > \frac{4}{3} \dots \dots \dots (1)$$

$$\text{Similarly } T_n = \frac{1}{n} \left[f(0) + f\left(\frac{1}{n}\right) + \dots + f\left(\frac{n-1}{n}\right) \right]$$

$$T_n < \int_0^1 (x^2 + 1) dx \Rightarrow T_n < \frac{4}{3} \dots \dots \dots (2)$$

Clearly from graph, we conclude that

$$T_n < \frac{4}{3} < S_n.$$



102. If $f(\theta) = \lim_{n \rightarrow \infty} \sum_{r=0}^{n\theta} \frac{2r}{n\sqrt{(3\theta n - 2r)(n\theta + 2r)}}$ then :

(A) $f(1) = \frac{\pi}{6}$

(B) $f(\theta) = \frac{\theta}{2} \int_0^\theta \frac{dx}{\sqrt{\theta^2 - \left(x - \frac{\theta}{2}\right)^2}}$

(C) $f(\theta)$ is a constant function

(D) $y = f(\theta)$ is invertible

यदि $f(\theta) = \lim_{n \rightarrow \infty} \sum_{r=0}^{n\theta} \frac{2r}{n\sqrt{(3\theta n - 2r)(n\theta + 2r)}}$ है, तो :

(A) $f(1) = \frac{\pi}{6}$

(B) $f(\theta) = \frac{\theta}{2} \int_0^\theta \frac{dx}{\sqrt{\theta^2 - \left(x - \frac{\theta}{2}\right)^2}}$

(C) $f(\theta)$ एक अचर फलन है

(D) $y = f(\theta)$ एक व्युत्क्रमणीय फलन है

Ans. ABD

Sol. $f(\theta) = \lim_{n \rightarrow \infty} \sum_{r=0}^{n\theta} \frac{2r}{n\sqrt{(3\theta n - 2r)(n\theta + 2r)}}$

$$= \lim_{n \rightarrow \infty} \sum_{r=0}^{n\theta} \frac{2\left(\frac{r}{n}\right)}{n\sqrt{\left(3\theta - 2\frac{r}{n}\right)\left(\theta + \frac{2r}{n}\right)}}$$

$$= \int_0^\theta \frac{2x}{\sqrt{(3\theta - 2x)(\theta + 2x)}} dx$$

$$= \int_0^\theta \frac{2x}{\sqrt{3\theta^2 + 4\theta x - 4x^2}} = \int_0^\theta \frac{2x}{2\sqrt{\theta x - x^2 + \frac{3}{2}\theta^2}}$$

$$I = \int_0^\theta \frac{x}{\sqrt{\theta^2 - (x - \theta/2)^2}} dx$$

$\Rightarrow$ Apply king prop.

$$I = \int_0^\theta \frac{(\theta - x)dx}{\sqrt{\theta^2 - (x - \theta/2)^2}}$$

$$\Rightarrow 2I = \theta \int_0^\theta \frac{dx}{\sqrt{\theta^2 - (x - \theta/2)^2}} = \frac{\theta}{2} \left[\frac{\sin^{-1}(x - \theta/2)}{\theta} \right]_0^\theta$$

$$\Rightarrow I = \frac{\theta}{2} \left[\frac{\pi}{6} + \frac{\pi}{6} \right] = \frac{\pi\theta}{6}$$

$$\Rightarrow f(I) = \frac{\pi}{6}$$

$$\Rightarrow f(\theta) = \frac{\theta}{2} \int_0^\theta \frac{dx}{\sqrt{\theta^2 - (x - \theta/2)^2}} = \frac{\pi\theta}{6}$$

$y = f(\theta)$ is invertible

103. Let $J = \int_{-1}^2 \left(\cot^{-1} \frac{1}{x} + \cot^{-1} x \right) dx$, $K = \int_{-2\pi}^{7\pi} \frac{\sin x}{|\sin x|} dx$. Then which of the following alternative(s) is/are correct?

माना $J = \int_{-1}^2 \left(\cot^{-1} \frac{1}{x} + \cot^{-1} x \right) dx$, $K = \int_{-2\pi}^{7\pi} \frac{\sin x}{|\sin x|} dx$ है, तो निम्न में से कौनसा/कौनसे विकल्प सत्य है?

(A) $2J + 3K = 8\pi$ (B) $4J^2 + K^2 = 26\pi^2$ (C) $2J - K = 3\pi$ (D) $\frac{J}{K} = \frac{2}{5}$

Ans. AB

Sol. $J = \int_{-1}^2 \left(\cot^{-1} \frac{1}{x} + \cot^{-1} x \right) dx$

$$= \int_{-1}^0 (\pi + \tan^{-1} x + \cot^{-1} x) dx + \int_0^2 (\tan^{-1} x + \cot^{-1} x) dx$$

$$= \pi + \frac{\pi}{2} + \pi = \frac{5\pi}{2}$$

$$K = \int_{-2\pi}^{7\pi} \frac{\sin x}{|\sin x|} dx$$

$$= \int_{-2\pi}^{-\pi} dx - \int_{-\pi}^0 dx + \int_0^\pi dx - \int_\pi^{2\pi} dx + \int_{2\pi}^{3\pi} dx - \int_{3\pi}^{4\pi} dx + \int_{4\pi}^{5\pi} dx - \int_{5\pi}^{6\pi} dx + \int_{6\pi}^{7\pi} dx$$

$$= \pi - \pi + \pi - \pi + \pi - \pi + \pi - \pi + \pi = \pi$$

$$2J + 3K = 8\pi \text{ and } 4J^2 + K^2 = 26\pi^2$$

$$2J - 3K = 2\pi \text{ and } \frac{J}{K} = \frac{5}{2}$$

104. Which of the following function(s) is/are even?

निम्न में से कौनसा/कौनसे सम फलन है?

$$(A) f(x) = \int_0^x \ln(t + \sqrt{1+t^2}) dt$$

$$(B) g(x) = \int_0^x \frac{(2^t + 1)t}{2^t - 1} dt$$

$$(C) h(x) = \int_0^x (\sqrt{1+t+t^2} - \sqrt{1-t+t^2}) dt$$

$$(D) l(x) = \int_0^x \ln\left(\frac{1-t}{1+t}\right) dt$$

Ans. ACD

Sol. (A) $f(x) = \int_0^x \ln(t + \sqrt{1+t^2}) dt$

$$= \int_0^x \ln\left(\frac{1}{-t + \sqrt{1+t^2}}\right) dt$$

$$f(-x) = \int_0^{-x} \ln\left(\frac{1}{\sqrt{1+t^2} - t}\right) dt$$

let $t = -v$

$$= \int_0^x \ln\left(\frac{1}{\sqrt{1+v^2} + v}\right) dv$$

$$= \int_0^x \ln(v + \sqrt{1+v^2}) dv$$

$$\Rightarrow f(-x) = f(x)$$

$\therefore f$ is an even function.

$$(B) g(-x) = \int_0^{-x} \frac{(2^t + 1)t}{2^t - 1} dt$$

let $-t = v$

$$\Rightarrow g(-x) = -1 \int_0^x \frac{(1 + 2^v) - v}{(1 - 2^v)} dv$$

$$g(-x) = - \int_0^x \frac{(2^v + 1)v}{2^v - 1} dv$$

$$g(-x) = -g(x)$$

$\therefore g$ is odd

$$(C) h(x) = \int_0^x (\sqrt{1+t+t^2} - \sqrt{1-t+t^2}) dt$$

$$= \int_0^x \left(\frac{2t}{\sqrt{1+t+t^2} + \sqrt{1-t+t^2}} \right) dt$$

$$h(-x) = \int_0^{-x} \frac{2t \, dt}{\left(\sqrt{1+t+t^2} + \sqrt{1-t+t^2}\right)}$$

$$\text{let } t = -v$$

$$h(-x) = + \int_0^x \frac{2v \, dv}{\left(\sqrt{1-v+v^2} + \sqrt{1+v+v^2}\right)} = h(x)$$

$$(D) \, l(x) = \int_0^x \ln\left(\frac{1-t}{1+t}\right) dt$$

$$l(-x) = \int_0^{-x} \ln\left(\frac{1-t}{1+t}\right) dt$$

$$\text{let } -t = v$$

$$l(-x) = \int_0^{-x} \ln\left(\frac{1+V}{1-V}\right) dv = l(x)$$

105. Let $S_n = \sum_{k=1}^n \frac{n}{n^2 + kn + k^2}$ and $T_n = \sum_{k=0}^{n-1} \frac{n}{n^2 + kn + k^2}$, for $n = 1, 2, 3, \dots$. Then : [JEE-2008]

माना $S_n = \sum_{k=1}^n \frac{n}{n^2 + kn + k^2}$ तथा $T_n = \sum_{k=0}^{n-1} \frac{n}{n^2 + kn + k^2}$, के लिए $n = 1, 2, 3, \dots$ तब : [JEE-2008]

(A) $S_n < \frac{\pi}{3\sqrt{3}}$ (B) $S_n > \frac{\pi}{3\sqrt{3}}$ (C) $T_n < \frac{\pi}{3\sqrt{3}}$ (D) $T_n > \frac{\pi}{3\sqrt{3}}$

Ans. AD

Sol. $S_n = \frac{1}{n} \sum_{k=1}^n \frac{1}{1 + \frac{k}{n} + \frac{k^2}{n^2}}$

$$S_n < \int_0^1 \frac{dx}{x^2 + x + 1}$$

($\because$ the function is decreasing)

$$S_n < \int_0^1 \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$S_n < \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{2x+1}{\sqrt{3}} \right]_0^1$$

$$S_n < \frac{2}{\sqrt{3}} \left[\frac{\pi}{3} - \frac{\pi}{6} \right]$$

$$S_n < \frac{\pi}{3\sqrt{3}}$$

$$\text{Now } T_n - S_n = 1 - \frac{1}{3n} \Rightarrow T_n - S_n > \frac{2}{3}$$

106. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 + \pi^x) \sin x} dx$, $n = 0, 1, 2, \dots$, then: [JEE-2009]

यदि $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 + \pi^x) \sin x} dx$, $n = 0, 1, 2, \dots$, तो: [JEE-2009]

(A) $I_n = I_{n+2}$ (B) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$ (C) $\sum_{m=1}^{10} I_{2m} = 0$ (D) $I_n = I_{n+1}$

Ans. ABC

Sol. $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 + \pi^x) \sin x} dx$

Apply king property

$$I_n = \int_{-\pi}^{\pi} \frac{\pi^x \sin nx}{(1 + \pi^x) \sin x} dx$$

$$2I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{\sin x} dx \quad \dots(i)$$

$$2I_{n+2} = \int_{-\pi}^{\pi} \frac{\sin(n+2)x}{\sin x} dx \quad \dots(ii)$$

(ii) - (i)

$$\Rightarrow 2(I_{n+2} - I_n) = \int_{-\pi}^{\pi} \cos(n+1)x = 0 \Rightarrow I_{n+2} = I_n$$

$$\sum_{m=1}^{10} I_{2m} = 10 \sum_{m=1}^{10} I_2 = \frac{10}{2} \int_{-\pi}^{\pi} \frac{\sin 2x}{\sin x} dx = 0$$

Put $n = 1$ in equation (i)

$$2I_1 = \int_{-\pi}^{\pi} \frac{\sin x dx}{\sin x} = 2\pi$$

$$I_1 = \pi$$

$$\sum_{m=1}^{10} I_{2m+1} = 10\pi$$

MATCH THE COLUMN TYPE QUESTION(S)

107. Match the following columns :

निम्न कॉलम को सुमेलित कीजिए :

Column-I

(A) $\int \frac{x^4 - 1}{x^2 \sqrt{x^4 + x^2 + 1}} dx$

(B) $\int \frac{x^2 - 1}{x \sqrt{1 + x^4}} dx$

(C) $\int \frac{1 + x^2}{(1 - x^2) \sqrt{1 + x^4}} dx$

(D) $\int \frac{1}{(1 + x^4) \sqrt{\sqrt{1 + x^4} - x^2}} dx$

Column-II

(P) $\ln \left(\frac{(x^2 + 1) + \sqrt{x^4 + 1}}{x} \right) + C$

(Q) $C - \frac{1}{\sqrt{2}} \ln \left(\frac{\sqrt{x^4 + 1} - \sqrt{2}x}{(x^2 - 1)} \right)$

(R) $C - \tan^{-1} \left(\sqrt{\sqrt{1 + \frac{1}{x^4}} - 1} \right)$

(S) $\frac{\sqrt{x^4 + x^2 + 1}}{x} + C$

Ans. (A) S; (B) P; (C) Q; (D) R

Sol. (A) $\int \frac{x^4 - 1}{x^2 \sqrt{x^4 + x^2 + 1}} dx = \int \frac{x^2 - \frac{1}{x^2}}{\sqrt{x^4 + x^2 + 1}} dx$
 $= \int \frac{\left(x - \frac{1}{x^3} \right) dx}{\sqrt{x^2 + 1 + \frac{1}{x^2}}}$

Let $x^2 + \frac{1}{x^2} = t$

$\Rightarrow 2 \left(x - \frac{1}{x^3} \right) dx = dt$

$= \frac{1}{2} \int \frac{dt}{\sqrt{t+1}} = \sqrt{t+1} + C = \sqrt{\frac{x^4 + x^2 + 1}{x^2}} + C$

(B) $\int \frac{x^2 - 1}{x \sqrt{1 + x^4}} dx = \int \frac{1 - \frac{1}{x^2}}{\sqrt{\frac{1}{x^2} + x^2}} dx$

$$\int \frac{\left(1 - \frac{1}{x^2}\right)}{\sqrt{\left(x + \frac{1}{x}\right)^2 - 2}} dx$$

$$\text{Let } x + \frac{1}{x} = t \Rightarrow \left(1 - \frac{1}{x^2}\right) dx = dt$$

$$= \int \frac{dt}{\sqrt{t^2 - 2}} = \frac{1}{2\sqrt{2}} \ln \left| \frac{t - \sqrt{2}}{1 + \sqrt{2}} \right| + C$$

$$= \frac{1}{2\sqrt{2}} \ln \left| \frac{x^2 - \sqrt{2}x + 1}{x^2 + \sqrt{2}x + 1} \right| + C$$

$$= \frac{1}{2\sqrt{2}} \ln \left| \frac{(x^2 + 1)^2 - 2\sqrt{2}(x^2 + 1) + 2x^2}{(x^2 + 1)^2 - 2x^2} \right| + C$$

$$= \frac{1}{2\sqrt{2}} \ln \left| \frac{(x^2\sqrt{2}x + 1)^2}{(x+1)^2(x-1)^2} \right| + C$$

$$= \frac{1}{2\sqrt{2}} \ln \left| \frac{x^2 - \sqrt{2} + 1}{x^2 - 1} \right| + C$$

$$(C) \int \frac{1+x^2}{(1-x^2)\sqrt{1+x^4}} dx = \int \frac{\left(1 + \frac{1}{x^2}\right) dx}{\left(\frac{1}{x} - x\right) \sqrt{x^2 + \frac{1}{x^2} + 2 - 2}}$$

$$= - \int \frac{\left(1 + \frac{1}{x^2}\right) dx}{\left(x - \frac{1}{x}\right) \sqrt{\left(x - \frac{1}{x}\right)^2 + 2}}$$

$$\text{let } x - \frac{1}{x} = t$$

$$= - \int \frac{dt}{t\sqrt{t^2 + 2}} \text{ let } \sqrt{t^2 + 2} = v$$

$$\Rightarrow \frac{t}{\sqrt{t^2+2}} dt = dv$$

$$= -\int \frac{1}{v^2-2} dv = \frac{-1}{2\sqrt{2}} \ln \left| \frac{v-\sqrt{2}}{v+\sqrt{2}} \right| + C$$

$$= \frac{-1}{2\sqrt{2}} \ln \left| \frac{\sqrt{t^2+2}-\sqrt{2}}{\sqrt{t^2+2}+\sqrt{2}} \right| + C$$

$$= \frac{-1}{2\sqrt{2}} \ln \left| \frac{\sqrt{x^4+1}-\sqrt{2}x}{\sqrt{x^4+1}+\sqrt{2}x} \right| + C$$

$$= \frac{+1}{2\sqrt{2}} \ln \left| \frac{\sqrt{x^4+1}+\sqrt{2}x}{\left(\sqrt{x^4+1}\right)^2 - \left(\sqrt{2}x\right)^2} \right| + C$$

$$(D) \int \frac{1}{(1+x^4)\sqrt{\sqrt{1+x^4}-x^2}} dx$$

$$= \int \frac{dx}{x^4 \left(1 + \frac{1}{x^4}\right) x \sqrt{\sqrt{1 + \frac{1}{x^4}} - 1}}$$

$$\text{Let } 1 + \frac{1}{x^4} = v^2$$

$$\Rightarrow \frac{dx}{x^5} = -\frac{v dv}{2}$$

$$= \frac{-1}{2} \int \frac{v dv}{v^2 \sqrt{v-1}}$$

$$= \frac{-1}{2} \int \frac{dv}{v \sqrt{v-1}}$$

$$\text{Let } v-1 = t^2$$

$$\Rightarrow dv = 2t dt$$

$$= \frac{-1}{2} \int \frac{2t dt}{(t^2+1)\sqrt{t^2}} = -\int \frac{dt}{t^2+1}$$

$$= -\tan^{-1}(t) + C$$

$$= -\tan^{-1}(\sqrt{v-1}) + C$$

$$= -\tan^{-1}\left(\sqrt{1+\frac{1}{x^4}-1}\right)+C$$

108. Let $I_1 = \int \tan x \tan(ax+b) dx$ and $I_2 = \int \cot x \cot(ax+b) dx$:

Column-I

Column-II

(A) value of I_1 for $a = 1$ is

(P) $x - \cot b \ln \frac{\cos(x-b)}{\cos x} + C$

(B) value of I_2 for $a = 1$ is

(Q) $\cot b \ln \frac{\sin x}{\sin(x+b)} - x + C$

(C) value of I_1 for $a = -1$ is

(R) $\cot b \ln \left(\frac{\cos x}{\cos(x+b)} \right) - x + C$

(D) value of I_2 for $a = -1$ is

(S) $x + \cot b \ln \left(\frac{\sin x}{\sin(b-x)} \right) + C$

माना $I_1 = \int \tan x \tan(ax+b) dx$

तथा $I_2 = \int \cot x \cot(ax+b) dx$:

स्तम्भ-I

स्तम्भ-II

(A) I_1 का मान $a = 1$ के लिये है

(P) $x - \cot b \ln \frac{\cos(x-b)}{\cos x} + C$

(B) I_2 का मान $a = 1$ के लिये है

(Q) $\cot b \ln \frac{\sin x}{\sin(x+b)} - x + C$

(C) I_1 का मान $a = -1$ के लिये है

(R) $\cot b \ln \left(\frac{\cos x}{\cos(x+b)} \right) - x + C$

(D) I_2 का मान $a = -1$ के लिये है

(S) $x + \cot b \ln \left(\frac{\sin x}{\sin(b-x)} \right) + C$

Ans. (A) R ; (B) Q ; (C) P ; (D) S

Sol. $I_1 = \int \tan x \tan(ax+b) dx$

For $a = 1$

(A) $I_1 = \int \tan x \left(\frac{\tan x + \tan b}{1 - \tan x \tan b} \right) dx$

$$= \int \frac{\tan^2 x}{1 - \tan x \tan b} dx + \tan b \int \frac{\tan x}{1 - \tan x \tan b} dx$$

$$= \int \frac{\sec^2 x - 1}{(1 - \tan x \tan b)} dx + \tan b \int \frac{\tan x}{1 - \tan x \tan b} dx$$

$$= \int \frac{\sec^2 x}{1 - \tan x \tan b} dx - \int dx$$

$$\cot b \int \frac{\sec^2 x dx}{\cot b - \tan x} - x + C$$

$$= -\cot b \ln |\cot b - \tan x| - x + C$$

$$= -\cot b \ln \left| \frac{\cos(b+x)}{\sin b \cos x} \right| - x + C$$

$$= \cot b \ln \left(\frac{\cos x}{\cos(b+x)} \right) - x + C$$

(B) For $a = 1$

$$I_2 = \int \cot x \cot(x+b) dx$$

$$= \int \cot x \left(\frac{\cot b \cot x - 1}{\cot b + \cot x} \right) dx$$

$$= \int \frac{\cot b (\cos^2 x - 1) - \cot x}{\cot b + \cot x} dx$$

$$= \cot b \int \frac{\cos^2 x}{\cot b + \cot x} dx - x + C$$

$$= -\cot b \ln \left| \frac{\cos b}{\sin b} + \frac{\cos x}{\sin x} \right| - x + C$$

$$= \cot b \ln \left| \frac{\sin x}{\sin(x+b)} \right| - x + C$$

(C) For $a = -1$

$$I_1 = \int \tan x \tan(b-x) dx = \int \tan x \frac{(\tan b - \tan c)}{1 + \tan b \tan c} dx$$

$$= \int \frac{\tan x \tan b - (\sec^2 x - 1)}{1 + \tan b \tan x} dx$$

$$= \int \left(1 - \frac{\sec^2 x}{1 + \tan x \tan b} \right) dx$$

$$= x - \cot b \ln |\tan x + \cot b| + C$$

$$= x - \cot b \ln \left| \frac{\cos(x-b)}{\cos x \sin b} \right| + C$$

$$= -\cot b \ln \left| \frac{\cos(x-b)}{\cos x} \right| + x + C$$

(D) For $a = -1$

$$I_2 = \int \cot x \left(\frac{\cot x \cot b + 1}{\cot x - \cot b} \right) dx$$

$$= \int \frac{(\cos^2 x - 1) \cot b + \cos x}{\cot x - \cot b} dx$$

$$= -\cot b \ln |\cot x - \cot b| + x + C$$

$$= x + \cot b \ln \left| \frac{\sin x}{\sin(b-x)} \right| + C$$

109. Let $I = \int \frac{e^x}{e^{4x} + 1} dx$ and $J = \int \frac{e^{-x}}{e^{-4x} + 1} dx$. Then for any arbitrary constant C, match the following :

यदि $I = \int \frac{e^x}{e^{4x} + 1} dx$ और $J = \int \frac{e^{-x}}{e^{-4x} + 1} dx$ तब किसी स्वेच्छ अचर C के लिए :

Column-I

(A) I

(B) J + I

(C) J - I

Column-II

(P) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{e^{2x} - 1}{\sqrt{2}e^x} \right) + C$

(Q) $\frac{1}{2\sqrt{2}} \ln \left(\frac{e^{2x} - \sqrt{2}e^x + 1}{e^{2x} + \sqrt{2}e^x + 1} \right) + C$

(R) $\frac{1}{2\sqrt{2}} \left(\tan^{-1} \left(\frac{e^{2x} - 1}{\sqrt{2}e^x} \right) - \frac{1}{2} \ln \left(\frac{e^{2x} - \sqrt{2}e^x + 1}{e^{2x} + \sqrt{2}e^x + 1} \right) \right) + C$

(S) $\frac{1}{2\sqrt{2}} \left(\tan^{-1} \left(\frac{e^{2x} - 1}{\sqrt{2}e^x} \right) + \frac{1}{2} \ln \left(\frac{e^{2x} - \sqrt{2}e^x + 1}{e^{2x} + \sqrt{2}e^x + 1} \right) \right) + C$

Ans. (A) R ; (B) P; (C) Q

Sol. Given $I = \int \frac{e^x}{1 + e^{4x}} dx$ and $J = \int \frac{e^{-x}}{1 + e^{-4x}} dx$

$$J = \int \frac{e^{4x} \cdot e^{-x}}{1+e^{4x}} dx = \int \frac{e^{3x}}{1+e^{4x}} dx$$

$$\therefore I + J = \int \frac{e^x(1+e^{2x})}{1+e^{4x}} dx$$

put $e^x = t \Rightarrow e^x dx = dt$

$$\therefore I + J = \int \frac{1+t^2}{1+t^4} dt = \int \frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}} dt = \int \frac{1+\frac{1}{t^2}}{\left(t-\frac{1}{t}\right)^2+2} dt$$

put $t - \frac{1}{t} = y \Rightarrow y = \left(t + \frac{1}{t^2}\right) dt = dy$

$$= \int \frac{dy}{y^2+2} = \frac{1}{\sqrt{2}} \tan^{-1} \frac{y}{\sqrt{2}} = \frac{1}{\sqrt{2}} \tan^{-1} \frac{t^2-1}{\sqrt{2}t}$$

$$J + I = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{e^{2x}-1}{\sqrt{2}e^x} \right) + C \quad \dots(1) \Rightarrow (P)$$

$$\text{||ly } J - I = \int \frac{e^x(-1+e^{2x})}{1+e^{4x}} dx = \int \frac{-1+t^2}{1+t^4} dt = + \int \frac{-1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}} dt = + \int \frac{-1+\frac{1}{t^2}}{\left(t+\frac{1}{t}\right)^2-2} dt \quad \left(t+\frac{1}{t}=y\right)$$

$$\therefore J - I = + \int \frac{dy}{y^2-2} = -\frac{1}{2\sqrt{2}} \ln \frac{y-\sqrt{2}}{y+\sqrt{2}}$$

$$J - I = -\frac{1}{2\sqrt{2}} \ln \frac{t+\frac{1}{t}-\sqrt{2}}{t+\frac{1}{t}+\sqrt{2}} = \frac{1}{2\sqrt{2}} \cdot \frac{t^2-\sqrt{2}t+1}{t^2+\sqrt{2}t+1} = \frac{1}{2\sqrt{2}} \cdot \frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1} \quad \dots(2) \Rightarrow (Q)$$

(1) - (2) gives

$$2I = \frac{1}{\sqrt{2}} \tan^{-1} \frac{e^{2x}-1}{\sqrt{2}e^x} - \frac{1}{2\sqrt{2}} \frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}$$

$$I = \frac{1}{2\sqrt{2}} \left[\tan^{-1} \frac{e^{2x}-1}{\sqrt{2}e^x} - \frac{1}{2} \cdot \frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1} \right] \Rightarrow (R)$$

110. Match the following column : (where C is the constant of integration.)

निम्न कॉलम को सुमेलित कीजिए : (जहाँ C समाकलन नियंताक है)

Column-I

Column-II

(A) $\int \frac{x^2(x^6 + x^5 - 1)dx}{(2x^6 + 3x^5 + 2)^2}$

(P) $\frac{1}{6} \left(\frac{1}{x^{-3}} + \frac{3}{x^{-2}} \right)^{\frac{1}{2}} + C$

(B) $\int \frac{(x^5 + x^4 + x^2)dx}{\sqrt{4x^7 + 5x^6 + 10x^4}}$

(Q) $\frac{1}{2} (1 + x^{-2} + x^{-5})^{-2} + C$

(C) $\int \frac{(2x^{12} + 5x^9)dx}{(x^5 + x^3 + 1)^3}$

(R) $-\frac{1}{6} (2x^3 + 3x^2 + 2x^{-3})^{-1} + C$

(S) $x \left(\frac{x^3}{25} + \frac{x^2}{20} + \frac{1}{10} \right)^{\frac{1}{2}} + C$

Ans. (A) R, (B) S, (C) Q

Sol. (A) Let $I_1 = \int \frac{x^2(x^6 + x^5 - 1)dx}{(2x^6 + 3x^5 + 2)^2} = \int \frac{x^2(x^6 + x^5 - 1)dx}{x^6 \left(2x^3 + 3x^2 + \frac{2}{x^3} \right)^2} = \int \frac{(x^2 + x - x^{-4})dx}{(2x^3 + 3x^2 + 2x^{-3})^2}$

Putting $2x^3 + 3x^2 + 2x^{-3} = t \Rightarrow 6 \left(x^2 + x - \frac{1}{x^4} \right) dx = dt \Rightarrow \frac{(x^6 + x^5 - 1)}{x^4} dx = \frac{1}{6} dt$

Hence $I_1 = \frac{1}{6} \int \frac{dt}{t^2} = \frac{-1}{6t} + C = \frac{-1}{6(2x^3 + 3x^2 + 2x^{-3})} + C = -\frac{1}{6} (2x^3 + 3x^2 + 2x^{-3})^{-1} + C.$

(B) Let $I_2 = \int \frac{(x^5 + x^4 + x^2)dx}{\sqrt{4x^7 + 5x^6 + 10x^4}} = \int \frac{(x^5 + x^4 + x^2)dx}{x \sqrt{4x^5 + 5x^4 + 10x^2}} = \int \frac{(x^4 + x^3 + x)dx}{\sqrt{4x^5 + 5x^4 + 10x^2}}$

Putting $4x^5 + 5x^4 + 10x^2 = t \Rightarrow 20(x^4 + x^3 + x)dx = dt$

Hence $I_2 = \frac{1}{20} \int \frac{dt}{\sqrt{t}} = \frac{1}{10} \sqrt{t} + C = \frac{1}{10} \sqrt{4x^5 + 5x^4 + 10x^2} + C = \frac{1}{10} x \sqrt{4x^3 + 5x^2 + 10} + C$

$= x \left(\frac{x^3}{25} + \frac{x^2}{20} + \frac{1}{10} \right)^{\frac{1}{2}} + C$

(C) Let $I_3 = \int \frac{2x^{12} + 5x^9}{(x^5 + x^3 + 1)^3} dx = \int \frac{2x^{12} + 5x^9}{x^{15} (1 + x^{-2} + x^{-5})^3} dx = \int \frac{(2x^{-3} + 5x^{-6})dx}{(1 + x^{-2} + x^{-5})^3}$

Putting $1 + x^{-2} + x^{-5} = t \Rightarrow (-2x^{-3} - 5x^{-6})dx = dt$

$$\therefore I_3 = -\int \frac{dt}{t^3} = \frac{1}{2t^2} + C = \frac{1}{2}(1+x^{-2}+x^{-5})^{-2} + C$$

111. Match the integrals in **Column I** with the values in **Column II**.

स्तम्भ का मिलान कीजिए।

Column I

Column II

(A) $\int_{-1}^1 \frac{dx}{1+x^2}$

(P) $\frac{1}{2} \log\left(\frac{2}{3}\right)$

(B) $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

(Q) $2 \log\left(\frac{2}{3}\right)$

(C) $\int_2^3 \frac{dx}{1-x^2}$

(R) $\frac{\pi}{3}$

(D) $\int_1^2 \frac{dx}{x\sqrt{x^2-1}}$

(S) $\frac{\pi}{2}$

Ans. (A) S, (B) S, (C) P, (D) R

Sol. (A) $\int_{-1}^1 \frac{dx}{1+x^2} = \left[\tan^{-1} x \right]_{-1}^1 = \frac{\pi}{2}$

(B) $\int_0^1 \frac{dx}{\sqrt{1-x^2}} = \left[\sin^{-1} x \right]_0^1 = \frac{\pi}{2}$

(C) $\int_2^3 \frac{dx}{1-x^2} = \left[\frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| \right]_2^3$

$= \frac{1}{2} \ln\left(\frac{2}{3}\right)$

(D) $\int_1^2 \frac{dx}{x\sqrt{x^2-1}} = \left(\sec^{-1} x \right)_1^2 = \frac{\pi}{3} - 0 = \frac{\pi}{3}$